

Policy learning “without” overlap: Pessimism and generalized empirical Bernstein’s inequality

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Abstract

This paper studies offline policy learning, which aims at utilizing observations collected a priori (from either fixed or adaptively evolving behavior policies) to learn an optimal individualized decision rule that achieves the best overall outcomes for a given population. Existing policy learning methods rely on a uniform overlap assumption, i.e., the propensities of exploring *all* actions for *all* individual characteristics must be lower bounded; put differently, the performance of the existing methods depends on the worst-case propensity in the offline dataset. As one has no control over the data collection process, this assumption can be unrealistic in many situations, especially when the behavior policies are allowed to evolve over time with diminishing propensities for certain actions.

In this paper, we propose Pessimistic Policy Learning (PPL), a new algorithm that optimizes lower confidence bounds (LCBs) — instead of point estimates — of the policy values. The LCBs are constructed using knowledge of the behavior policies for collecting the offline data. Without assuming any uniform overlap condition, we establish a data-dependent upper bound for the suboptimality of our algorithm, which only depends on (i) the overlap for the *optimal* policy, and (ii) the complexity of the policy class we optimize over. As an implication, for adaptively collected data, we ensure efficient policy learning as long as the propensities for optimal actions are lower bounded over time, while those for suboptimal ones are allowed to diminish arbitrarily fast. In our theoretical analysis, we develop a new self-normalized type concentration inequality for inverse-propensity-weighting estimators, generalizing the well-known empirical Bernstein’s inequality to unbounded and non-i.i.d. data. We complement our theory with an efficient optimization algorithm via Majorization-Minimization and policy tree search, as well as extensive simulation studies and real-world applications that demonstrate the efficacy of PPL.

1 Introduction

Policy learning aims at constructing an optimal individualized decision rule that achieves the best overall outcomes for a given population (Manski, 2004; Hirano and Porter, 2009; Stoye, 2009, 2012). Due to its broad application in healthcare (Murphy, 2003; Kim et al., 2011; Bertsimas et al., 2017), advertising (Bottou et al., 2013; Farias and Li, 2019), recommendation systems (Schnabel et al., 2016; Li et al., 2011; Swaminathan et al., 2017), etc., how to utilize pre-collected data for efficient policy learning has received extensive attention in various fields including economics, causal inference, and statistical machine learning.

This paper studies policy learning from an offline dataset $\mathcal{D} = \{(X_t, A_t, Y_t)\}_{t=1}^T$ collected by some agent a priori. To be precise, $X_t \sim \mathbb{P}_X$ are i.i.d. contexts (i.e., individual characteristics), and A_t is the action taken according to the *behavior policy* (also known as the propensity) $e_t(x, a) := \mathbb{P}(A_t = a | X_t = x)$; the outcome (reward) is $Y_t = \mu(X_t, A_t) + \epsilon_t$, where $\mu(\cdot, \cdot)$ is an unknown function, and $\{\epsilon_t\}_{t=1}^T$ are independent mean-zero random variables. We assume knowledge of $e_t(\cdot, \cdot)$ for all $t \in [T]$, but have no control over how it is specified by the experimenter. A policy π is a mapping from the context space to the action space, and its performance is measured by the *policy value* $Q(\pi) = \mathbb{E}[\mu(X, \pi(X))]$, i.e., the expected reward by executing π on future contexts. Given a policy class Π and the offline dataset $\mathcal{D}$, our goal is to select a policy $\pi \in \Pi$

whose performance is as close as possible to the optimal policy $\pi^* = \operatorname{argmax}_{\pi \in \Pi} Q(\pi)$. We consider two types of data collection processes that are ubiquitous in practice.

- (a) **Batched data:** the dataset is collected by executing a fixed and potentially suboptimal policy, so that $\{(X_t, A_t, Y_t)\}_{t=1}^T$ are i.i.d. tuples. Such datasets may come from randomized controlled trials using a pre-fixed treatment rule (Murphy, 2003; Athey and Imbens, 2017), A/B testing in online platforms (Kohavi et al., 2020), online recommendation systems with a static rule (Schnabel et al., 2016), to name a few.
- (b) **Adaptively collected data:** the dataset is collected by running an adaptive learning algorithm, where the behavior policies are progressively modified to achieve certain goals of the experimenter, e.g., improving statistical efficiency in adaptive experiments (Anscombe, 1963; Simon, 1977; Murphy, 2005; Collins et al., 2007), or maximizing cumulative performance by bandit algorithms in production systems (Lai et al., 1985; Thompson, 1933; Agrawal and Goyal, 2013). This setting strictly generalizes the first one by allowing for adaptivity. It is more challenging due to data dependence and possibly diminishing exploration, and is studied only very recently (Zhan et al., 2021; Bibaut et al., 2021).

Existing algorithms for offline policy learning are based on the principle of welfare maximization (Kitagawa and Tetenov, 2018), which typically consist of two components: evaluation and optimization (e.g., Athey and Wager, 2021; Zhou et al., 2022; Zhan et al., 2021; Bibaut et al., 2021). They first construct policy value estimators $\hat{Q}(\pi)$ for all $\pi \in \Pi$, and then optimize the point estimates to return $\hat{\pi} = \operatorname{argmax}_{\pi \in \Pi} \hat{Q}(\pi)$. As we shall discuss in Section 3.1, a good performance of such “greedy” learners relies crucially on accurate evaluation of *all* policies in Π . This further requires that $e_t(x, \pi(x))$ is sufficiently large for all $\pi \in \Pi$: intuitively, all actions that are of interest are taken sufficiently many times in the offline dataset. We call it the *uniform overlap assumption*, as $e_t(x, \pi(x))$ measures the *overlap* between the behavior policies and π . Accordingly, the worst-case overlap $\inf_{\pi \in \Pi} e_t(x, \pi(x))$ determines the learning performance of those methods.

Although uniform overlap is commonly assumed in the policy learning literature, it can be violated in many practical scenarios. When learning from rich policy classes, it essentially posits that $e_t(x, a)$ must be large for all $(x, a) \in \mathcal{X} \times \mathcal{A}$. In the adaptive setting, imagining the data is collected by a bandit algorithm that gradually decreases exploration (Thompson, 1933), the propensities $e_t(x, a)$ will diminish over time for some (x, a) pairs, often at a rate faster than required by existing policy learning methods (Bembom and van der Laan, 2008; Hadad et al., 2021). In the batched setting, physical constraints (e.g., some actions are expensive to take) may lead to highly unbalanced sampling of different actions. In these cases, the greedy algorithms may only provide vacuous guarantees due to the lack of uniform overlap. It is worth noting several recent efforts to overcome the overlap issue broadly in the causal inference literature; they either modify the evaluated policy (Kennedy, 2019) or clip tiny propensities (Swaminathan and Joachims, 2015), both introducing considerable bias. As such, the fundamental role of overlap in policy learning remains an open question. To fill in this gap, we aim to advance the understanding of the fundamental limit of policy learning in the absence of uniform overlap.

1.1 Pessimistic policy learning

In this paper, we propose a new framework, Pessimistic Policy Learning (PPL), to demonstrate that uniform overlap, while sufficient, is not a necessary condition for efficient policy learning from a given policy class. PPL learns a high-quality policy whose gap from the optimal policy π^* scales with the overlap of π^* , thereby strictly improving upon existing methods and automatically adapting to the quality of the offline data.

The crux of PPL is a first-of-the-kind instantiation of the *pessimism* principle (Jin et al., 2021) in policy learning. PPL optimizes a carefully designed class of confidence lower bounds (LCBs) – instead of point estimates – of the policy values, where the LCBs are constructed to reflect the estimation error of commonly-used inverse-propensity-weighting (IPW)-type estimators.

Table 1 summarizes the key features of PPL in comparison to prior works. The upper half of it contrasts PPL with representative works in the policy learning literature. Besides providing a comprehensive treatment for both batched and adaptive data, PPL achieves a so-called oracle property where the performance adapts to the overlap of π^* , and the general guarantees do not rely on any overlap condition. By introducing pessimism, PPL uniformly improve upon these methods Kitagawa and Tetenov (2018); Zhan et al. (2021);

Swaminathan and Joachims (2015) without paying a price. We note the comparison to Swaminathan and Joachims (2015) which relies on clipping tiny propensities: while clipping is central to facilitate their analysis of IPW-type estimators, it loses the oracle adaptivity provided by pessimism. This highlights the importance of careful treatment of unbounded propensities.

Moreover, PPL applies without structural assumptions on the regression function $\mu(x, a)$. This is the benefit of the unbiasedness and consistency of IPW-type estimators, which is in contrast to existing algorithms under the hood of pessimism listed in the lower half of Table 1 (these works mostly address offline reinforcement learning; see Section 2.4 for a summary). This line of works exclusively rely on structural assumptions on quantities similar to $\mu(x, a)$ to obtain valid quantification of uncertainty. However, as pointed by (Dimakopoulou et al., 2017), the risk of mis-specification is an important issue in policy learning especially in critical domains, where tradeoff between knowledge of behavior policy and structural assumptions becomes necessary. As such, our statistical techniques are in sharp contrast to these works.

	Adaptive data	Structure free	Oracle property	No overlap	Variance adaptive	Knowledge of behavior policy
Kitagawa and Tetenov (2018)	✗	✓	✗	✗	✗	assumed
Swaminathan and Joachims (2015)	✗	✓	✗	✓	✓	assumed
Zhan et al. (2021)	✓	✓	✗	✗	✗	assumed
Jin et al. (2021)	✓	✗	✓	✓	✓	not required
Xie et al. (2021a)	✗	✗	✓	✓	✓	not required
This work	✓	✓	✓	✓	✓	assumed

Table 1: Summary of related works. In each column, we denote whether the method achieves the property by ✓ = achievement, ✗ = no achievement, and ✓ = partial achievement. The first three methods are from the policy learning literature, all based on the greedy approach. The fourth and fifth lines are from the offline reinforcement learning literature which implement the pessimism principle by leveraging parametric/functional structures of the outcomes, highlighting the tradeoff between outcome modeling and design knowledge.

1.2 Preview of contributions

We summarize the contributions of this work as follows.

- (i) For both batched and adaptively collected data, we propose the generic recipe for PPL, which outputs

$$\hat{\pi} = \operatorname{argmax}_{\pi \in \Pi} \{ \hat{Q}(\pi) - R(\pi) \},$$

where $\hat{Q}(\pi)$ is a commonly-used policy value estimator, and $R(\pi)$ is a policy-dependent, data-dependent quantity that obeys $|\hat{Q}(\pi) - Q(\pi)| \leq R(\pi)$ simultaneously for all $\pi \in \Pi$ with high probability.

- (ii) We then propose to construct uniformly-valid upper bound $R(\pi)$ (hence the LCBs) via a novel concentration inequality for offline policy evaluation (OPE), which roughly states that with high probability,

$$|\hat{Q}(\pi) - Q(\pi)| \lesssim \text{Complexity}(\Pi) \cdot \sqrt{\widehat{\text{Var}}(\hat{Q}(\pi))}, \quad \text{for all } \pi \in \Pi.$$

Here $\hat{Q}(\pi)$ is the augmented-inverse-propensity-weighting (AIPW) estimator for $Q(\pi)$, and $\widehat{\text{Var}}(\hat{Q}(\pi))$ is an estimator for the variance of $\hat{Q}(\pi)$ (the definition is given in Section 3.3). We establish this result using tools from tree symmetrization (Rakhlin et al., 2015) and self-normalized processes (De la Pena et al., 2004). Its single-policy version can be viewed as a generalization of the well-known empirical Bernstein’s inequality (Bartlett et al., 2005, 2006; Koltchinskii, 2006) to unbounded and dependent data, which may be of independent interest for other reweighting-based methods (Jin et al., 2023; Gui

et al., 2022); we discuss the technical novelty in more details in Section 3.2 and compare our generalized empirical Bernstein’s inequality with existing ones in Section 2.4.

- (iii) With (i) as the key technical tool, we establish the particular benefit of PPL in terms of a generic data-dependent upper bound on its performance, in the form of

$$Q(\pi^*) - Q(\hat{\pi}) \lesssim \text{Complexity}(\Pi) \cdot \text{EstErr}(\pi^*) \lesssim \text{Complexity}(\Pi) \cdot \sqrt{\widehat{\text{Var}}(\hat{Q}(\pi^*))},$$

where $\hat{\pi}$ is the learned policy, and π^* is the optimal policy. The first term in the upper bound is a complexity measure of Π (see Section 2.3 for its precise definition). For any $\pi \in \Pi$, $\widehat{\text{Var}}(\hat{Q}(\pi))$ is an estimator for the policy evaluation error; it is small as long as the overlap for π is large. Thus, the learned policy $\hat{\pi}$ has a small suboptimality gap as long as the overlap for π^* is sufficiently good, while no assumption is needed for suboptimal policies. This shows an ‘oracle’ property of our approach (see Figure 1), i.e., it adapts to the overlap of π^* in the data, even if π^* is not known a priori.

- (iv) When specialized to batched data, in Section 4, we achieve the standard $O(1/\sqrt{T})$ rate for the learning performance when $e_t(X, \pi^*(X))$ is lower bounded by a constant almost surely. This rate is sharper than the existing results and hold under weaker conditions. We also provide a matching minimax lower bound in this setting, showing that pessimism is the best effort given the dataset.
- (v) For adaptively collected data, in Section 5, we show that our algorithm achieves polynomial learning rate when the propensity for the *optimal* policy is decaying at a polynomial rate. As a special case, we achieve $O(1/\sqrt{T})$ rate if $e_t(X, \pi^*(X))$ is uniformly lower bounded by a constant, while propensities of suboptimal policies are allowed to decay at a much faster (and potentially non-polynomial) rate. We complement our polynomial upper bound with a matching minimax lower bound; in the information-theoretic sense, the overlap of the *optimal* policy captures the fundamental difficulty in offline policy learning with adaptively collected data.
- (vi) Section 6 addresses practical challenges in implementing pessimistic policy learning. To tackle the challenge that our objective is non-convex, we devise an efficient approximate optimization algorithm that combines Majorization-Minimization (Van Den Burg and Groenen, 2016) and policy tree search (Sverdrup et al., 2020). Additionally, we develop a cross-validation strategy specialized to sequential data to choose penalty parameters with favorable performance. Finally, in Section 7, we evaluate our methods via extensive simulation studies and application to 33 real-world datasets. Our results confirm the benefit of pessimism in policy learning, as well as the efficacy of our optimization algorithms.

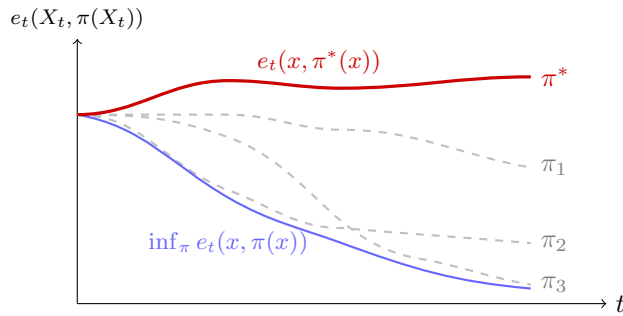


Figure 1: An illustration of the oracle property. The performance of pessimistic policy learning only depends on the overlap for the optimal policy π^* (red), while the worst-case overlap $\inf_{\pi} e_t(x, \pi(x))$ is allowed to be small (or vanish over time). See Sections 4 and 5 for more detailed discussion.

2 Background

Let $\mathcal{X}$ denote the space of the contexts and $\mathcal{Y}$ that of the reward. A contextual bandit model $\mathcal{C}$ is specified by an action set $\mathcal{A} = \{a_1, \dots, a_K\}$ for $K = |\mathcal{A}|$, a context distribution $\mathbb{P}_X$ and a set of reward distributions

$\{\mathbb{P}_Y \mid X=x, A=a\}_{x \in \mathcal{X}, a \in \mathcal{A}}$. We consider policy learning from an offline dataset that is collected from $\mathcal{C}$.

2.1 Data collecting process

At each time $t = 1, 2, \dots, T$, a context X_t is independently drawn from $\mathbb{P}_X$. The experimenter then takes an action $A_t \in \mathcal{A}$ according to the behavior policy, and receives a reward $Y_t = \mu(X_t, A_t) + \epsilon_t$, where ϵ_t are independent mean-zero noise. Throughout, we denote $\mathcal{H}_t = (X_1, A_1, Y_1, \dots, X_{t-1}, A_{t-1}, Y_{t-1})$ as the history *before* time t , so that $\mathbb{P}(A_t = a \mid X_t = x, \mathcal{H}_t) = e_t(x, a \mid \mathcal{H}_t)$. We use $e_t(\cdot, \cdot \mid \mathcal{H}_t)$ to additionally emphasize that the behavior policy is potentially adaptive. In the context of causal inference (Imbens and Rubin, 2015), A_t can be understood as the treatment option taken for X_t , and the reward is $Y_t = Y_t(A_t)$, where $\{Y_t(a)\}_{a \in \mathcal{A}}$ are potential outcomes with $(X_t, Y_t(a)_{a \in \mathcal{A}})$ following a population joint distribution, and we implicitly assume the conditional independence condition: $Y_t(a) \perp\!\!\!\perp A_t \mid X_t, \mathcal{H}_t$. The two types of data generating processes we introduced in Section 1 are formally defined as follows.

- (a) Batched data: the behavior policy is fixed and known, where $\mathbb{P}(A_t = a \mid X_t = x, \mathcal{H}_t) \equiv e(x, a)$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$ and all $t \in [T]$.
- (b) Adaptively collected data: at each time $t \in [T]$, the action is taken according to a history-dependent rule, where $\mathbb{P}(A_t = a \mid X_t = x, \mathcal{H}_t) = e_t(x, a \mid \mathcal{H}_t)$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$. As the behavior policy may depend on previous observations, $\{(X_t, A_t, Y_t)\}_{t=1}^T$ are no longer i.i.d.

For simplicity, in both cases we write $e_t(x, a \mid \mathcal{H}_t)$ as the treatment probability at time $t \in [T]$, with the convention that $e_t(x, a \mid \mathcal{H}_t) \equiv e(x, a)$ for batched data. Throughout, We impose the following minimal assumption on the data generating process.

Assumption 2.1. *The data generating process satisfies the following properties.*

- (a) *Bounded reward: the rewards satisfy $Y_t \in [0, 1]$ almost surely.*
- (b) *Known behavior policy: the behavior policies $e_t(x, a \mid \mathcal{H}_t)$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$ are known to the learner.*

The bounded reward assumption is standard in the literature and the upper bound 1 can be simply achieved by normalization. Assumption 2.1(b) is typically satisfied for data resulting from running bandit algorithms or online experiments, since the agent/experimenter often has access to the policy she uses to collect the data (see e.g., Collins et al. (2007); Li et al. (2010); Offer-Westort et al. (2021)).

2.2 Policy learning and the performance metric

Let Π be a collection of policies, with each element in Π being a mapping $\pi: \mathcal{X} \rightarrow \mathcal{A}$. For any $\pi \in \Pi$, we define its average policy value under the contextual bandit model $\mathcal{C}$ as

$$Q(\pi; \mathcal{C}) = \mathbb{E}_{\mathcal{C}}[\mu(X, \pi(X))] = \sum_{a \in \mathcal{A}} \mathbb{E}_{\mathcal{C}}[\mathbb{1}\{\pi(X) = a\}Y(a)],$$

where the expectation is taken with respect to the context distribution.

The optimal policy $\pi^* \in \Pi$ maximizes the policy value, i.e.,

$$\pi^*(\mathcal{C}, \Pi) = \operatorname{argmax}_{\pi \in \Pi} Q(\pi; \mathcal{C}). \quad (1)$$

Our goal is to learn from the offline data a policy $\hat{\pi} \in \Pi$, where the performance of $\hat{\pi}$ is measured by its *suboptimality*, defined as

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) := Q(\pi^*(\mathcal{C}, \Pi); \mathcal{C}) - Q(\hat{\pi}; \mathcal{C}).$$

The above notations emphasize the dependence of our performance measure on the underlying contextual bandit $\mathcal{C}$ and policy class Π . In the sequel, we omit $\mathcal{C}$ and Π when no confusion arises from the context.

2.3 Policy class complexity

We use the Natarajan dimension (Natarajan, 1989) to quantify the complexity of a policy class Π , which plays an important role in characterizing the performance of policy learning algorithms.

Definition 2.2 (Natarajan dimension). *Given a policy class $\Pi = \{\pi : \mathcal{X} \mapsto \mathcal{A}\}$, we say $\{x_1, x_2, \dots, x_m\}$ is shattered by Π if there exist two functions $f_1, f_2 : \{x_1, x_2, \dots, x_m\} \mapsto \mathcal{A}$ such that the following holds:*

- (1) *For any $j \in [m]$, $f_1(x_j) \neq f_2(x_j)$;*
- (2) *For any subset $S \subseteq [m]$, there exists some $\pi \in \Pi$ such that $\pi(x_j) = f_1(x_j)$ for all $j \in S$ and $\pi(x_j) = f_2(x_j)$ for all $j \notin S$.*

The Natarajan dimension of Π , denoted as $\text{N-dim}(\Pi)$, is the largest size of a set of points shattered by Π .

The Natarajan dimension is a generalization of the well-known Vapnik-Chervonenkis (VC) dimension (Vapnik and Chervonenkis, 2015) in the multi-action classification problem. Finite upper bounds for the Natarajan dimension have been established for many commonly used policy classes including linear function classes, reduction trees, decision trees, random forests and neural networks (Daniely et al., 2011; Jin, 2023).

2.4 Related work

Policy learning with a policy class. This work adds to the literature of policy learning, where existing works have primarily focused on i.i.d. data collected by a fixed behavior policy that is known (Zhang et al., 2012; Zhao et al., 2012; Swaminathan and Joachims, 2015; Zhou et al., 2017; Kitagawa and Tetenov, 2018), or unknown but estimable (Zhao et al., 2015; Athey and Wager, 2021; Zhou et al., 2022; Kallus, 2018). These works all assume a uniform overlap condition, i.e., the propensity for all actions in the behavior policy is uniformly lower bounded by a constant. The overlap assumption is standard in causal inference for estimating average treatment effects (Imbens and Rubin, 2015), and is essential for unknown behavior policy due to the necessity of controlling the error in its estimation (Athey and Wager, 2021; Zhou et al., 2022). However, we show that when the behavior policy is known, the uniform overlap assumption is not necessary for efficient policy learning; our approach is based on an algorithmic change to the standard greedy approach in this literature. Recently, there has been an increasing interest in policy learning from adaptively collected data (Zhan et al., 2021; Bibaut et al., 2021). This setting is more challenging because the observations are no longer i.i.d., and further, adaptively collected data often have diminishing propensities of certain arms, which causes huge variance in the IPW-type estimators and translates to slow rates for the learning performance. To tackle these challenges, existing works assume the propensities for *all* actions are deterministically lower bounded over time; as we introduced, this may be unrealistic for some commonly adopted algorithms. In contrast to the existing literature, we introduce a novel algorithm to policy learning and show that the overlap of the *optimal* policy is sufficient for efficient policy learning. Our results improve the learning performance and broaden the scenarios where efficient policy learning is feasible in both settings.

Inference and learning without overlap. This work adds to the literature on relaxing the overlap condition, a fundamental assumption in causal inference, policy evaluation and learning. Earlier works use strategies like sample trimming to rule out extreme propensities in treatment effect estimation and study asymptotic inference (Yang and Ding, 2018; Branson et al., 2023). There have also been recent works on policy evaluation and learning without overlap, concurrent with or after the first appearance of this work. For example, Khan et al. (2023) studies policy evaluation (i.e., without learning) by extrapolating under smoothness assumptions; Liu et al. (2023) shifts the target to overlap weighted average treatment effect on the treated; Mou et al. (2023) studies policy evaluation via regularized kernel regression, focusing on the minimax optimality under certain structural assumptions. For policy learning, Zhao et al. (2023) overcomes the overlap issue by considering an “incremental” policy that slightly deviates from the observational policy (Kennedy, 2019), and study asymptotic, doubly robust policy learning among the class of incremental policies. In contrast, our work is not limited to the incremental policy class, and finite-sample analysis with pessimism and empirical Bernstein’s inequality is also drastically different from the strategies in those works. Chen et al. (2023) considers the problem under a Markov Decision Process, where the trajectory is assumed stationary, and induced by

a fixed behavior policy (similar to our batched data setting). The main problem therein is the unidentifiable part of policy value due to singularity, i.e., part of the space that has zero probability under the behavior policy; They employ distributionally robust optimization ideas to optimize the worst-case performance of policies, and handle the weak overlap in the estimation of the identifiable part via functional approximation of the conditional mean function. Both the problem and method therein are different from ours.

Pessimism in offline RL and bandits. The principle of pessimism was initially proposed in offline reinforcement learning (RL) (Buckman et al., 2020; Jin et al., 2021) for learning an optimal policy in Markov decision processes (MDPs) using pre-collected datasets (Zanette et al., 2021; Xie et al., 2021a; Chen and Jiang, 2022; Rashidinejad et al., 2021; Yin and Wang, 2021; Shi et al., 2022; Yan et al., 2022; Xie et al., 2021b; Uehara and Sun, 2021) and has been applied to contextual bandits as a special case of MDPs. To the best of our knowledge, this work is the first to instantiate the pessimism principle for learning individualized decision rules that requires no assumptions on the context space or the mean outcome model.

To be specific, existing pessimism-based algorithms optimize LCBs for conditional value functions (which reduce to $\mu(x, a) := \mathbb{E}[Y(a) | X = x]$ in our setting), whose construction either relies on finite cardinality of the context space (Buckman et al., 2020; Rashidinejad et al., 2021) or strong modeling assumptions (Jin et al., 2021; Xie et al., 2021a). In contrast, our proposal does not rely on any modeling assumption on the conditional mean function. It can be viewed as a ‘design-based’ version of the pessimism principle which uses the knowledge of the sampling propensities. This is particularly suitable for continuously-valued contexts and for high-stakes applications where model misspecification is of concern.

Variance-regularized ERM and empirical Bernstein’s inequality. The idea of pessimism also appears in empirical risk minimization (ERM), where the principle is implemented by taking into account the estimation uncertainty of empirical risks. Among them, Maurer and Pontil (2009) proposed to optimize a variance-regularized empirical risk, and was extended by Duchi and Namkoong (2016) to a computationally tractable formulation. The variance regularization is justified by empirical Bernstein’s inequality studied by a series of works (Bartlett et al., 2005, 2006; Koltchinskii, 2006). Recently, Xu and Zeevi (2020) studied instance-dependent risk without explicitly using a regularization term. These works all consider i.i.d. data and bounded loss. Our method extends this line of work to *unbounded* random variables and adaptively collected hence mutually *dependent* data. Closer to our setting, Swaminathan and Joachims (2015) studies counterfactual risk minimization using i.i.d. data and unbounded weights. However, they used clipping to exclude extremely small weights and reduced the analysis to bounded random variables. The clipping causes bias, so the “oracle” property of pessimism (the sole dependence on the *optimal* instance) may not hold. Our work can also be of interest for providing a novel solution to counterfactual risk minimization with unbounded weights.

To deal with non-i.i.d. data and unbounded weights, we develop a new self-normalized-type uniform concentration inequality for empirical risks. When applied to one single policy (i.e., without uniform concentration), it can be viewed as a generalization of the empirical Bernstein’s inequality (Maurer and Pontil, 2009). Several recent independent works also generalize on empirical Bernstein’s inequality, including Waudby-Smith and Ramdas (2020) for bounded data and Waudby-Smith et al. (2022) for estimating a policy value using adaptively collected data. While their methods and analysis are complementary to ours, it will be interesting to see whether these bounds are applicable to providing uniform concentration over policy classes to enable policy learning in our approach.

Other reweighting-based methods. Since we use an IPW-type estimator, this work is generally related to reweighting-based approaches for off-policy evaluation (OPE) in offline RL and contextual bandits such as marginal importance sampling (Liu et al., 2018; Xie et al., 2019; Uehara et al., 2020; Zhang et al., 2020), whose goal is to estimate the value of a policy from a pre-collected data set. Several works in this strand also consider policy optimization (Nachum et al., 2019; Lee et al., 2021; Zhan et al., 2022; Rashidinejad et al., 2022) with some pessimism (or conservativeness) features. These works operate under uniformly bounded weights, i.e., the uniform overlap condition. Our empirical Bernstein’s inequality applies to unbounded weights, which may also be of interest for extending these works to broader settings. Finally, similar to

our motivation, Kuzborskij et al. (2021) studies estimation of $Q(\pi)$ for a small number of policies π with i.i.d. data without uniform overlap between π and the behavior policy. They use self-normalized importance weighting to trade bias for reduced variance. Compared with them, our work uses a different estimator, applies to adaptively collected data, and studies policy optimization instead of policy evaluation. It would be interesting to see whether their ideas may lead to other construction of LCBs for similar ideas as PPL that works also for the adaptive case.

We summarize our contributions and their relation to existing literature in Figure 2.

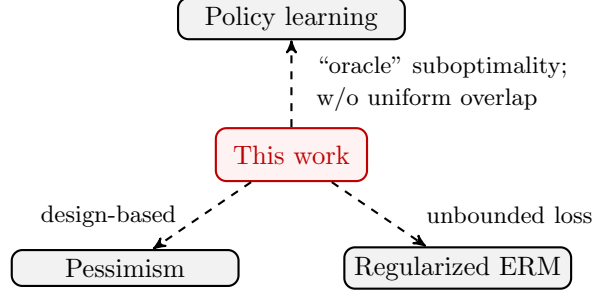


Figure 2: Our contributions and their relation to existing literature.

3 Pessimistic policy learning

3.1 What makes offline policy learning difficult?

Before presenting our method, we first discuss the necessity of controlling the worst-case estimation error for greedy approaches, which explains the dependence on stringent uniform overlap assumptions.

As introduced earlier, a greedy learner returns $\hat{\pi}_{\text{greedy}} = \arg\max_{\pi \in \Pi} \hat{Q}(\pi)$. Here, $\hat{Q}(\pi)$ is typically an IPW-type (unbiased) point estimator, which takes the form $\hat{Q}(\pi) = \frac{1}{T} \sum_{t=1}^T \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} Y_t$. We can decompose the suboptimality of $\hat{\pi} = \hat{\pi}_{\text{greedy}}$ into

$$Q(\pi^*) - Q(\hat{\pi}) = \underbrace{Q(\pi^*) - \hat{Q}(\pi^*)}_{\text{(i) intrinsic uncertainty}} + \underbrace{\hat{Q}(\pi^*) - \hat{Q}(\hat{\pi})}_{\text{(ii) optimization error}} + \underbrace{\hat{Q}(\hat{\pi}) - Q(\hat{\pi})}_{\text{(iii) greedy uncertainty}}. \quad (2)$$

Above, (i) is the estimation uncertainty for the optimal policy π^* , (ii) is the nonpositive optimization error, and (iii) is the estimation uncertainty of the greedily learned policy. Among these three sources of uncertainty, it is most challenging to control (iii), since it might well be the case that $\hat{\pi}_{\text{greedy}}$ appears to be optimal *as a result of* a large estimation error.

In Figure 3, we consider a simple case with two policies $\Pi = \{\pi_1, \pi_2\}$ and a fixed behavior policy $e(x, a)$ to illustrate how poor estimation of suboptimal policies confuses the greedy learner. In this example, $\pi^* = \pi_1$ is well-explored by the behavior policy — i.e., its propensities $e(X_t, \pi^*(X_t))$ are all large — and as a result, $\hat{Q}(\pi_1) - Q(\pi_1)$ is of a small scale; in contrast, the suboptimal policy π_2 is less explored by the behavior policy and has small propensities, which translates to a large scale of $\hat{Q}(\pi_2) - Q(\pi_2)$. Thus, even though $Q(\pi_2)$ is smaller than $Q(\pi_1)$, a greedy learner may still choose $\hat{\pi}_{\text{greedy}} = \pi_2$ since there is a considerable chance that $\hat{Q}(\pi_2)$ appears to be larger than $\hat{Q}(\pi_1)$. In other words, to ensure a small suboptimality of $\hat{\pi}_{\text{greedy}}$, we need both policies to be accurately estimated.

The standard approach to controlling (iii) is by bounding the worst-case estimation error $V_{\text{worst}} := \sup_{\pi \in \Pi} |\hat{Q}(\pi) - Q(\pi)|$, which is indeed necessary for the greedy approach as illustrated by the above example. Since the estimation error (or roughly, the standard deviation) of $\hat{Q}(\pi)$ depends on the inverse propensity for π , controlling V_{worst} often relies on assuming sufficiently large $e_t(X_t, \pi(X_t) | \mathcal{H}_t)$ for all $\pi \in \Pi$ and all $t \in [T]$ (Kitagawa and Tetenov, 2018; Zhan et al., 2021; Bibaut et al., 2021). As we discussed earlier,

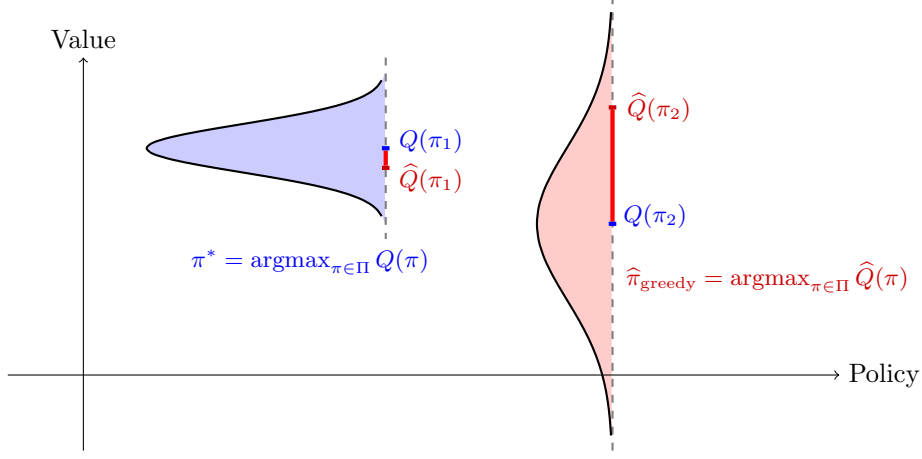


Figure 3: An illustration of how poor estimation of a suboptimal policy π_2 (red) leads to large suboptimality of the greedy approach. Here, $Q(\pi_i)$ is the actual policy value, and $\hat{Q}(\pi_i)$ the IPW estimator for $i = 1, 2$. The shaded area represents the distribution of two estimators. In this case, the greedy learner picks π_2 because $\hat{Q}(\pi_2)$ appears to be large due to its large estimation error.

such assumptions often fail to hold in practice. In contrast, term (i) is much easier to control since it is the estimation error of one single policy. Indeed, we will show that for both batched and adaptively collected data, (i) is the key quantity that captures the intrinsic difficulty of policy learning, as it arises in the information-theoretic lower bound.

3.2 Pessimistic policy learning

To eliminate the reliance on the uniform overlap assumption, we make an algorithmic change to the greedy learning paradigm by introducing the pessimism principle (Jin et al., 2021).

For every policy $\pi \in \Pi$, we complement the point estimator $\hat{Q}(\pi)$ with a policy-dependent regularization term $R(\pi)$, and optimize the *pessimistic policy value*

$$\hat{\pi} := \operatorname{argmax}_{\pi \in \Pi} \{\hat{Q}(\pi) - R(\pi)\}. \quad (3)$$

Roughly speaking, we expect $R(\pi)$ to reflect the estimation error of $\hat{Q}(\pi)$, and therefore (3) can be viewed as optimizing the lower confidence bounds (LCBs) for the policy values. The core merit of the pessimism principle is captured by the following proposition.

Proposition 3.1. *Let $\hat{\pi}$ be obtained as in (3), and let $\pi^* = \pi^*(\mathcal{C}, \Pi)$ be the optimal policy among Π . Then*

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq 2R(\pi^*)$$

holds on the event

$$\mathcal{E} := \{|\hat{Q}(\pi) - Q(\pi)| \leq R(\pi), \forall \pi \in \Pi\}. \quad (4)$$

Proof of Proposition 3.1. The uniform concentration on $\mathcal{E}$ ensures $Q(\pi^*) \leq \hat{Q}(\pi^*) + R(\pi^*)$ and $Q(\hat{\pi}) \geq \hat{Q}(\hat{\pi}) - R(\hat{\pi})$. On the other hand, the optimality of $\hat{\pi}$ implies $\hat{Q}(\hat{\pi}) - R(\hat{\pi}) \geq \hat{Q}(\pi^*) - R(\pi^*)$. Collectively,

$$Q(\hat{\pi}) \geq \hat{Q}(\pi^*) - R(\pi^*) \geq Q(\pi^*) - 2R(\pi^*).$$

By the definition of suboptimality, we conclude the proof of Proposition 3.1. $\square$

A crucial consequence of Proposition 3.1 is that the suboptimality of $\hat{\pi}$ in (3) only depends on the regularization term (roughly, the estimation uncertainty) of the *optimal* policy. This result shows the particular role of pessimism in eliminating the most challenging term (iii) in (2). We will see in Sections 4 and 5 that with a proper construction, $R(\pi^*)$ is small as long as π^* has sufficiently large propensities, and thus we achieve efficient policy learning under a significantly weaker condition than the uniform overlap.

Both the learning performance guarantee and the uniform concentration event $\mathcal{E}$ in our approach significantly differ from the greedy algorithms in the literature. For the greedy algorithm, the theoretical guarantee often relies on the fact that $\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq 2R$ on the event

$$\mathcal{E}_{\text{greedy}} := \{|\hat{Q}(\pi) - Q(\pi)| \leq R, \forall \pi \in \Pi\},$$

where $R \in \mathbb{R}$ is a constant that upper bounds the worst-case estimation error but does not appear in the optimization problem. In contrast, we seek a data-dependent bound $R(\pi)$ for every policy $\pi \in \Pi$. Incorporating this piece of information into our optimization problem leads to an improved data-dependent suboptimality upper bound.

Proper construction of $R(\pi)$ is key to efficient policy learning bounds as in Proposition 3.1: we need it to tightly reflect the estimation quality of $\hat{Q}(\pi)$ for each individual policy $\pi \in \Pi$. In the rest of this paper, our efforts are devoted to constructing the regularization term and proving the uniform concentration result in the form of (4). The key technical milestone is a high-probability upper bound on the self-normalized-type quantity

$$\sup_{\pi \in \Pi} \frac{|\frac{1}{T} \sum_{t=1}^T \hat{\Gamma}_t(\pi) - Q(\pi)|}{V(\pi)} \leq \beta$$

for both i.i.d. and adaptively collected data without overlap conditions, where $V(\pi)$ is an empirical proxy of the estimation error that does not rely on unknown quantities.

3.3 Construction of the estimator and the regularizer

The key contribution of PPL is to instantiate the generic idea in Section 3.2 into a concrete algorithm by constructing the estimator $\hat{Q}(\pi)$ and the regularizer $R(\pi)$ for all $\pi \in \Pi$. We begin with a standard AIPW-type (Robins et al., 1994) estimator for the policy value:

$$\hat{Q}(\pi) = \frac{1}{T} \sum_{t=1}^T \hat{\Gamma}_t(\pi), \quad \hat{\Gamma}_t(\pi) = \hat{\mu}_t(X_t, \pi(X_t)) + \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} \cdot (Y_t - \hat{\mu}_t(X_t, \pi(X_t))), \quad (5)$$

and we set $\hat{Q}(\pi) = -\infty$ if $e_t(X_t, \pi(X_t) | \mathcal{H}_t) = 0$ for any $t \in [T]$. Here $\hat{\mu}_t(\cdot, \cdot): \mathcal{X} \times \mathcal{A} \rightarrow [0, 1]$ is an estimator for $\mu(x, a)$. With batched data, we assume for simplicity that $\hat{\mu}_t \equiv \hat{\mu}(\cdot, \cdot): \mathcal{X} \times \mathcal{A} \rightarrow [0, 1]$ is an estimator that is independent of the data. This is only for the convenience of exposition; in practice, this can be achieved by a cross-fitted version of our algorithm (Schick, 1986; Chernozhukov et al., 2018) described in Section A of the supplementary material with the same theoretical guarantee. With adaptively collected data, we allow $\hat{\mu}_t(x, a)$ to be constructed using $\{X_i, A_i, Y_i\}_{i=1}^{t-1}$. The estimator (5) is standard in the policy learning literature. Taking $\hat{\mu}(\cdot, \cdot) \equiv 0$ reduces (5) to the IPW estimator used in Zhao et al. (2012); Kitagawa and Tetenov (2018); Swaminathan and Joachims (2015); Bibaut et al. (2021). For general $\hat{\mu}_t(\cdot, \cdot)$, it is similar to the AIPW-type estimator in Athey and Wager (2021); Zhou et al. (2022); Zhan et al. (2021).

The second ingredient is the regularizer $R(\pi)$. PPL takes inspirations from empirical Bernstein's inequality, in that $R(\pi)$ shall capture the standard deviation of $\hat{Q}(\pi)$. However, naively taking $R(\pi)$ to be the empirical standard deviation as in the classical empirical Bernstein's inequalities (Bartlett et al., 2005, 2006; Koltchinskii, 2006) does not lead to the desired bound. There remains significant challenges in establishing the inequalities in our context, including (i) unbounded $\hat{\Gamma}_t(\pi)$, (ii) dependent data, and (iii) uniform convergence.

We construct $R(\pi) = \beta \cdot V(\pi)$ for a scaling constant $\beta > 0$ to be decided later (with additional log-terms

for adaptive data), and

$$V(\pi) := \max \{V_s(\pi), V_p(\pi), V_h(\pi)\} \quad (6)$$

with the three data-dependent components

$$\begin{aligned} \text{(sample deviation)} \quad V_s(\pi) &= \frac{1}{T} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e(X_t, \pi(X_t) | \mathcal{H}_t)^2} \right)^{1/2}, \\ \text{(population deviation)} \quad V_p(\pi) &= \frac{1}{T} \left(\sum_{t=1}^T \frac{1}{e(X_t, \pi(X_t) | \mathcal{H}_t)} \right)^{1/2}, \\ \text{(higher-order deviation)} \quad V_h(\pi) &= \frac{1}{T} \left(\sum_{t=1}^T \frac{1}{e(X_t, \pi(X_t) | \mathcal{H}_t)^3} \right)^{1/4}. \end{aligned} \quad (7)$$

As we will see later, the scaling factor $\beta > 0$ depends on the complexity of the policy class Π , and the choice of β differs slightly for batched and adaptively collected data.

Let us discuss the intuitions in these three terms. The idea is to leverage knowledge of the behavior policy to estimate the fluctuation of $\widehat{Q}(\pi)$ around $Q(\pi)$. The uncertainty term $V(\pi)$ can be interpreted as a proxy for the standard deviation of $\widehat{Q}(\pi)$. To be more specific, $V_s(\pi)$ roughly upper bounds the uncertainty of $\widehat{Q}(\pi)$ conditional on all but $\{Y_t\}_{t=1}^T$:

$$\frac{1}{T^2} \sum_{t=1}^T \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2} \cdot \text{Var}(Y_t | X_t, A_t) \leq V_s(\pi)^2.$$

Similarly, $V_p(\pi)$ approximately upper bounds the uncertainty conditional on $\{X_t\}_{t=1}^T$:

$$\frac{1}{T^2} \sum_{t=1}^T \frac{\text{Var}(Y_t | X_t, \pi(X_t))}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} \leq V_p(\pi)^2.$$

Finally, $V_h(\pi)$ upper bounds the standard deviation of $V_s(\pi)$ conditional on $\{X_t\}_{t=1}^T$, and is a higher-order term dominated by $V_p(\pi)$ in general, in the sense that $V_h(\pi) \leq V_p(\pi)$ if $e(x, a) \geq 1/T$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$.

In Sections 4 and 5, we will demonstrate that $R(\pi)$ is valid uncertainty quantifier obeying (4). This technical route diverges significantly from existing works in the policy learning literature (Zhao et al., 2012; Zhou et al., 2022; Zhan et al., 2021), which rely heavily on lower bounded propensities – hence upper bounded $\widehat{\Gamma}_t(\pi)$ – to establish uniform concentration of $\widehat{Q}(\pi)$. Without imposing lower bounds on $e_t(x, a | \mathcal{H}_t)$, it is unclear whether, and at what rate, $\widehat{Q}(\pi)$ would converge to $Q(\pi)$. In contrast, our approach seeks for a data-dependent characterization of the estimation quality of $\widehat{Q}(\pi)$. This strategy not only facilitates analysis in the absence of overlap but also enables instantiating the pessimism principle for IPW-type policy value estimators.

Our proof leverages self-normalization inequalities to derive the deviation bound of the martingale $\widehat{\Gamma}_t(\pi) - \mu(X_t, A_t)$. Importantly, the unbounded $\widehat{\Gamma}_t(\pi)$ and dependent data structure requires going beyond a straightforward application of existing self-normalized concentration inequalities. This is because many exponential-tail self-normalized inequalities (see, e.g., (Peña et al., 2009, Section 9.3)) require constant bounds on $|\widehat{\Gamma}_t(\pi)|$ or summation of the second moments of $|\widehat{\Gamma}_t(\pi)|$, which is not infeasible in the absence of uniform overlap. To address these challenges, we employ a suite of techniques to obtain a specialized class of martingales, which then enables the application of a particular self-normalized concentration inequality that does not require any bounded constraints (De la Pena et al., 2004). Further details on these techniques are provided in our discussion of the theoretical results.

Finally, we highlight PPL as the first-of-the-kind instantiation of the general pessimism principle in optimizing AIPW-type estimates over a policy class. While the pessimism principle has led to various

algorithms in value-based offline RL (Jin et al., 2021; Rashidinejad et al., 2021; Xie et al., 2021a), these results rely on well-specified function classes for the value function ($\mu(x, a)$ in our context). Instead, by leveraging knowledge of the sampling policies, we enable a design-based, model-free implementation. As such, the analysis techniques and theoretical guarantees are quite different.

4 Learning from a fixed behavior policy

In this section, we consider the i.i.d. setting where the offline data are collected by a fixed behavior policy, i.e., $e_t(x, a | H_t) \equiv e(x, a)$ for a *known* policy $e(\cdot, \cdot)$ and all $(x, a) \in \mathcal{X} \times \mathcal{A}$, $t \in [T]$. We begin in Section 4.1 with an upper bound on the suboptimality of our pessimistic learning algorithm, and provide a matching lower bound in Section 4.2.

4.1 Suboptimality upper bound

We set the scaling constant β to be approximately $\sqrt{N \cdot \dim(\Pi)}$ in order to achieve efficient learning. The suboptimality of the resulting algorithm is characterized by Theorem 4.1, whose proof is in Appendix B.1.

Theorem 4.1. *Fix $\delta \in (0, 1)$. For any policy class Π , suppose we set $\hat{Q}(\pi)$ as in (5), $V(\pi)$ as in (6), and*

$$R(\pi) = \beta \cdot V(\pi), \quad \text{for } \beta \geq 10\sqrt{2(N \cdot \dim(\Pi) \log(TK^2) + \log(16/\delta))}.$$

Then with probability at least $1 - \delta$, it holds that

$$|\hat{Q}(\pi) - Q(\pi)| \leq \beta \cdot V(\pi) \quad \text{for all } \pi \in \Pi.$$

Furthermore, with such a choice, we have

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq \min \{2\beta \cdot V(\pi^*), 1\}$$

with probability at least $1 - \delta$, where $\hat{\pi}$ is defined in (3) and $\pi^ = \operatorname{argmax}_{\pi \in \Pi} Q(\pi; \mathcal{C})$.*

The constant term in the above upper bound is trivial since $0 \leq Q(\pi; \mathcal{C}) \leq 1$ for all $\pi \in \Pi$ by assumption; in the term $2\beta \cdot V(\pi^*)$, the regularizer $V(\pi^*)$ is roughly the estimation error for $\hat{Q}(\pi^*)$, and the choice of β arises from uniform concentration over the policy class Π . Theorem 4.1 implies that the suboptimality gap of pessimistic learning is small when $V(\pi^*)$ is small — that is, when the value of $Q(\pi^*)$ can be well estimated. This is the case, for example, when there is enough overlap for the optimal policy π^* .

Theorem 4.1 also shows an “oracle” property (Donoho and Johnstone, 1994; Fan and Li, 2001; Zou, 2006) of PPL, in the sense that it automatically “adapts” to the overlap condition of π^* , even though π^* is not known a priori (Jin et al., 2021).

Remark 4.2. Theorem 4.1 holds without any assumptions on the behavior policy. Instead, the burden of assumptions is “shifted” to the upper bound, which honestly reflects how well the offline data collecting process is for enabling learning the optimal policy.

To better understand the upper bounds in Theorem 4.1, we consider in the following corollary a specific setting where there is sufficient overlap for the *optimal* policy, but not necessarily for all $\pi \in \Pi$.

Corollary 4.3. *Fix any $\delta \in (0, \exp(-1))$. Assume there exists some $C_* > 0$ (allowed to depend on T) such that $e(x, \pi^*(x)) \geq C_*$ for $\mathbb{P}_X$ -almost all $x \in \mathcal{X}$. Suppose we choose $\beta = c \cdot \sqrt{N \cdot \dim(\Pi) \log(TK^2) + \log(16/\delta)}$ for some $c \geq 10\sqrt{2}$, and suppose $T \geq 3$. Then with probability at least $1 - 2\delta$,*

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq \min \left\{ 2c \cdot \sqrt{\frac{N \cdot \dim(\Pi) \log(TK^2)}{C_* T}} \cdot \log(2/\delta)^{3/2}, 1 \right\},$$

where $\hat{\pi}$ is defined in (3) and $\pi^* = \operatorname{argmax}_{\pi \in \Pi} Q(\pi; \mathcal{C})$.

The proof is provided in Appendix C.4. Corollary 4.3 implies that efficient policy learning with a rate of $T^{-1/2}$ is feasible as long as the optimal policy is explored well. It is helpful to compare Corollary 4.3 to state-of-the-art results in the literature to gain more intuitions. Kitagawa and Tetenov (2018) show that the greedy policy obeys $\mathcal{L}(\hat{\pi}_{\text{greedy}}; \mathcal{C}, \Pi) \lesssim \sqrt{\text{N-dim}(\Pi)/(\underline{C}T)}$ under the condition that $\bar{C} > 0$, where $\lesssim$ hides logarithmic terms, and $\underline{C} = \inf_{x,a} e(x, a)$. Compared with this result, PPL broadens the scenarios where efficient policy learning is feasible, as it could happen that $\bar{C} = 0$ but $C_* > 0$. In addition, PPL achieves sharper statistical rates as it holds deterministically that $C_* \geq \bar{C}$.

Finally, we extend Theorem 4.1 and show that the performance of PPL is comparable to any reference policy under consideration. Its proof slightly modifies that of Theorem 4.1, and can be found in Appendix C.5.

Corollary 4.4. *Under the same setup as in Theorem 4.1, with probability at least $1 - \delta$,*

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq \inf_{\pi \in \Pi} \{ \mathcal{L}(\pi; \mathcal{C}, \Pi) + 2\beta \cdot V(\pi) \}.$$

Corollary 4.4 can be viewed as a “softened” version of Theorem 4.1, which further broadens the possibility of efficient policy learning: even if the optimal policy does not have sufficient overlap so that $V(\pi^*)$ is large, the suboptimality of PPL would still be small as long as there is some near-optimal policy with good overlap. This is an additional “oracle” property that PPL is comparable to any well-covered policy in the class.

Remark 4.5 (Extension to unknown propensities). We have assumed knowledge of the true propensities $e_t(\cdot, \cdot)$. For batched data, this is a standard setting considered in the literature (Kitagawa and Tetenov, 2018; Zhao et al., 2012). For future work, it would be interesting to extend PPL to observational studies where the fixed behavior policy $e(\cdot, \cdot)$ has to be estimated from data (Athey and Wager, 2021). We expect this to be a nontrivial task, especially when it is estimated via nonparametric regression or machine learning algorithms that do not rely on strong modeling assumptions (Chernozhukov et al., 2018). We defer a detailed discussion on the technical challenges to Appendix A.5.

4.2 Pessimistic policy learning is minimax optimal

We now show that PPL is the “best effort” given the fixed behavior policy. To be specific, given an action set $\mathcal{A}$, a policy class Π , a (possibly T -dependent) constant $C_* > 0$, and a sample size $T \in \mathbb{N}_+$, we let $\mathcal{R}(C_*, T, \mathcal{A}, \Pi)$ denote the set of instances we consider — each element of $\mathcal{R}(C_*, T, \mathcal{A}, \Pi)$ corresponds to a contextual bandit model $\mathcal{C}$ defined on $\mathcal{A}$, a behavior policy $e(\cdot, \cdot)$ obeying

$$\inf_{x \in \mathcal{X}} e(x, \pi^*(x)) \geq C_*, \tag{8}$$

which together generates the offline data $\mathcal{D} = \{(X_t, Y_t, A_t)\}_{t=1}^T$. Here $\pi^* = \pi^*(\mathcal{C}, \Pi)$ is the optimal policy among Π for the contextual bandit model $\mathcal{C}$ defined in (1).

The next theorem establishes the minimax lower bound for policy learning in Π from any behavior policy obeying (8).

Theorem 4.6. *For any action set $\mathcal{A}$, sample size $T \in \mathbb{N}_+$, any policy class Π and any $C_* > 0$ with $\frac{\text{N-dim}(\Pi)}{C_* T} \leq 1.5$, one has*

$$\inf_{\hat{\pi}} \sup_{(\mathcal{C}, e) \in \mathcal{R}(C_*, T, \mathcal{A}, \Pi)} \mathbb{E}_{\mathcal{C}, e} [\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi)] \geq 0.12 \sqrt{\frac{\text{N-dim}(\Pi)}{C_* T}},$$

where $\text{N-dim}(\Pi)$ is the Natarajan dimension of Π , $\hat{\pi}$ is any data-dependent policy, and $\mathbb{E}_{\mathcal{C}, e}$ means taking expectation with respect to the randomness of the offline data generated under $\mathcal{C}$ and $e(\cdot, \cdot)$.

The lower bound in Theorem 4.6 matches the upper bound in Corollary 4.3 up to logarithm factors of T and K : when $\text{N-dim}(\Pi) \geq C_*T$, the upper and lower bound are both of the constant order; when $\text{N-dim}(\Pi) < C_*T$, both bounds scale as $\sqrt{\text{N-dim}(\Pi)/C_*T}$ (up to logarithmic factors). Through matching upper and lower bounds, we have shown that pessimistic policy learning is the best effort given offline data from a fixed behavior policy. Theorem 4.6 can in fact be implied by (Zhan et al., 2021, Theorem 1), and we provide its proof in Appendix C.6 for completeness.

5 Learning from adaptive behavior policies

In this section, we turn our attention to pessimistic policy learning from adaptively collected data. Here, the behavior policy is allowed to change over time depending on previous observations.

5.1 Suboptimality upper bound

The following theorem provides a data-dependent upper bound for the suboptimality of pessimistic policy learning, and its proof can be found in Appendix B.2.

Theorem 5.1. *Fix any $\delta \in (0, 1)$. For any policy class Π , we set $\hat{Q}(\pi)$ as in (5), $V(\pi)$ as in (6), and let*

$$R(\pi) = \beta \cdot V(\pi) \sqrt{2 \log(30T \cdot V(\pi))}, \quad \text{for } \beta \geq 6 \sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(16/\delta)}.$$

Then with probability at least $1 - \delta$, it holds that

$$|\hat{Q}(\pi) - Q(\pi; \mathcal{C})| \leq \beta \cdot V(\pi), \quad \text{for all } \pi \in \Pi. \quad (9)$$

Furthermore,

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq \min \{2\beta \cdot V(\pi^*) \sqrt{2 \log(30T \cdot V(\pi^*))}, 1\}$$

holds with probability at least $1 - \delta$, where $\hat{\pi}$ is defined in (3) and $\pi^ = \arg\max_{\pi \in \Pi} Q(\pi; \mathcal{C})$.*

As before, the term $V(\pi^*)$ in the upper bound roughly reflects the estimation error for $\hat{Q}(\pi^*)$, and β arises from uniform concentration over the policy class Π . The implication of Theorem 5.1 is that the suboptimality is small as long as the value of π^* can be well estimated, which generalizes the “oracle” property we stated for the batched case to the much more challenging adaptive setting. Again, similar to Corollary 4.4, we can derive a generalized oracle property which states that with probability at least $1 - \delta$, $Q(\hat{\pi}) - Q(\pi) \leq \min\{1, 2\beta \cdot V(\pi) \sqrt{2 \log(30T \cdot V(\pi))}\}$ simultaneously for all $\pi \in \Pi$. This means the policy learned by PPL is comparable to any policy that is explored well in the adaptively collected dataset.

The key result in Theorem 5.1 is $|\hat{Q}(\pi) - Q(\pi)| \preceq V(\pi) \sqrt{\log(V(\pi))}$. We call it a “generalized empirical Bernstein’s inequality” because $V(\pi)^2$ is an empirical estimator for the variance of $\hat{Q}(\pi)$, and it generalizes existing empirical Bernstein’s inequalities (Bartlett et al., 2005, 2006; Koltchinskii, 2006) to adaptive data and unbounded loss. Without requiring any lower bounds on the adaptive sampling probability $e_t(x, a | \mathcal{H}_t)$, this result is applicable to any adaptive data collection process.

Compared with Theorem 4.1 for non-adaptive data, the additional $\log(V(\pi))$ term arises from applying a special self-normalized inequality (c.f. Lemma F.2 in the supplementary material) we use to address unbounded loss. As discussed in Section 3.3, we utilize the martingale structure of $\hat{\Gamma}_t(\pi) = \mathbf{1}\{A_t = \pi(X_t)\}/e_t(X_t, A_t | \mathcal{H}_t) \cdot Y_t$ to address the adaptive nature of data. However, existing self-normalized inequalities often need to rely on constant bounds on the moments of martingale differences De la Pena et al. (2004), which is not available as we allow $1/e_t(X_t, A - t | \mathcal{H}_t)$ to grow arbitrarily and adaptively as t proceeds. To circumvent this limitation, we transform the original problem to bounding another class of martingales. Derived through a series of techniques, including symmetrization, their randomness is solely from a tree Rademacher process while conditioned on the data. This structure enables the application of Lemma F.2 to unbounded martingales; a detailed proof based on these ideas is available in Appendix B.2.

Remark 5.2. The $\log V(\pi)$ term in Theorem 5.1 is small compared with $V(\pi)$, and can be upper bounded by $\text{poly}(\log(T))$ terms under mild conditions. For instance, if there exists a fixed constant $\alpha \geq 1$ such that $\log e_t(x, a | \mathcal{H}_t) \geq -(\log T)^\alpha$ deterministically for all $t \in [T]$ and all $(x, a) \in \mathcal{X} \times \mathcal{A}$, then it holds that $\log V(\pi) \leq (\log T)^\alpha$ and the upper bound in Theorem 5.1 becomes $O(V(\pi) \cdot (\log T)^{\alpha/2})$. Taking $\alpha = 2$, this is to require $e_t(x, a | \mathcal{H}_t) \geq 1/T^{\log T}$, which is minimal for obtaining any meaningful exploration.

Remark 5.3. Since Theorem 5.1 does not require any assumption on the adaptive data collecting process, we obtain concrete upper bounds on the suboptimality even in situations where the conditions in the literature do not hold (Zhan et al., 2021; Bibaut et al., 2021). The data-dependent upper bound frees us from imposing stringent assumptions on the adaptive propensities; as a result, the learning performance directly depends on the “quality” of the offline data.

To gain a more quantitative understanding of our upper bound, we study a special case where the overlap of the *optimal* policy admits a polynomial-decaying lower bound.

Assumption 5.4 (Polynomial decay). *Assume $e_t(X, \pi^*(X_t) | \mathcal{H}_t) \geq g_t$ almost surely for a deterministic sequence $g_t := \bar{c} \cdot t^{-\gamma}$ for some constant $\bar{c} > 0$ and all $t \in [T]$.*

We emphasize that Assumption 5.4 does not require any of the suboptimal policies to have lower bounded sampling probability, and the standard assumption that $e_t(x, a | \mathcal{H}_t) \geq \bar{c} \cdot t^{-\lambda}$ for some constant $\lambda \in (0, 1)$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$ almost surely (Zhan et al., 2021; Bibaut et al., 2021) is strictly stronger than ours. Under Assumption 5.4, we show below that the upper bound on the suboptimality of pessimistic policy learning is polynomial in T . Its proof is in Appendix D.5.

Corollary 5.5. *Suppose Assumption 5.4 holds, and we set*

$$\beta \geq c \cdot (\log T)^{\alpha/2} \cdot \sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(8/\delta)}$$

for some $c \geq 67$, and suppose $TK^2 \geq 8/\delta$. Then with probability at least $1 - 2\delta$,

$$\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq \min \left\{ 12c \cdot \sqrt{\frac{\text{N-dim}(\Pi)}{T^{1-\gamma}}} \cdot \frac{(\log(TK^2))^{(1+\alpha)/2} \cdot \log(1/\delta)}{\max\{1, \bar{c}^{3/4}\}}, 1 \right\}.$$

Our upper bound in Corollary 5.5 improves upon the results developed in Zhan et al. (2021); Bibaut et al. (2021), where they show the greedy approach ensures a suboptimality upper bound of order $\sqrt{\text{N-dim}(\Pi)}/T^{1-\lambda}$ when $\inf_{x,a} e_t(x, a | \mathcal{H}_t) \geq \bar{c} \cdot t^{-\lambda}$ for some $\lambda \in (0, 1)$. We discuss two examples where the improvement is significant. First, when $\inf_{x,a} e_t(x, a | \mathcal{H}_t) \geq \bar{c} \cdot t^{-\lambda}$ does hold, and π^* is explored better than this lower bound, so that Assumption 5.4 holds for some $\gamma < \lambda$. By Corollary 5.5, pessimistic policy learning achieves a strictly sharper rate than greedy learning. Second, when the assumptions in greedy learning do not hold (so that they cannot provide any guarantee), our method may still enable efficient learning. For instance, in Thompson sampling (Thompson, 1933; Russo et al., 2018), an arm that is considered suboptimal may have a vanishing sampling propensity. However, if the adaptive experiment turns out to successfully preserve the arms for the optimal policy π^* , Assumption 5.4 may still be true, and potentially even true for $\gamma = 0$, leading to an efficient upper bound of order $O(\sqrt{\text{N-dim}(\Pi)}/T)$.

Remark 5.6. *With Theorems 4.1 and 5.1, the use of PPL may extend beyond contextual bandits to dynamic treatment regimes (Chakraborty and Murphy, 2014). In particular, suppose there are T units, each subject to a sequence of treatments (actions) $A_{t,k} \in \mathcal{A}_k$, $k = 1, \dots, N$ and receiving an outcome Y_t after all actions are taken. The sampling probability of $A_{t,k}$ depends on $\mathcal{H}_{t,k} := \{X_{s,\ell}, A_{s,\ell}, Y_s\}_{s \leq t-1} \cup \{A_{t,\ell}\}_{\ell \leq k-1}$ and is assumed to be known. There are similar IPW-type estimators for evaluating a policy π over the action sequence $(A_1, \dots, A_N)$, yet due to the long sequence of actions the lack of overlap is even more exacerbated. PPL with our data-dependent concentration inequality naturally applies to such estimators to enable efficient time-varying policy learning. On the other hand, IPW-type estimators are also used in offline RL (Xie et al., 2019); however, we anticipate that our ideas are only applicable when the state transition probabilities can be accurately estimated.*

5.2 Minimax optimality

Finally, we investigate the minimax optimality of pessimistic policy learning when data is adaptively collected. As the adaptive propensities are difficult to characterize in general, we work under the polynomial decay setting for clear interpretability. As in the fixed behavior policy case, our minimax lower bound unveils that the fundamental difficulty of policy learning with adaptive behavior policy is actually captured by the sampling propensities for the optimal policy.

Given an action set $\mathcal{A}$, a policy class Π , a sample size $T \in \mathbb{N}_+$ and some constants $\bar{c} > 0$ and $\gamma \in (0, 1)$, we denote $\mathcal{R}(\bar{c}, \gamma, T, \mathcal{A}, \Pi)$ as the set of instances we consider. Each element of the instance set corresponds to a contextual bandit model $\mathcal{C}$ defined on $\mathcal{A}$ and an adaptive sampling process obeying

$$e_t(X, \pi^*(X) | \mathcal{H}_t) \geq \bar{c} \cdot t^{-\gamma}, \quad a.s., \quad (10)$$

which together generates the offline data $\mathcal{D} = \{(X_t, Y_t, A_t)\}_{t=1}^T$ as outlined in Section 2. Here, $\pi^* = \pi^*(\mathcal{C}, \Pi)$ is the optimal policy among Π for the contextual bandit $\mathcal{C}$ defined in (1). The next theorem provides the minimax lower bound for learning any policy in Π from an adaptive data collecting process obeying (10), which matches our upper bound in Corollary 5.5 up to logarithm factors.

Theorem 5.7. *For any action set $\mathcal{A}$, any policy class Π , any sample size $T \in \mathbb{N}_+$, any $\bar{c} > 0$ and any $\gamma \in (0, 1)$ with $(1 - \gamma) \cdot \text{N-dim}(\Pi) \leq 1.5\bar{c} \cdot T^{1-\gamma}$, one has*

$$\inf_{\hat{\pi}} \sup_{(\mathcal{C}, e) \in \mathcal{R}(\bar{c}, \gamma, T, \mathcal{A}, \Pi)} \mathbb{E}_{\mathcal{C}, e} [\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi)] \geq 0.12 \cdot \sqrt{\frac{\text{N-dim}(\Pi)}{T^{1-\gamma}}} \cdot \sqrt{\frac{1-\gamma}{\bar{c}}},$$

where $\text{N-dim}(\Pi)$ is the Natarajan dimension of Π , $\hat{\pi}$ is any data-dependent policy, and $\mathbb{E}_{\mathcal{C}, e}$ means taking expectation over the randomness of the offline data under $\mathcal{C}$ and $e_t(\cdot, \cdot | \mathcal{H}_t)$.

Theorem 5.7 matches our upper bound in Corollary 5.5, which shows that PPL is minimax optimal for adaptively collected data. Our lower bound is implied by the lower bound in Zhan et al. (2021) established for a family of problems with the uniform lower bound assumption $\inf_{x,a} e_t(x, a | \mathcal{H}_t) \geq \bar{c} \cdot t^{-\gamma}$. Indeed, we take the worst-case suboptimality over a larger set of instances obeying (10). We include the proof in Appendix D.6 for completeness.

In summary, our theoretical results (1) describe the fundamental difficulty of offline policy learning in terms of the overlap of the optimal policy (Theorem 4.6 for the fixed case and Theorem 5.7 for the adaptive case), and (2) show that PPL is minimax optimal (Theorem 4.1 for the fixed case and Theorem 5.1 for the adaptive case). More specifically, these results concern a family of contextual bandit problems whose propensity of the *optimal policy* is lower bounded. While this family is larger than existing ones in the literature for both cases, our lower bounds coincide with them. At a high level, this reveals the key role of the overlap of the *optimal policy* in the fundamental limit of learning from an offline dataset. By providing an optimal solution to a broader family of problems, we show that some problems are not as hard as they have been perceived in the literature, especially when the uniform overlap is weak yet the overlap for the optimal policy is strong.

Remark 5.8. The improvement of PPL relative to the greedy approach depends on the unknown overlap of the optimal policy π^* , and our minimax lower bound shows that PPL is the best effort for a given dataset. When one has the power to collect data, an important follow-up question is how to design the behavior policy to exploit the benefit of pessimism in subsequent policy learning. By Theorem 4.1 and 5.1, one should aim to ensure a good overlap for π^* . In the batched setting with a fixed behavior policy, one may utilize external data to guess and assign higher probability to the optimal arms. In the adaptive setting, one may use online algorithms to learn the optimal arms; one may also leverage the quantification of uncertainty in (9) to guide the online exploration to avoid strong modeling assumptions on the reward regression function. We leave these aspects for future investigation.

6 Practical algorithms

Our theoretical results so far have addressed the minimax optimal rates of policy learning under weak overlap achieved by PPL. In complement, we now devise practical PPL algorithms, whose efficacy will be demonstrated in the next section via numerical experiments. In Section 6.1, we propose concrete optimization algorithms to approximate our initial non-convex program (3). In Section 6.2, we discuss a cross-validation scheme for choosing the hyperparameter in practice. It is worth noting that while these methods are heuristic and non-exact by nature, similar optimization and computational challenges are present in existing methods (Zhou et al., 2022; Swaminathan and Joachims, 2015; Jin et al., 2021) (see Appendix A.4 of the supplementary material for a detailed discussion). Consequently, we defer exact optimization and the theoretical aspects of PPL algorithm to future work. Nonetheless, our extensive experiments indicate that the proposed algorithms exhibit robust, superior performance and remain a practical choice.

6.1 Optimization for PPL

The first practical challenge we address is that for a given parameter $\beta > 0$, the penalty term $R(\pi)$ is non-convex in π , adding to the well-recognized non-convexity of greedy empirical welfare maximization (i.e., without our penalty term) (Zhou et al., 2022). To this end, we leverage a Majorization-Maximization scheme inspired by SVM approximation algorithms (Van Den Burg and Groenen, 2016; Swaminathan et al., 2017) that progressively approximate the non-convex, non-additive penalty term (7) with an additive objective that is compatible with the widely adopted tree search algorithm for policy learning (Sverdrup et al., 2020).

Specifically, tree search (Sverdrup et al., 2020) works for optimizing any additive objective of the form $\sum_{t=1}^T \Gamma_t(X_t, \pi(X_t))$ such as that in Zhan et al. (2021) over a tree policy class. However, it is not readily applicable to PPL since our penalty term $R(\pi) = \beta \cdot V(\pi)$ is non-additive. To begin with, we ignore the higher order deviation in (7) since it is usually of a smaller scale, and further replace $V(\pi)$ by $V_s(\pi) + V_p(\pi)$, noting that $V(\pi) \leq V_s(\pi) + V_p(\pi) \leq 2V(\pi)$. The following lemma finally bounds it by an additive form based on any starting point π_0 . The proof adapts that of Swaminathan et al. (2017) and is omitted here.

Lemma 6.1. Denote $\Gamma_t^s(\pi) = \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e(X_t, \pi(X_t) | \mathcal{H}_t)^2}$ and $\Gamma_t^p(\pi) = \frac{1}{e(X_t, \pi(X_t) | \mathcal{H}_t)}$, so that $V_s(\pi) = \frac{1}{T} \sqrt{\sum_{t=1}^T \Gamma_t^s(\pi)}$ and $V_p(\pi) = \frac{1}{T} \sqrt{\sum_{t=1}^T \Gamma_t^p(\pi)}$. Then for any $\pi_0: \mathcal{X} \rightarrow \mathcal{A}$ such that $V_s(\pi_0) > 0$,

$$V_s(\pi) \leq G_s(\pi; \pi_0) := C_s + \frac{B_s}{T} \sum_{t=1}^T \Gamma_t^s(\pi)^2, \quad V_p(\pi) \leq G_p(\pi; \pi_0) := C_p + \frac{B_p}{T} \sum_{t=1}^T \Gamma_t^p(\pi)^2,$$

where $C_s = V_s(\pi_0)/2$, $C_p = V_p(\pi_0)/2$, $B_s = (2TV_s(\pi_0))^{-1}$, and $B_p = (2TV_p(\pi_0))^{-1}$.

The Majorization-Minimization scheme is proved to converge to local training minimum for cost functions in SVM methods (Van Den Burg and Groenen, 2016). Inspired by this fact, we use the following Majorization-Maximization scheme: starting with some π_0 , $i = 0$, we iteratively replace our original objective $\frac{1}{T} \sum_{t=1}^T \hat{\Gamma}_t(\pi) - \beta \cdot V(\pi)$ by its lower bound $\frac{1}{T} \sum_{t=1}^T \{\hat{\Gamma}_t(\pi) - \beta \cdot G_s(\pi; \pi_i) - \beta \cdot G_p(\pi; \pi_i)\}$, and use tree search to find the optimal solution π_t , and continue this process until a convergence criterion is met. We summarize the approximation algorithm in Algorithm 1, where we use $\text{Tree}(L, \Pi_D)$ to denote the tree search algorithm that maximizes the empirical welfare L over the policy class Π_D , which is the collection of depth- D trees (Sverdrup et al., 2020). In our numerical experiments, we set π_0 as the greedy policy from tree search, and set the convergence criterion to be either $i > 50$ or $\pi_i(X_t) = \pi_{i+1}(X_t)$ for all $t \in [T]$.

6.2 Cross-validation for hyper-parameter

Having addressed the optimization with a fixed $\beta > 0$, the second challenge we tackle is the choice of the parameter in practice. Note that Theorems 4.1 and 5.1 provide recipes for choosing β that achieves the minimax optimal statistical rates, but the specific value of β that leads to the actual optimal policy value

Algorithm 1 Approximate Iterative Majorization-Maximization for PPL

Input: Training data $\{X_t, A_t, Y_t\}_{t=1}^T$, convergence criterion, tree search algorithm $\text{Tree}(\cdot; \Pi)$ and depth of tree class D , parameter $\beta > 0$.

- 1: Set $i = 0$, and compute $\pi_0 = \text{Tree}(\sum_{t=1}^T \hat{\Gamma}_t(\pi); \Pi_D)$ according to (5). // Initialization
- 2: **while** Convergence criterion not met **do**
- 3: Compute $B_s = (2TV_s(\pi_i))^{-1}$ and $B_p = (2TV_p(\pi_i))^{-1}$. // Iterative update
- 4: Set $\Gamma_t^s(\pi) = \mathbb{1}\{A_t = \pi(X_t)\}/e(X_t, \pi(X_t) | \mathcal{H}_t)^2$.
- 5: Set welfare function $\hat{\Gamma}_{t,i}(\pi) = \hat{\Gamma}_t(\pi) - \beta \frac{B_s}{T} \Gamma_t^s(\pi) - \beta \frac{B_p}{T} \Gamma_t^p(\pi)$.
- 6: Compute $\pi_{i+1} = \text{Tree}(\sum_{t=1}^T \hat{\Gamma}_{t,i}(\pi); \Pi_D)$.
- 7: $i \leftarrow i + 1$.
- 8: **end while**

Output: Learned policy π_i .

may depend on the problem context. Also, the necessity of computing the Natarajan dimension can be undesirable in spite of existing bounds (Daniely et al., 2011; Jin, 2023).

A natural idea is to choose the parameter by cross-validation. For batched data, one can directly use classical cross-validation methods (Stone, 1974; Hastie et al., 2009). In addition, here we develop a heuristic cross-validation strategy for adaptively collected data, summarized in Algorithm 2. We find it effective in our numerical experiments, whose theoretical properties are left for future work.

Algorithm 2 Cross validation for PPL with adaptive data

Input: Training data $\{X_t, A_t, Y_t\}_{t=1}^T$, candidate parameter list $\mathcal{B}$, number of folds N , convergence criterion, tree search algorithm $\text{Tree}(\cdot; \Pi)$ and depth of tree class D .

- 1: Split $\{1, \dots, T\}$ into N folds $\mathcal{I}_j = \{\lfloor (j-1)T/N \rfloor + 1, \dots, \lfloor jT/N \rfloor\}$, $j = 1, \dots, N$. // Data split
- 2: **for** β in $\mathcal{B}$ **do**
- 3: **for** $j = 1, \dots, \lfloor 3N/4 \rfloor$ **do**
- 4: Set training fold $\mathcal{I}_j^{\text{train}} = \cup_{\ell \leq j} \mathcal{I}_\ell$.
- 5: Set evaluation fold $\mathcal{I}_j^{\text{eval}} = \cup_{\ell > j} \mathcal{I}_\ell$.
- 6: Learn policy $\hat{\pi}_{j,\beta}$ with Algorithm 1 applied to training data $\{X_t, A_t, Y_t\}_{t \in \mathcal{I}_j^{\text{train}}}$. // Training
- 7: Evaluate $\hat{Q}_{j,\beta} = \frac{1}{|\mathcal{I}_j^{\text{eval}}|} \sum_{t \in \mathcal{I}_j^{\text{eval}}} \hat{\Gamma}_t(\hat{\pi}_{j,\beta})$. // Evaluation
- 8: **end for**
- 9: Compute average evaluated reward $\hat{Q}_\beta = \frac{1}{\lfloor 3N/4 \rfloor} \sum_{j=1}^{\lfloor 3N/4 \rfloor} \hat{Q}_{j,\beta}$.
- 10: **end for**
- 11: Compute $\hat{\beta}^{\text{cv}} = \arg\max\{\hat{Q}_\beta : \beta \in \mathcal{B}\}$.

Output: Cross-validated hyper-parameter $\hat{\beta}^{\text{cv}}$.

The adaptive setting requires additional care due to its sequential nature: with dependent data, one cannot randomly split the data points. Instead, we are to split the data into N folds without permutation, so that the folds containing consecutive data points preserve the sequential structure. Then, we repeatedly use the first j folds to learn the policy with a specific parameter, use the remaining data to evaluate the learned policy, and aggregate the evaluation to select the parameter. In this way, the martingale structure ensures unbiased evaluation of the policy learned with earlier folds. Furthermore, for the stability in weak overlap settings, we only take $j \leq 3N/4$ to ensure sufficiently large evaluation folds.

7 Numerical experiments

In this part, we evaluate the proposed PPL algorithms via extensive numerical studies. In Section 7.1, we use several stylized scenarios with non-adaptive propensities to build intuitions for the behavior of PLL.

Section 7.2 evaluate the finite-sample performance of PPL and shows its efficacy in a wide range of complex scenarios where data are adaptively collected by Thompson Sampling. Finally, in Section 7.3, we apply PPL to 33 real-world datasets on OpenML (Vanschoren et al., 2013) which shows its practical relevance.¹

7.1 Simulations with non-adaptive propensities

We first consider offline datasets collected by non-adaptive propensities, i.e., $e_t(x, a | \mathcal{H}_t)$ may change with t but does not depend on historical data. We use such well-controlled settings to build intuitions on what should be expected from the algorithms and verify our theoretical results.

7.1.1 Multi-armed bandits

We first consider 5-armed bandits without covariates. For a fixed vector $\mu = (\mu_1, \dots, \mu_5) \in \mathbb{R}^5$, we sample T i.i.d. data from a fixed behavior policy with probability e_k for each arm $k \in [5]$. To ensure that we are in an interesting asymptotic regime, in each experiment with sample size T , the mean vector for the arms is set to be $\mu/\sqrt{T}$. We suppose the data are sampled from a fixed behavior policy with probability e_k for arm k , $k \in [5]$. Given the data, a greedy learner (called GPL hereafter for simplicity) takes $\arg\max_k \{\hat{\mu}_k\}$, where $\hat{\mu}_k$ is the mean reward for data with pulled arm k . In contrast, PPL takes $\arg\max_k \{\hat{\mu}_k - \hat{\sigma}_k/\sqrt{T}\}$, where $\hat{\sigma}_k$ is the standard deviation of the observed rewards for arm k . This is a simplified implementation of PPL, which nevertheless delivers essential messages. We design three data-generating processes:

- *Setting 1: Optimal.* Set $\mu = (0, 0.05, 0.01, 0, -0.01)$ and $e = (0.07, 0.9, 0.01, 0.01, 0.01)$. The optimal arm is sampled with high probability, while the second-optimal is sampled with a small probability. According to Section 3.1, GPL would be confused by the poor estimation of the second-optimal arm.
- *Setting 2: Suboptimal.* Set $\mu = (0, 0.05, 0.04, 0.01, -0.01)$ and $e = (0.07, 0.1, 0.8, 0.02, 0.01)$. With weak overlap, GPL might confuse the two, but our theory implies that PPL would learn an arm that is very close to arm 3 (as the estimation error of arm 3 is small).
- *Setting 3: Uniform.* Set $\mu = (0, 0.05, 0.03, 0.01, -0.01)$ and $e_k \equiv 0.2$. This means all arms are sampled equally well, and we expect both GPL and PPL to perform well.

Figure 4 plots the rescaled suboptimality $\sqrt{T} \cdot (\mu^* - \hat{\mu})$ for GPL and PPL with various choices of β and sample sizes, where $\mu^* = \max_k \{\mu_k\}$ and $\hat{\mu}$ is the true reward for the learned arm. Setting 3 confirms the comparable performance of greedy and PPL given high-quality data. However, once the overlap is not uniformly good (settings 1 and 2), PPL show superior performance especially with reasonably large β . The behavior of PPL and GPL is demonstrated in Appendix A.3.2. In setting 1, GPL misses the optimal arm even if it is pulled many times due to large estimation uncertainty of the second-optimal one (see our discussion in Section 3.1), while PPL overcomes this issue. In setting 2, although both methods miss the optimal arm due to poor overlap, PPL can still find the second-optimal arm whose overlap is sufficiently good (see Corollary 4.4). Finally, GPL and PPL perform comparably in setting 3 where all arms are sampled uniformly well.

In Appendix A.3.1, we compare PPL with Swaminathan and Joachims (2015) where the inverse propensities are first clipped to a maximum of $M = 5$ (which can be viewed as another hyper-parameter to tune in their framework) and then penalized welfare maximization similar to PPL is applied. In setting 1-2, we observe inferior performance of their method due to the bias such clipping introduces, which highlight the importance of properly dealing with weak overlap even in the simplest i.i.d. settings. In general, tuning the parameter M may introduce another layer of complexity in practice, whereas PPL achieves automatic tradeoff between bias and variance.

7.1.2 Contextual bandits with tree search

We then study 10-arm contextual bandits, with policy learned by tree search (Sverdrup et al., 2020). In this part, we still control the propensity sequence $e_t(x, a)$, but the policy search is more complex.

¹Reproduction code is available in the GitHub repository <https://github.com/ying531/pess-policy-learning>.

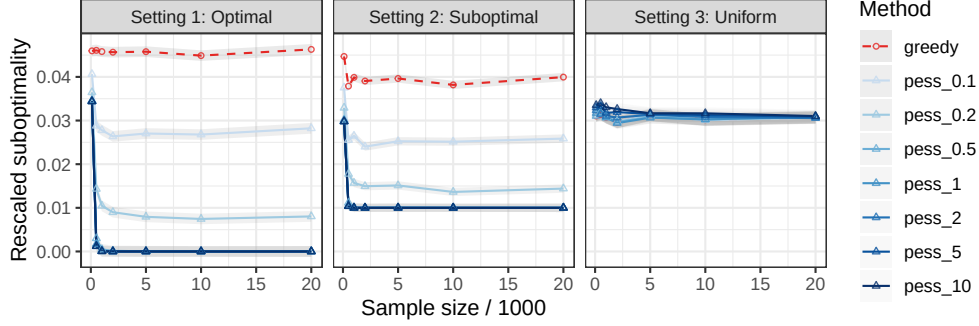


Figure 4: Average rescaled suboptimality of the learned policy from GPL and PPL with $\beta \in \{0.1, 0.2, 1, 10\}$ at various sample sizes. The confidence bands shows mean $\pm$ two times the empirical standard deviation over $N = 1000$ independent runs.

We fix the number of arms at $K = 10$ and data dimension at $p = 2$, extending the setup of Zhan et al. (2021) to a larger problem scale. Let $\{a_k\}_{k=1}^{10}$ be equally spaced within $[-1, 1]$. At each time t , one randomly pulls an arm A_t with probability $e_t(x, a)$ for $a \in \mathcal{A}$, and the reward is generated via $Y_t = \mu(X_t, A_t) + \epsilon_t$, with $X_{t,j} \stackrel{\text{i.i.d.}}{\sim} \text{Unif}[-2, 2]$ and $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, 0.1^2)$. We set nonlinear rewards $\mu(x, a_k) = 1 - \alpha_k/2 + \alpha_k x_1^2/2$, and vary the time-dependent propensity sequence in three ways:

- Setting 1 (Optimal overlap): We fix $e_t(x, a_k) = 0.1 \cdot t^{-\alpha}$ if $k \neq \arg\max_{\ell} \{\mu(x, a_{\ell})\}$ and $e_t(x, a_k) = 1 - 0.9 \cdot t^{-\alpha}$ for the optimal arm, while varying $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$. Thus, the optimal arms has good overlap, but the suboptimal ones are pulled with decaying rates.
- Setting 2 (Half-half overlap): We split the entire trajectory into 5 batches, and alternately in each batch, we let $e_t(x, a_k) = 0.2(1 - t^{-\alpha})$ if $\mu(x, a_k) \geq \text{Median}(\{\mu(x, a_{\ell})\}_{\ell})$ and $e_t(x, a_k) = 0.2 \cdot t^{-\alpha}$ otherwise, or vice versa, varying $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$. This mimics a case where the agent alternates between sampling high-valued and low-valued arms.
- Setting 3 (Worst overlap): We fix $e_t(x, a_k) = 0.2 \cdot t^{-\alpha}$ if $\mu(x, a_k) \geq \text{Median}(\{\mu(x, a_{\ell})\}_{\ell})$ and $e_t(x, a_k) = 0.2(1 - t^{-\alpha})$ otherwise, varying $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$. In this case, the overlap for arms with large rewards decays quickly: for $\alpha = 0.2$ and $T = 1000$, the optimal arms are only pulled about 25 times.

Besides four scenarios with various decay rates α , we additionally study a “pure exploration” scenario where $e_t(x, a_k) = 0.001$ for low-propensity arms and $e_t(x, a) = 0.99$ for high-propensity arms in each setting, in order to show the most drastic violation of uniform overlap. Given the data, we apply GPL (the method of Zhan et al. (2021)) and PPL with Π being the hybrid policy tree with depth 5, optimized using PolicyTree (Sverdrup et al., 2020). As the tree search algorithm is not exact, there might be mis-specification in the policy class.

In addition to GPL and PPL, we evaluate the linear PEVI method in Jin et al. (2021) that addresses the overlap issue via the pessimism principle with linear function approximation $\mu(x, a) = \phi(x, a)^{\top} \theta$, where θ is an unknown parameter (the method of Swaminathan and Joachims (2015) evaluated in the preceding part is not applicable to changing behavior policies). Following Jin et al. (2021), we set the feature vector as $\phi(x, a) \in \mathbb{R}^{20}$ where the $(2i-1)$ -th and $(2i)$ -th elements in $\phi(x, a)$ are $(1, x)$ for the i -th action $a \in \mathcal{A}$, so the linear model is mis-specified. Then, a pessimistic point-wise estimation $\hat{\mu}(x, a) = \phi(x, a)^{\top} \hat{\theta} - \beta \cdot |\phi(x, a)^{\top} \Lambda^{-1} \phi(x, a)|^{1/2}$ is obtained, where $\hat{\theta}$ is the ridge regression estimator, Λ is the augmented covariance matrix, and $\beta > 0$ is a tuning parameter. The learned policy is given by $\hat{\pi}(x) = \arg\max_{a \in \mathcal{A}} \hat{\mu}(x, a)$. This method differs from PPL in both the policy class searched over and the theoretical assumptions needed for efficient learning. The performance of the three methods across all settings is shown in Figure 5.

In setting 1 with optimal overlap, PPL stably outperforms both GPL and PEVI. A moderate decay rate like 0.4 and 0.6 already imposes challenges to a greedy learner. GPL performs better under a higher decay rate of 0.8: in this case, suboptimal arms are rarely sampled for any x , thus the AIPW estimator

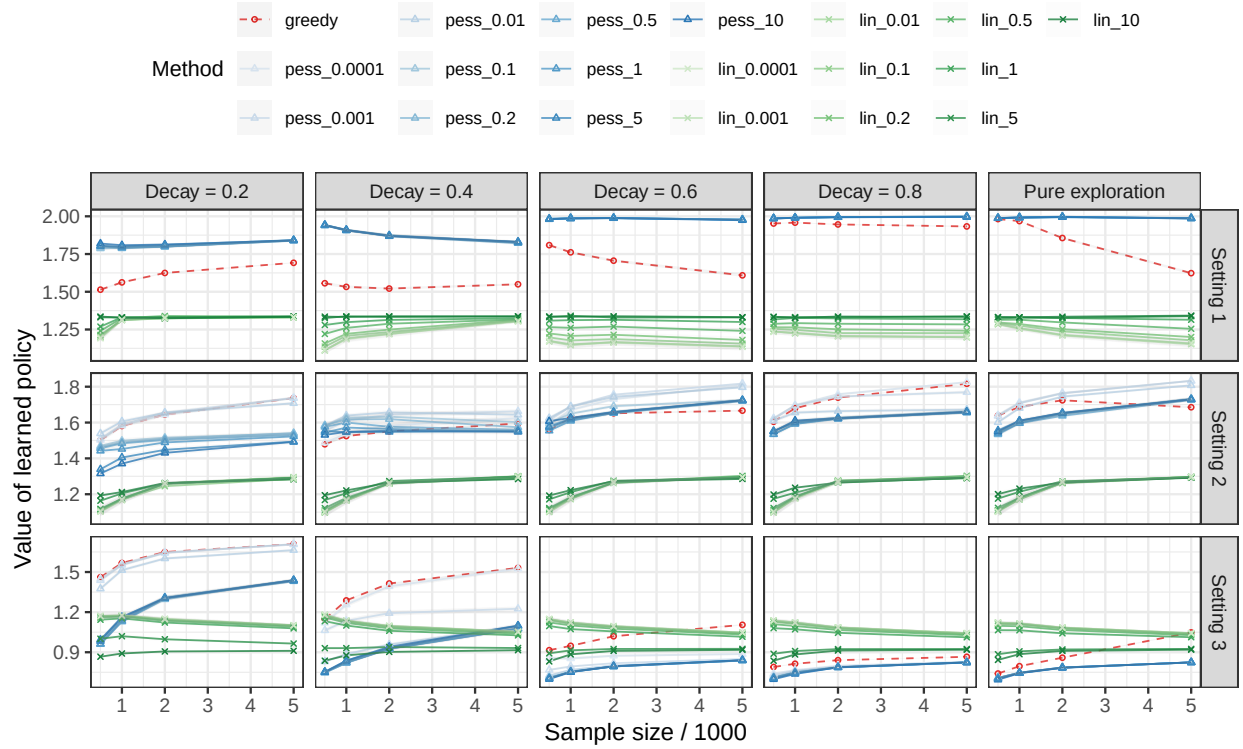


Figure 5: Evaluated value of policy learned from GPL (**greedy**), PPL (**pess**), and the linear method of [Jin et al. \(2021\)](#) (**lin**) with $\beta \in \{0.1, 0.2, 0.5, 1, 5, 10\}$, averaged over $N = 200$ runs. Each subplot shows one sampling scheme.

favors optimal ones. However, in the pure exploration setting, with tiny propensities and large sample sizes, suboptimal arms that occur in the offline data by chance could incur a huge estimation error in the AIPW estimator. Due to mis-specification in the linear model, PEVI suffers from a large bias. In setting 2 where each arm alternates between good overlap and poor overlap, both methods start to struggle, yielding lower values than setting 1. In this case, PPL with a small value of β outperforms GPL, but a large penalty can be harmful. PEVI still suffers from mis-specification bias. Finally, in setting 3 where optimal arms are very rarely taken, both GPL and PPL lead to poor learning performance (PPL appears comparable to GPL especially with small penalty), while PEVI slightly outperforms the two tree-based methods despite reduced performance compared with settings 1 and 2. While PEVI is only slightly better than random guess (a uniform policy over $\mathcal{A}$ would yield a reward of 1), these results demonstrate the stability of linear methods in the most challenging—nearly hopeless—scenarios.

7.2 Simulations with Thompson Sampling

We then move on to contextual bandits with data collected by an adaptive linear Thompson Sampling (TS) agent (Agrawal and Goyal, 2013), following the simulations in Zhan et al. (2021). As we now have less control over the sampling process, our goal is to demonstrate the *finite-sample* performance of PPL in relatively complex settings with weaker overlap and various levels of mis-specification in exploration.

We consider 10-arm contextual bandits similar to Section 7.1.2. Let $\{a_k\}_{k=1}^{10}$ be equally spaced within $[-1, 1]$. At each time t , the linear TS agent randomly pulls arm A_t , and the reward is generated via $Y_t = \mu(X_t, A_t) + \epsilon_t$, with $X_{t,j} \stackrel{\text{i.i.d.}}{\sim} \text{Unif}[-2, 2]$ for $j = 1, 2$, and $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, 0.1^2)$. We design three data generating processes with varied adaptivity of the agent:

- *Setting 1: well-specified exploration.* We set $\mu(x, a_k) = 1 - \alpha_k/2 + x_1/2 - x_2$, so that the linear TS is well-specified and the sampling process settles down to the optimal arms, but the overlap of sub-optimal arms can be poor unless a lower bound is enforced.
- *Setting 2: misspecified with optimal overlap.* We set nonlinear rewards $\mu(x, a_k) = 1 - \alpha_k/2 + \alpha_k x_1^2/2$, so that the linear TS is mis-specified. After calculating the adaptive propensities $\bar{e}(x, a | \mathcal{H}_t)$ by TS, we linearly rescale it to make sure the optimal arm is pulled with probability at least 0.1.
- *Setting 3: misspecified exploration.* We set nonlinear rewards $\mu(x, a_k) = 1 - \alpha_k/2 + \alpha_k x_1^2/2$, so that the linear TS is mis-specified, and the sampling process might not concentrate on optimal arms.

In each setting, we enforce deterministic lower bounds on $e(x, a | \mathcal{H}_t)$ in three ways: (i) Pure exploration: no lower bound, so $e(x, a | \mathcal{H}_t)$ can decay very fast. (ii)-(iv): Polynomial decay: linearly rescaling the raw propensity by linear TS so that $e(x, a | \mathcal{H}_t) \geq T^{-\alpha}/K$, where $\alpha \in \{0.2, 0.5, 0.8\}$. The decay rate of 0.5 is studied in the simulations of Zhan et al. (2021), and we investigate a broader class of settings. The larger α is, the more imbalanced the overlap is. In addition, we set the agent to update its TS sampler (i.e., linear estimators for arm means) with batch sizes in $\{10, 100\}$, where a smaller batch size allows more frequent updates hence stronger adaptivity. The tree-based policy search is the same as Section 7.1.2. The experiments are repeated for $N = 200$ times.

7.2.1 Well-specified exploration

Figure 6 shows the average reward for GPL and PPL when the linear TS sampler is well-specified, across the eight configurations of propensity lower bound (column) and batch size (row). In this setting, the true optimal arms have a good overlap, while uniform overlap might not be as good.

The results confirm the *superior finite-sample performance* of PPL across all settings. Consistent with Setting 1 (optimal actions have good overlap) in Section 7.1.2, a polynomial decay rate of 0.5 seems most challenging for GPL with TS sampler, which is also the setting where PPL yields the largest improvement. In specific, even though the linear TS sampler is well-specified, decaying overlap for suboptimal arms still confuses the greedy learner, and its performance even decreases with the sample size. The pure exploration setting is also challenging for GPL, but the performance of PPL improves with the sample size. In addition,

PPL slightly improves as the exploration becomes more aggressive, where we observe that the overlap for optimal arms improves. Both algorithms perform stably across batch sizes.

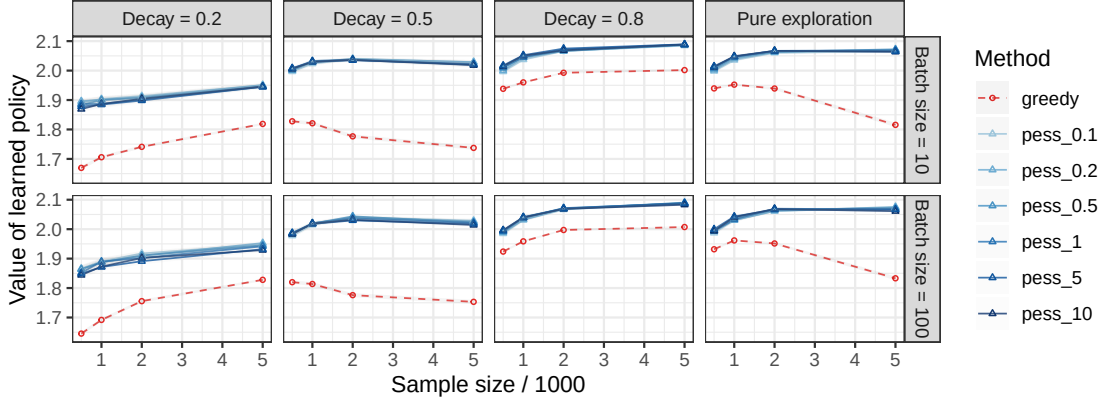


Figure 6: Evaluated value of policy learned from GPL and PPL with $\beta \in \{0.1, 0.2, 0.5, 1, 5, 10\}$ for well-specified exploration, averaged over $N = 200$ runs. Shaded bands are ± 2 times standard deviation. Each column corresponds to one exploration scheme, and each row corresponds to one batch size.

7.2.2 Mis-specified with optimal overlap

Figure 7 shows the results when the linear TS sampler is mis-specified but the optimal action is always well-explored, meaning that $V(\pi^*)$ is small. The pattern we observe is still consistent with Setting 1 in Section 7.1.2. This setting is strictly more challenging than the preceding one, but PPL maintains its superior finite-sample performance for appropriate choice of β . We also observe that PPL is now more sensitive to the choice of β . The decay rate of 0.5 still seems most challenging for GPL, under which PPL also yields the most significant improvement.

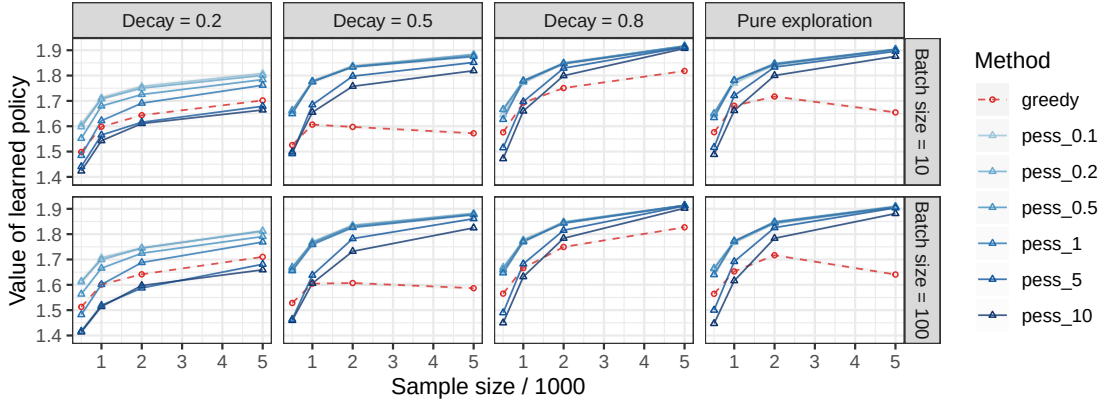


Figure 7: Evaluated value of policy learned from GPL and PPL with mis-specified exploration yet optimal overlap. Details are otherwise the same as Figure 6.

7.2.3 Entirely mis-specified exploration

Figure 8 shows the reward for GPL and PPL when the TS sampler is entirely mis-specified. This is intuitively the most challenging setting for both, since the exploration can be highly imbalanced. However, we empirically find that the optimal arms are still sampled sufficiently many times, hence the overlap for the optimal policy is still good, yet not for all arms. As such, we still observe a similar pattern as Setting 1 in Section 7.1.2: GPL performs relatively well with a decay rate of 0.8 compared with other cases; we

conjecture implicit regularization in the tree search algorithm. GPL can again fail with pure exploration when the uniform overlap further worsens. In contrast, PPL yields reliable performance in all settings. In addition, GPL already performs well when the decay rate is 0.2, yet PPL still yields slight improvement in finite-sample, which requires a smaller penalty parameter β . We also note that the benefit of pessimism is more significant for larger sample sizes, under which the uniform overlap is worse.

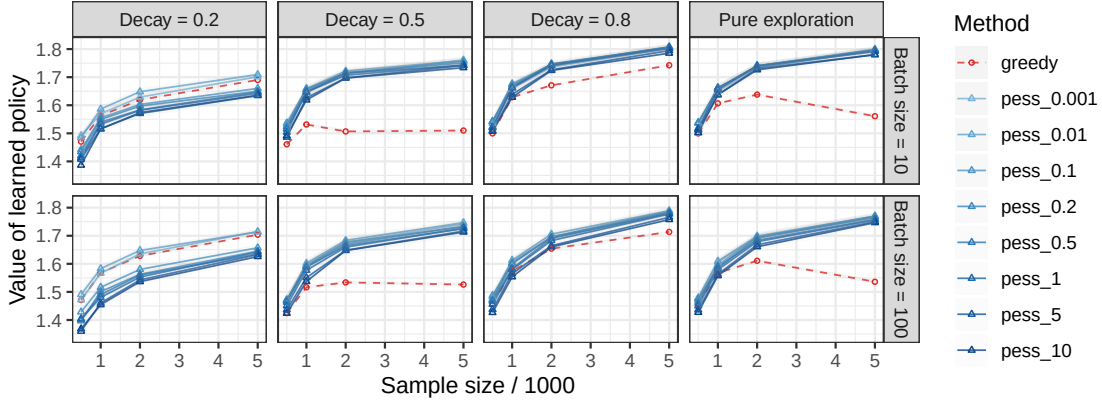


Figure 8: Evaluated value of policy learned from GPL and PPL with entire mis-specified exploration. Details are otherwise the same as Figure 6.

7.2.4 Results for cross-validation

Finally, we evaluate whether cross-validation performs reasonably well in our simulations. We focus on Setting 2 (misspecified with optimal overlap) where the choice of β makes a significant difference, and only show the results with batch size equal to 10 for brevity. We apply the 5-fold cross-validation in Algorithm 2 to determine the parameter $\hat{\beta}^{cv}$, and run PPL with $\hat{\beta}^{cv}$ on the same training data. The policy value is again computed on the evaluation fold.

Figure 9 shows the average evaluated reward when the hyper-parameter β are chosen by 5-fold cross-validation. We observe that our cross-validation strategy is effective in selecting the hyper-paramter with near-optimal choice, leading to significant improvement upon GPL.

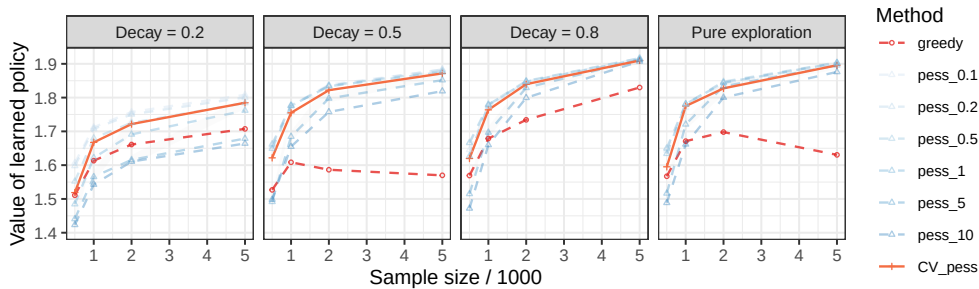


Figure 9: Evaluated value of policy learned from GPL (red, dashed line) and PPL with 5-fold cross-validation in Algorithm 2 (orange, solid line) under the setting of Section 7.2.2. The light blue curves are PPL with various fixed hyper-parameters shown for reference. Details are otherwise the same as Figure 6.

7.3 Real data experiments

Finally, we test the performance of PPL on 33 real-world classification datasets on OpenML (Vanschoren et al., 2013) (a subset of datasets used in Zhan et al. (2021) subject to availability at the time of this paper), following the classification-to-bandit transformation in Zhan et al. (2021).

In specific, we treat each class as an arm, and generate $\mu(X_t, a) = 1$ if the original label is a and 0 otherwise, as well as the semi-synthetic rewards $Y_t = \mu(X_t, A_t) + \epsilon_t$, where $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, 0.1^2)$. We randomly split the data into training and evaluation folds. We focus on the adaptive setting where a linear Thompson Sampling agent adaptively collects data on the training fold similar to Section 7.2. Finally, we apply GPL and PPL to the training fold ($\beta \in \{0.1, 0.2, 0.5, 1, 2, 5, 10, 15\}$ chosen by 5-fold cross validation in Algorithm 2), and compute the average reward of the learned policy on the evaluation fold. We plot the absolute improvement in the evaluated policy value of PPL upon that of GPL in Figure 10.

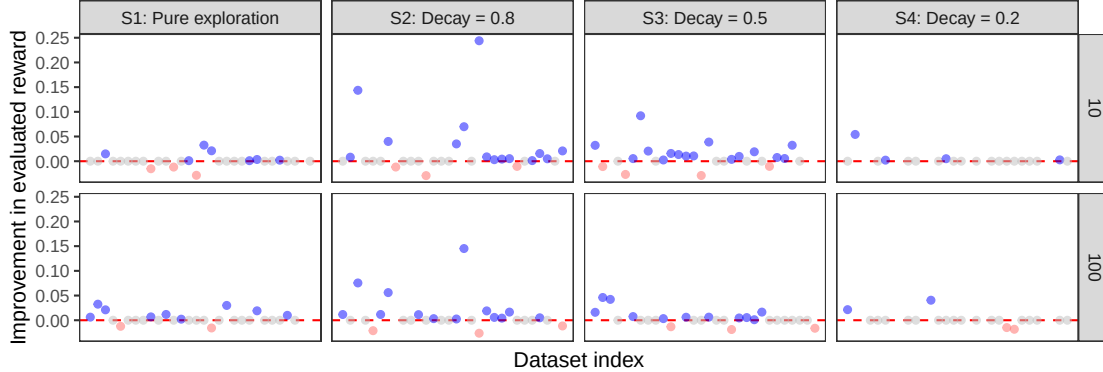


Figure 10: Improvement in the evaluated policy value learned from PPL (with β chosen by 5-fold cross validation) compared with that from the greedy algorithm for all real datasets. Each column corresponds to one exploration scheme, and each row corresponds to one batch size (10 or 100). The x -axis is the index for a dataset.

The message from the real datasets is largely consistent with our simulation studies. PPL outperforms GPL in many of the real datasets (marked in blue). In particular, the margin is significant when the decay rate is moderately large, such as 0.5 and 0.8. GPL and PPL achieve comparable performance at a decay rate of 0.2, where there is no huge gap between the uniform overlap and optimal overlap. The results also show that our cross-validation strategy is able to find hyper-parameters that lead to favorable performance.

Discussion

In this paper, we introduce a new algorithm based on the pessimism principle for policy learning from offline data (collected by either fixed or adaptive behavior policies). We show that the proposed procedure promises efficient learning *as long as* the optimal policy is sufficiently covered by the offline data set, in stark contrast to existing methods in the literature that require uniform coverage for all policies. Our theoretical guarantee builds upon self-normalized processes; as the key technical tool, we develop a generalized empirical Bernstein inequality that provides uncertainty quantification for estimators with unbounded empirical losses. We also devise practical algorithms and demonstrate their relevance via extensive experiments. We close the paper by pointing out several potential future directions.

Extension to ERM. In this work, we have primarily focused on policy learning, which is a special case within the general ERM framework. An interesting future research avenue is to extend our method to more general ERM problems such as regression (Swaminathan and Joachims, 2015). The challenge there is to establish uniform concentration inequalities for more complex function classes, for which additional techniques might be needed.

Efficient and scalable algorithms. We present both theoretical aspects of pessimistic policy learning and practical implementation with policy tree search. A natural follow-up step is to improve the optimization component of our algorithm and develop variants that are applicable to more general policy classes while preserving similar theoretical properties.

Extension to sequential decision making. This work considers policy optimization in contextual bandits using IPW-type estimators, where we design a “model-free” pessimistic penalty without referring to parametric forms or function approximation. It will be interesting to see if this approach can be applied to reweighting-based methods in sequential decision making such as offline reinforcement learning.

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A Deferred details

A.1 Real datasets

The names of real datasets on the OpenML platform used in Section 7.3 are:

‘waveform-5000’, ‘Long’, ‘cmc’, ‘artificial-characters’,
‘Click_prediction_small’, ‘skin-segmentation’, ‘allrep’,
‘mfeat-morphological’, ‘satellite_image’,
‘jungle_chess_2pcs_endgame_elephant_elephant’, ‘wilt’, ‘Satellite’, ‘ringnorm’, ‘mammography’,
‘delta_ailerons’, ‘PhishingWebsites’, ‘splice’, ‘pendigits’, ‘texture’, ‘cardiotocography’,
‘volcanoes-d4’, ‘volcanoes-b3’, ‘dis’, ‘optdigits’, ‘electricity’, ‘kr-vs-kp’, ‘bank-marketing’,
‘satimage’, ‘MagicTelescope’, ‘houses’, ‘eeg-eye-state’, ‘car’, ‘segment’

A.2 Cross-fitted pessimistic policy learning

In this section, we provide a cross-fitted version of our algorithm for the fixed behavior policy case. We first randomly split the index set $[T]$ into two disjoint folds $\mathcal{T}_1$ and $\mathcal{T}_2$ of size $T/2$, where we assume $T/2 \in \mathbb{N}$ without loss of generality. For $k = 1, 2$ we use $\{X_t, A_t, Y_t\}_{t \notin \mathcal{T}_k}$ to obtain an estimator $\hat{\mu}^{(k)}$ for $\mu(x, a) = \mathbb{E}[Y(a) | X = x]$, and construct the cross-fitted estimator: for $t \in \mathcal{T}_k$,

$$\hat{Q}(\pi) = \frac{1}{T} \sum_{t=1}^T \hat{\Gamma}_t(\pi), \quad \hat{\Gamma}_t(\pi) = \hat{\mu}^{(k)}(X_t, \pi(X_t)) + \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e(X_t, \pi(X_t))} \cdot (Y_t - \hat{\mu}^{(k)}(X_t, \pi(X_t))).$$

We then have $\hat{\mu}^{(k)}$ is independent of $\{X_t, A_t, Y_t\}_{t \in \mathcal{T}_k}$. The estimation error of $\hat{Q}(\pi)$ can be bounded as

$$|\hat{Q}(\pi) - Q(\pi)| \leq \frac{|\hat{Q}^{(1)}(\pi) - Q(\pi)|}{2} + \frac{|\hat{Q}^{(2)}(\pi) - Q(\pi)|}{2},$$

where each $|\hat{Q}^{(k)}(\pi) - Q(\pi)|$ reduces to the case in Section 4 with sample size $|\mathcal{T}_k| = T/2$. Using similar ideas as those in Section 4, we construct the regularization term $R(\pi) = (R^{(1)}(\pi) + R^{(2)}(\pi))/2$, where $R^{(k)}(\pi) = \beta \cdot V^{(k)}(\pi)$, and

$$V^{(k)}(\pi) := \max \{V_s^{(k)}(\pi), V_p^{(k)}(\pi), V_h^{(k)}(\pi)\}$$

for

$$V_s^{(k)}(\pi) = \frac{2}{T} \left(\sum_{t \in \mathcal{T}_k} \frac{\mathbb{1}\{A_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} \right)^{1/2},$$

$$V_p^{(k)}(\pi) = \frac{2}{T} \left(\sum_{t \in \mathcal{T}_k} \frac{1}{e(X_t, \pi(X_t))} \right)^{1/2}, \quad V_h^{(k)}(\pi) = \frac{2}{T} \left(\sum_{t \in \mathcal{T}_k} \frac{1}{e(X_t, \pi(X_t))^3} \right)^{1/4}.$$

The uniform concentration results can be similarly obtained as in Theorem 4.1 by a union bound over $k = 1, 2$ and taking

$$\beta \geq 10\sqrt{2(\text{N-dim}(\Pi) \log(TK^2/2) + \log(4/\delta))}.$$

We omit the details for brevity.

A.3 Deferred numerical experiments

A.3.1 Comparison with clipped method in MAB settings

Figure 11 compares PPL with the method of Swaminathan and Joachims (2015) for i.i.d. data in MAB, where the inverse propensities are clipped before applying variance-based penalized welfare maximization.

We see that clipping introduces a bias which deteriorates the performance (suboptimality).

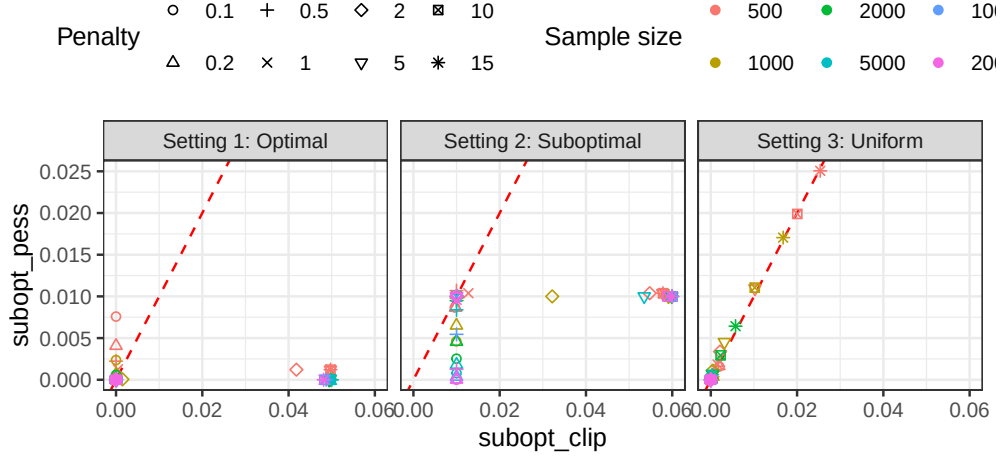


Figure 11: Average rescaled suboptimality of the learned policy from PPL (y -axis) and the method of Swaminathan and Joachims (2015) (x -axis) with various penalty parameter β at various sample sizes in the three settings of Section 7.1.1 in the main text.

A.3.2 Selection frequency in MAB experiments

Figure 12 plots the frequency of selecting each arm using the learned policy from both GPL and PPL in the simulations in Section 7.1.1 of the main text.

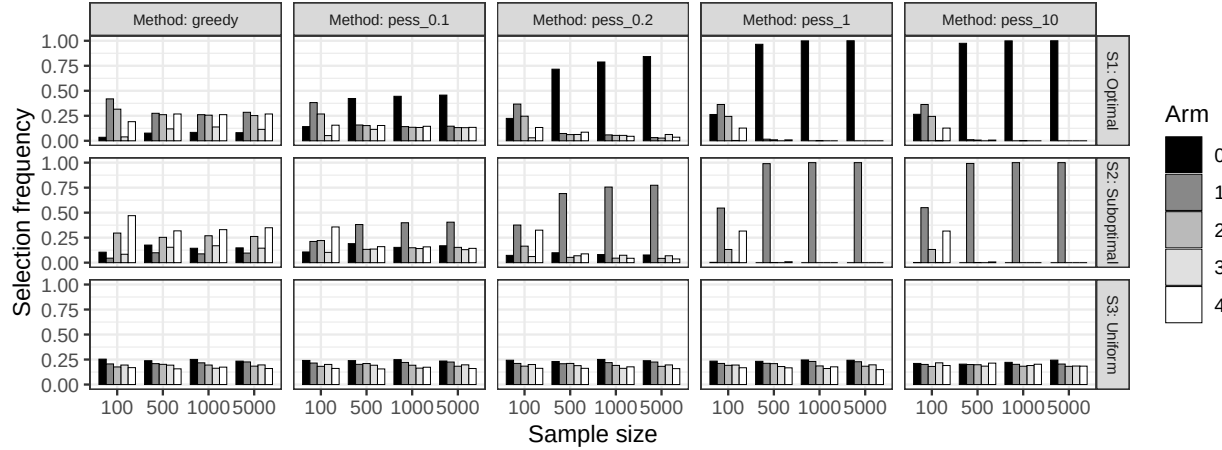


Figure 12: Frequency of selecting each arm using the learned policy from GPL and PPL with $\beta \in \{0.1, 0.2, 1, 10\}$, averaged over $N = 1000$ independent runs. Arms are colored from darkest to lightest according to highest to lowest rewards. Each row corresponds to one setting, and each column is a method.

A.4 Deferred discussion on optimization

In the following, we compare the optimization aspects of PPL with several genres of related methods:

- **Learning over a policy class.** When using IPW estimator to learn over a policy class, state-of-the-art methods that come with optimization schemes (Athey and Wager, 2021; Zhou et al., 2022; Zhan et al., 2021) all rely on tree-based search algorithms to learn policies based on flexible decision trees

or random forests. Notably, they need an exhaustive search over all possible trees of a fixed depth, with is the same as ours. To facilitate efficient computation, they also need to rely on approximations without clear guarantees (though these methods tend to work well in practice). As such, our strategy is comparable to them.

- **Penalized ERM-based policy learning.** Swaminathan and Joachims (2015) uses a similar variance-penalized ERM (with clipped propensities) for policy learning. Our optimization algorithm is indeed inspired by their strategies, where they also need to use approximate optimization that can only converge to a local minima.
- **Other structure-based pessimism approach.** Other pessimism-based algorithms that rely on structured reward functions are also either purely theoretical exploration without a practical implementation (Xie et al., 2021a; Yin and Wang, 2021), or admit a closed-form solution that is similarly nonconvex (such as with linear function classes (Jin et al., 2021)). For the latter, the optimization aspects of the nonconvex problem have also inspired a line of research that aims to approximately reduce (but not fully address) the computational issue via e.g., random perturbations (Nguyen-Tang and Arora, 2023). As such, we view the optimization issue as pertinent to pessimism-based methods whose exact solution goes way beyond the scope of the current work.

A.5 Deferred discussion on unknown propensity scores

Remark A.1 (Extension to unknown propensities, continued). We have assumed knowledge of the true propensities $e_t(\cdot, \cdot)$. For batched data, this is a standard setting considered in the literature (Kitagawa and Tetenov, 2018; Zhao et al., 2012). For future work, it would be interesting to extend PPL to observational studies where the fixed behavior policy $e(\cdot, \cdot)$ has to be estimated from data (Athey and Wager, 2021). We expect this to be a highly nontrivial task, especially when it is estimated via nonparametric regression or machine learning algorithms that do not rely on strong modeling assumptions (Chernozhukov et al., 2018). In specific, standard analysis of AIPW estimators with estimated propensity scores $\hat{e}(\cdot, \cdot)$ heavily relies on the uniform overlap condition, under which the impact of estimation error (which is often characterized by the L_2 -norm or sup-norm of $\hat{e}(\cdot, \cdot) - e(\cdot, \cdot)$) becomes negligible in inverse-propensity weighting. However, without uniform overlap, it is unclear how to (i) characterize the estimation error (since actions with very small propensity rarely appear), (ii) characterize the impact of estimation error on the AIPW estimator, and (iii) translate them into a data-dependent, policy-dependent, and finite-sample valid upper bound $R(\pi)$. Even if (i) and (ii) are plausible to be addressed with assumptions on the model of $\hat{e}(\cdot, \cdot)$, the last step (iii) remains challenging. We thus leave this extension for future work. Finally, we note that knowledge of $e_t(\cdot, \cdot)$ is necessary for learning from adaptively collected data, which we study in Section 5. Estimating $e_t(\cdot, \cdot)$ is infeasible for adaptive data with one trajectory without strong assumptions (Zhan et al., 2021), and the issues for uncertainty quantification with estimated propensities we discuss above still persist.

B Proof sketch of main results

This section provides proof sketches for the main results. Section B.1 focuses on the upper bound in the fixed policy case and Section B.2 that in the adaptive policy case. Throughout, we omit the dependence on $\mathcal{C}$ to simplify the notation when no confusion can arise given the context.

B.1 Proof of upper bound for the fixed policy case

Proof of Theorem 4.1. We start by showing that the (rescaled) estimation error $\frac{\hat{Q}(\pi) - Q(\pi)}{V(\pi)}$ is uniformly bounded over $\pi \in \Pi$. Recall that $\mu(x, a) := \mathbb{E}[Y_t(a) | X_t = x]$ for any $(x, a) \in \mathcal{X} \times \mathcal{A}$, and we decompose the error as follows.

$$\left| \frac{\hat{Q}(\pi) - Q(\pi)}{V(\pi)} \right| = \left| \frac{1}{T} \sum_{t=1}^T \frac{\hat{\Gamma}_t(\pi) - Q(\pi)}{V(\pi)} \right|$$

$$\leq \underbrace{\left| \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi)} \right|}_{\text{term(i)}} + \underbrace{\left| \frac{1}{T} \sum_{t=1}^T \frac{\mu(X_t, \pi(X_t)) - Q(\pi)}{V(\pi)} \right|}_{\text{term(ii)}}.$$

We show in steps I and II how to bound term (i), and establish the upper bound for term (ii) in step III.

Step I: Symmetrization via a Rademacher process. For each $t \in [T]$, we let (A'_t, Y'_t) be an independent copy of (A_t, Y_t) conditional on X_t , i.e.,

$$(A'_t, Y'_t) \perp\!\!\!\perp (A_t, Y_t) \mid X_t \quad \text{and} \quad (A'_t, Y'_t) \mid X_t \stackrel{\text{d}}{=} (A_t, Y_t) \mid X_t.$$

Based on these independent copies, we define

$$\begin{aligned} \widehat{\Gamma}'_t(\pi) &= \widehat{\mu}(X_t, \pi(X_t)) + \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))} \cdot (Y'_t - \widehat{\mu}(X_t, \pi(X_t))), \\ \text{and } V'_s(\pi) &= \frac{1}{T} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))} \right)^{1/2}. \end{aligned}$$

We are to use the exchangeability between $\{A_t, Y_t\}_{t=1}^T$ and $\{A'_t, Y'_t\}_{t=1}^T$ conditional on $\{X_t\}_{t=1}^T$ to turn the uniform concentration to tail bounds on a Rademacher process.

To this end, we let $x = (x_1, \dots, x_T)$, $a = (a_1, \dots, a_T)$, $a' = (a'_1, \dots, a'_T)$, $y = (y_1, \dots, y_T)$ and $y' = (y'_1, \dots, y'_T)$ denote any realized values of $\{X_t\}_{t=1}^T$, $\{A_t\}_{t=1}^T$, $\{A'_t\}_{t=1}^T$, $\{Y_t\}_{t=1}^T$, and $\{Y'_t\}_{t=1}^T$, respectively. We then define

$$\begin{aligned} \gamma_t(\pi) &= \widehat{\mu}(x_t, \pi(x_t)) + \frac{\mathbb{1}\{a_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \cdot \{y_t - \widehat{\mu}(x_t, \pi(x_t))\} \\ \text{and } \gamma'_t(\pi) &= \widehat{\mu}(x_t, \pi(x_t)) + \frac{\mathbb{1}\{a'_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \cdot \{y'_t - \widehat{\mu}(x_t, \pi(x_t))\} \end{aligned}$$

as the corresponding realization of $\widehat{\Gamma}_t(\pi)$ and $\widehat{\Gamma}'_t(\pi)$. The following lemma bounds the probability of term (i) exceeding a constant $\xi \geq 4$ by the tail probability of a Rademacher process, and its proof is in Appendix C.1.

Lemma B.1. *For any constant $\xi \geq 4$, it holds that*

$$\begin{aligned} &\mathbb{P} \left(\sup_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi)} \right| \geq \xi \right) \\ &\leq \sup_{x, a, a', y, y'} 2\mathbb{P}_\varepsilon \left(\left| \sum_{t=1}^T \varepsilon_t \cdot (\widehat{\gamma}_t(\pi) - \widehat{\gamma}'_t(\pi)) \right| \right. \\ &\quad \left. \geq \frac{\xi}{8} \cdot \left\{ \sum_{t=1}^T \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2} \right\}^{1/2} \text{ for all } \pi \in \Pi \right), \end{aligned}$$

where $\mathbb{P}_\varepsilon$ is the probability over i.i.d. Rademacher random variables $\varepsilon_1, \dots, \varepsilon_T \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{\pm 1\}$ while holding other quantities fixed at their realized values.

Step II: Uniform concentration. The next lemma establishes a uniform upper bound for the tail probability of the Rademacher process in terms of the sample size T and the complexity of Π , whose proof is provided in Appendix C.2.

Lemma B.2. *For any fixed $\delta \in (0, 1)$ and any fixed realizations x, a, a', y, y' of $\{X_t\}_{t=1}^T$, $\{A_t\}_{t=1}^T$, $\{A'_t\}_{t=1}^T$,*

$\{Y_t\}_{t=1}^T$, and $\{Y'_t\}_{t=1}^T$, the event

$$\left| \sum_{t=1}^T \varepsilon_t \cdot (\hat{\gamma}_t(\pi) - \hat{\gamma}'_t(\pi)) \right| \leq 2\sqrt{2(\text{N-dim}(\Pi) \log(TK^2) + \log(2/\delta))} \\ \times \left\{ \sum_{t=1}^T \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2} \right\}^{1/2}, \quad \text{for all } \pi \in \Pi$$

holds with probability at least $1 - \delta$ over the randomness of $\varepsilon_1, \dots, \varepsilon_T$.

Taking $\xi = 16\sqrt{2(\text{N-dim}(\Pi) \log(TK^2) + \log(8/\delta))}$, combining Lemmas B.1 and B.2, we know that

$$\sup_{\pi \in \Pi} \frac{1}{T} \cdot \frac{|\sum_{t=1}^T \hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))|}{V(\pi)} \leq 8\sqrt{2 \cdot \text{N-dim}(\Pi) \cdot \log(TK^2) + 2\log(8/\delta)} \quad (11)$$

with probability at least $1 - \delta/2$. This controls term (i).

Step III: Controlling term (ii). Next, we use similar techniques as in Step I and Step II to develop a uniform bound for term (ii). The proof of the next lemma is deferred to Appendix C.3.

Lemma B.3. *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, there is*

$$\sup_{\pi \in \Pi} \frac{|\sum_{t=1}^T Q(X_t, \pi) - Q(\pi)|}{V(\pi)} \leq 2\sqrt{2 \cdot \text{N-dim}(\Pi) \cdot \log(TK^2) + 2\log(8/\delta)}.$$

Finally, applying a union bound to (11) and Lemma B.3, we have with probability at least $1 - \delta$ that

$$\left| \frac{\hat{Q}(\pi) - Q(\pi)}{V(\pi)} \right| \leq 10\sqrt{2 \cdot \text{N-dim}(\Pi) \cdot \log(TK^2) + 2\log(16/\delta)} \leq \beta,$$

which concludes the proof of the first claim. Combining this result with Lemma 3.1 by taking $R(\pi) = \beta \cdot V(\pi)$, we conclude the proof of Theorem 4.1. $\square$

B.2 Proof of upper bound for the adaptive policy case

Proof of Theorem 5.1. As before, we begin with the triangle inequality

$$\sup_{\pi \in \Pi} \left| \frac{\hat{Q}(\pi) - Q(\pi)}{V(\pi)} \right| \\ \leq \underbrace{\sup_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi)} \right|}_{\text{term(i)}} + \underbrace{\sup_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\mu(X_t, \pi(X_t)) - Q(\pi)}{V(\pi)} \right|}_{\text{term(ii)}},$$

for which we will control terms (i) and (ii) separately. With adaptive behavior policy, bounding term (ii) is similar to the i.i.d. case while controlling term (i) is much more challenging. The general tool for controlling term (i) is still symmetrization. Due to the adaptivity of the data-collecting processes, however, we are to use a decoupled tangent sequence (De la Pena and Giné, 2012; Rakhlin et al., 2015) which possesses a “weaker” symmetry property than the symmetric copies generated in the i.i.d. case. The decoupled tangent sequence allows us to turn the tail bound on term (i) to that of a tree-Rademacher process (Rakhlin et al., 2015), whose definition will be provided shortly; from there, we shall utilize a conditional symmetry property of the tree-Rademacher process to invoke a self-normalized concentration inequality to control term (i).

Step I: Symmetrization. For every $t \in [T]$, we let (A'_t, Y'_t) be an independent copy of (A_t, Y_t) conditional on $\{X_t\} \cup \{X_i, A_i, Y_i\}_{i=1}^{t-1}$, and $\{A'_t, Y'_t\}_{t=1}^T$ are conditionally independent of each other given $\{X_t, A_t, Y_t\}_{t=1}^T$. This can be achieved as follows. For $t = 1, 2, \dots, T$, we sequentially sample $A'_t \sim e_t(X_t, \pi(X_t) | \mathcal{H}_t)$ and $Y'_t \sim \mathbb{P}_{Y(a) | X=x}$ for $a = A'_t$ and $x = X_t$, all independently of everything else. We then define

$$\hat{\Gamma}'_t(\pi) = \hat{\mu}_t(X_t, \pi(X_t)) + \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} \cdot (Y'_t - \hat{\mu}_t(X_t, \pi(X_t))).$$

For notational simplicity, for any $t \in [T]$, we also define

$$\hat{U}_t(\pi) = \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2}.$$

The following lemma moves the supremum over $\pi \in \Pi$ out of the probability, and turns the tail bound of term (i) into that of a process symmetric in (A_t, Y_t) and (A'_t, Y'_t) . The proof is in Appendix D.1.

Lemma B.4. *Let $\Pi(X) = \{(\pi(X_1), \dots, \pi(X_T)) \in \mathcal{A}^T : \pi \in \Pi\}$ be the set of all possible actions on $\{X_t\}_{t=1}^T$ from $\pi \in \Pi$. For any $\xi \geq 4$ that may depend on $\{X_t\}_{t=1}^T$, it holds that*

$$\begin{aligned} & \mathbb{P}\left(\sup_{\pi \in \Pi} \frac{1}{T} \cdot \frac{|\sum_{t=1}^T \hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))|}{V(\pi) \sqrt{2 \log(30T \cdot V(\pi))}} \geq \xi \mid \{X_t\}_{t=1}^T\right) \\ & \leq 2|\Pi(X)| \cdot \sup_{\pi \in \Pi} \mathbb{P}\left(\frac{|\sum_{t=1}^T \{\hat{\Gamma}'_t(\pi) - \hat{\Gamma}_t(\pi)\}|}{\sqrt{\sum_{t=1}^T \hat{U}_t(\pi)} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(\pi))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T\right). \end{aligned} \quad (12)$$

It then suffices to control the probability in (12) for any fixed $\pi \in \Pi$. In the following, we are to turn the upper bound in Lemma B.4 into the tail probability of a tree Rademacher process (Rakhlin et al., 2015).

Step II: Tree Rademacher process. We define a superset for all possible realizations $z_t = (a, a', y, y', f, h)$ for any $t \in [T]$ as

$$\mathcal{Z} = \mathcal{A} \times \mathcal{A} \times [0, 1] \times [0, 1] \times \mathcal{F}_e \times \mathcal{F}_\mu, \quad (13)$$

where $\mathcal{F}_e$ is the set of all possible sampling rule, i.e., all mappings of the form $f(\cdot, \cdot): \mathcal{X} \times \mathcal{A} \rightarrow [0, 1]$ such that $\sum_{a \in \mathcal{A}} f(x, a) = 1$ for all $x \in \mathcal{X}$ and $\log f(x, a) \geq -(\log T)^\alpha$ for all $(x, a) \in \mathcal{X} \times \mathcal{A}$; $\mathcal{F}_\mu$ is the set of all fitted value functions, i.e., all mappings of the form $h(\cdot, \cdot): \mathcal{X} \times \mathcal{A} \rightarrow [0, 1]$. Note that $\mathcal{Z}$ is a superset of all possible realization of $(A_t, A'_t, Y_t, Y'_t, e_t(\cdot, \cdot), \hat{\mu}_t(\cdot, \cdot))$ because in $\mathcal{Z}$ we allow the sampling rule and fitted mean functions to be arbitrary. We further let x_t denote any realization of X_t , and then for any x_t and $z_t = (a_t, a'_t, y_t, y'_t, f_t, h_t) \in \mathcal{Z}_t$, we define the corresponding realized value of $\hat{\Gamma}_t(\pi)$ and $\hat{\Gamma}'_t(\pi)$ as

$$\begin{aligned} \hat{\gamma}_t(\pi; x_t, z_t) &= h_t(x_t, \pi(x_t)) + \frac{\mathbb{1}\{a_t = \pi(x_t)\}}{f_t(x_t, \pi(x_t))} \cdot (y_t - h_t(x_t, \pi(x_t))), \\ \hat{\gamma}'_t(\pi; x_t, z_t) &= h_t(x_t, \pi(x_t)) + \frac{\mathbb{1}\{a'_t = \pi(x_t)\}}{f_t(x_t, \pi(x_t))} \cdot (y'_t - h_t(x_t, \pi(x_t))), \end{aligned}$$

and the realized value of $\hat{U}_t(\pi)$ as

$$\hat{u}_t(\pi; x_t, z_t) = \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{f_t(x_t, \pi(x_t))^2}.$$

Definition B.5 ($\mathcal{Z}$ -valued tree (Rakhlin et al., 2015)). *A $\mathcal{Z}$ -valued tree z of depth T is a rooted complete binary tree with nodes labeled by elements of $\mathcal{Z}$ defined in (13). We identify the tree z with the sequence $(z_1, \dots, z_T)$ of labeling functions $z_t: \{\pm 1\}^{t-1} \rightarrow \mathcal{Z}$ which provide the label for each node. Here, $z_1 \in \mathcal{Z}$ is the label of the root of the tree, while $z_t(\varepsilon_1, \dots, \varepsilon_{t-1})$ for $t > 1$ and $\varepsilon_i \in \{\pm 1\}$ is the label of the node obtained*

by following the path of length $t - 1$ from the root, with $+1$ indicating ‘right’ and -1 indicating ‘left’.

More intuitively, each $\mathcal{Z}$ -valued tree of depth T can be equivalently represented by $2^T - 1$ elements in $\mathcal{Z}$:

$$\underbrace{z_1}_{1 \text{ element}}, \underbrace{z_2(-1), z_2(1)}_{2 \text{ elements}}, \dots, \underbrace{z_T(-1, \dots, -1), \dots, z_T(1, \dots, 1)}_{2^{T-1} \text{ elements}}.$$

See Figure 13 below for an illustration of a tree with depth $T = 3$.

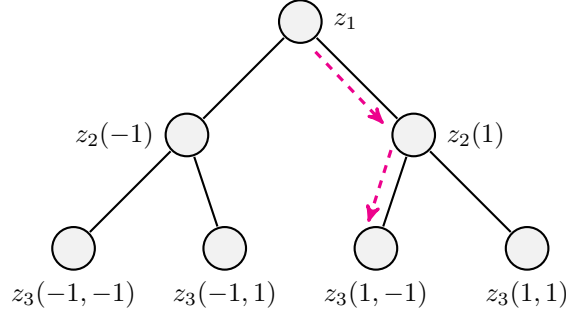


Figure 13: A $\mathcal{Z}$ -valued tree of depth $T = 3$. The pink line shows a path of $\varepsilon_1 = 1$ and $\varepsilon_2 = -1$.

We denote the collection of such trees of depth T as $\mathbb{T}_T(\mathcal{Z})$. For simplicity, we write $\varepsilon_{1:t} = (\varepsilon_1, \dots, \varepsilon_t)$, and write $\mathbf{z} = (z_1, \dots, z_T)$. The following lemma turns the tail bound (12) in Step I into the tail probability of a tree Rademacher process, whose proof is in Appendix D.2.

Lemma B.6. *Let $\varepsilon_1, \dots, \varepsilon_T$ be independent Rademacher random variables. Then for any $\pi \in \Pi$,*

$$\begin{aligned} & \mathbb{P} \left(\frac{|\sum_{t=1}^T \{\hat{\Gamma}'_t(\pi) - \hat{\Gamma}_t(\pi)\}|}{\sqrt{\sum_{t=1}^T \hat{U}_t(\pi)} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(\pi))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T \right) \\ & \leq \sup_{\mathbf{z} \in \mathbb{T}_T(\mathcal{Z})} \mathbb{P}_\varepsilon \left(\left| \frac{\sum_{t=1}^T \varepsilon_t \cdot \{\hat{\gamma}_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)})) - \hat{\gamma}'_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)}))\}}{\sqrt{\sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)}))} \sqrt{\log(56 \sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)}))}} \right| \geq \frac{\xi}{8} \right), \end{aligned}$$

where $\mathbb{P}_\varepsilon$ denotes the probability over $\varepsilon_1, \dots, \varepsilon_T$ conditional on everything else.

Step III: Self-normalized concentration. Next, we invoke the self-normalized concentration inequality in Lemma F.2 to control the tail bound in Step II. A crucial property that allows us to apply this inequality is the conditional symmetry of the tree Rademacher process $\sum_{t=1}^T \varepsilon_t \cdot \{\hat{\gamma}_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)})) - \hat{\gamma}'_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)}))\}$, coupled with the self-normalized-type regularization term $\sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t(\varepsilon_{1:(t-1)}))$ we design. The following lemma is proved in Appendix D.3.

Lemma B.7. *For any constant $\xi \geq \sqrt{2}/2$, we have*

$$\mathbb{P}_\varepsilon \left(\frac{|\sum_{t=1}^T \varepsilon_t \cdot \{\hat{\gamma}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1})) - \hat{\gamma}'_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))\}|}{\sqrt{\sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))} \sqrt{\log[56 \sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))]}} \geq \xi \right) \leq 2e^{-\xi^2/8}.$$

The above lemma, together with Lemma B.4 and Lemma B.6 subsequently leads to

$$\mathbb{P} \left(\sup_{\pi \in \Pi} \frac{1}{T} \cdot \frac{|\sum_{t=1}^T \hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))|}{V(\pi) \sqrt{2 \log(30T \cdot V(\pi))}} \geq \xi \mid \{X_t\}_{t=1}^T \right) \leq 2|\Pi(X)| \cdot \exp \left(-\frac{\xi^2}{8} \right),$$

Invoking the Natarajan lemma (c.f. Lemma F.1), we further have

$$\begin{aligned} \mathbb{P}\left(\sup_{\pi \in \Pi} \frac{1}{T} \cdot \frac{|\sum_{t=1}^T \widehat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))|}{V(\pi)\sqrt{2\log(30T \cdot V(\pi))}} \geq \xi \mid \{X_t\}_{t=1}^T\right) \\ \leq 2T^{\text{N-dim}(\Pi)} K^{2\text{N-dim}(\Pi)} \cdot \exp\left(-\frac{\xi^2}{8}\right) \leq \delta/2 \end{aligned}$$

by taking $\xi = 3\sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(4/\delta)}$. We have thus controlled term (i).

Step IV: Controlling term (ii). The following lemma is similar to Step III of the fixed policy results, whose proof is in Appendix D.4.

Lemma B.8. *For any $\delta \in (0, 1)$, with probability at least $1 - \delta/2$, there is*

$$\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \frac{\mu(X_t, \pi(X_t)) - Q(\pi)}{V(\pi)} \right| \leq 2\sqrt{2 \cdot \text{N-dim}(\Pi) \cdot \log(TK^2) + 2\log(16/\delta)}.$$

Finally, taking a union bound over terms (i) and (ii), we know that

$$\left| \frac{\widehat{Q}(\pi) - Q(\pi)}{V(\pi)\sqrt{2\log(30T \cdot V(\pi))}} \right| \leq 6\sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(16/\delta)} \leq \beta$$

holds with probability at least $1 - \delta$. Combining this result with Lemma 3.1 by taking $R(\pi) = \beta \cdot V(\pi)$, we conclude the proof of Theorem 5.1. $\square$

C Proof of fixed-policy results

This section collects the proofs of all intermediate lemmas in the proof of Theorem 4.1, and the proof of the lower bound in Theorem 4.6.

C.1 Proof of Lemma B.1

Proof of Lemma B.1. To begin with, for any $\xi \geq 4$, we define the event we wish to control as

$$\mathcal{E} := \left\{ \sup_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi)} \right| \geq \xi \right\} = \left\{ \left| \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t))}{V(\pi^\dagger)} \right| \geq \xi \right\},$$

where we define

$$\pi^\dagger = \operatorname{argmax}_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\widehat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi)} \right|.$$

We assume this supremum is attained; otherwise, a limiting argument yields exactly the same conclusion. Here, both $\mathcal{E}$ and $\pi^\dagger$ are measurable with respect to $\{X_t, A_t, Y_t\}_{t=1}^T$. Next, we define two additional events

$$\begin{aligned} \mathcal{E}_1 &= \left\{ \left| \frac{1}{T} \sum_{t=1}^T \widehat{\Gamma}'_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| \geq 2V_p(\pi^\dagger) \right\}, \\ \mathcal{E}_2 &= \left\{ V'_s(\pi^\dagger) \geq 2 \cdot \max \{V_h(\pi^\dagger), V_p(\pi^\dagger)\} \right\}. \end{aligned}$$

The general idea of our proof is to control the probability of $\mathcal{E}$ by proving

$$\mathbb{P}(\mathcal{E} \mid \{X_t\}_{t=1}^T) \leq 2 \cdot \mathbb{P}(\mathcal{E} \cap \mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t\}_{t=1}^T). \quad (14)$$

From there, we will employ a symmetrization trick using Rademacher random variables to further bound $\mathbb{P}(\mathcal{E} \cap \mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t\}_{t=1}^T)$ by the tail probability of a Rademacher process.

A key step in the proof of (14) is to show that

$$\mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t, A_t, Y_t\}_{t=1}^T) \geq \frac{1}{2}. \quad (15)$$

The next lemma proves (15), which controls the deviation of $\frac{1}{T} \sum_{t=1}^T \widehat{\Gamma}'_t(\pi)$ from its (conditional) expectation $\frac{1}{T} \sum_{t=1}^T \mu(X_t, \pi(X_t))$, and the sample uncertainty term $V'_s(\pi)$ for the copies from that of the original data. Its proof is deferred to Appendix E.1.

Lemma C.1. *For any fixed $\pi \in \Pi$, we have*

$$\mathbb{P}\left(\left|\frac{1}{T} \sum_{t=1}^T \widehat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right| \geq 2V_p(\pi) \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \leq \frac{1}{4},$$

and

$$\mathbb{P}\left(V'_s(\pi)^2 \geq 4 \cdot \max\{V_h(\pi)^2, V_p(\pi)^2\} \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \leq \frac{1}{4}.$$

By Lemma C.1 and a union bound we get (15). The tower property subsequently leads to

$$\mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t\}_{t=1}^T, \mathcal{E}) \geq \frac{1}{2}.$$

Multiplying both sides of the above by $\mathbb{P}(\mathcal{E} \mid \{X_t\}_{t=1}^T)$ and applying Bayes' rule, we have

$$\frac{1}{2} \cdot \mathbb{P}(\mathcal{E} \mid \{X_t\}_{t=1}^T) \leq \mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t\}_{t=1}^T, \mathcal{E}) \times \mathbb{P}(\mathcal{E} \mid \{X_t\}_{t=1}^T) = \mathbb{P}(\mathcal{E} \cap \mathcal{E}_1^c \cap \mathcal{E}_2^c \mid \{X_t\}_{t=1}^T). \quad (16)$$

Note that on the event $\mathcal{E}_1^c \cap \mathcal{E}$, it holds for any $\xi \geq 4$ that

$$\begin{aligned} \frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \widehat{\Gamma}'_t(\pi^\dagger) \right| &\geq \frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| - \frac{1}{T} \left| \sum_{t=1}^T \mu(X_t, \pi^\dagger(X_t)) - \widehat{\Gamma}'_t(\pi^\dagger) \right| \\ &\geq \frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| - 2V_p(\pi^\dagger) \\ &\geq \frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| - \frac{\xi}{2} \cdot \max\{V_s(\pi^\dagger), V_p(\pi^\dagger), V_h(\pi^\dagger)\} \\ &\geq \frac{\xi}{2} \cdot \max\{V_s(\pi^\dagger), V_p(\pi^\dagger), V_h(\pi^\dagger)\}, \end{aligned}$$

where the first step uses the triangle inequality, the third step uses the fact that $\xi \geq 4$, and the final step follows from the fact that $\frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| \geq \xi \cdot \max\{V_s(\pi^\dagger), V_p(\pi^\dagger), V_h(\pi^\dagger)\}$ on the event $\mathcal{E}$. Then, on the event $\mathcal{E}_1^c \cap \mathcal{E}_2^c \cap \mathcal{E}$, the above inequality be further lower bounded as

$$\begin{aligned} \frac{1}{T} \left| \sum_{t=1}^T \widehat{\Gamma}_t(\pi^\dagger) - \widehat{\Gamma}'_t(\pi^\dagger) \right| &\geq \frac{\xi}{4} \cdot \max\{2V_s(\pi^\dagger), V'_s(\pi^\dagger)\} \geq \frac{\xi}{8} \cdot \{V_s(\pi^\dagger) + V'_s(\pi^\dagger)\} \\ &\geq \frac{\xi}{8T} \cdot \left(\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi^\dagger(X_t)\} + \mathbb{1}\{A'_t = \pi^\dagger(X_t)\})^2}{e(X_t, \pi^\dagger(X_t))^2} \right)^{1/2}, \end{aligned}$$

where the last step is due the triangle inequality $\|x\| + \|y\| \geq \|x + y\|$ for $x, y \in \mathbb{R}^T$ and noting that $V_s(\pi^\dagger) =$

$\|x\|$ for $x_t = \mathbb{1}\{A_t = \pi^\dagger(X_t)\}/e(X_t, \pi^\dagger(X_t))$ and $V'_s(\pi^\dagger) = \|y\|$ for $y_t = \mathbb{1}\{A'_t = \pi^\dagger(X_t)\}/e(X_t, \pi^\dagger(X_t))$. The above relation then leads to the following inequality:

$$\begin{aligned}
(16) &\leq \mathbb{P}\left(\left|\sum_{t=1}^T \{\widehat{\Gamma}_t(\pi^\dagger) - \widehat{\Gamma}'_t(\pi^\dagger)\}\right|\right. \\
&\quad \left.\geq \frac{\xi}{8} \cdot \left\{\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi^\dagger(X_t)\} + \mathbb{1}\{A'_t = \pi^\dagger(X_t)\})^2}{e(X_t, \pi^\dagger(X_t))^2}\right\}^{1/2} \left|\{X_t\}_{t=1}^T\right|\right) \\
&\leq \mathbb{P}\left(\sup_{\pi \in \Pi} \left\{\left|\sum_{t=1}^T \{\widehat{\Gamma}_t(\pi) - \widehat{\Gamma}'_t(\pi)\}\right|\right.\right. \\
&\quad \left.\left.\times \left[\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e(X_t, \pi(X_t))^2}\right]^{-1/2}\right\} \geq \frac{\xi}{8} \left|\{X_t\}_{t=1}^T\right|\right) \\
&= \mathbb{P}\left(\sup_{\pi \in \Pi} \left\{\left|\sum_{t=1}^T \varepsilon_t \{\widehat{\Gamma}_t(\pi) - \widehat{\Gamma}'_t(\pi)\}\right|\right.\right. \\
&\quad \left.\left.\times \left[\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e(X_t, \pi(X_t))^2}\right]^{-1/2}\right\} \geq \frac{\xi}{8} \left|\{X_t\}_{t=1}^T\right|\right),
\end{aligned}$$

where the probability is taken over $\{A_t, Y_t, A'_t, Y'_t\}_{t=1}^T$ and the Rademacher variables $\varepsilon_1, \dots, \varepsilon_T$ that are i.i.d. from $\text{Unif}(\{\pm 1\})$. Above, the last step uses the exchangeability between $\{A_t, Y_t\}_{t=1}^T$ and $\{A'_t, Y'_t\}_{t=1}^T$ conditional on $\{X_t\}_{t=1}^T$. By the tower property, we know that

$$\begin{aligned}
&\mathbb{P}\left(\sup_{\pi \in \Pi} \left\{\left|\sum_{t=1}^T \varepsilon_t \{\widehat{\Gamma}_t(\pi) - \widehat{\Gamma}'_t(\pi)\}\right|\right.\right. \\
&\quad \left.\left.\times \left[\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e(X_t, \pi(X_t))^2}\right]^{-1/2}\right\} \geq \frac{\xi}{8} \left|\{X_t\}_{t=1}^T\right|\right) \\
&= \mathbb{E}\left[\mathbb{P}\left(\sup_{\pi \in \Pi} \left\{\left|\sum_{t=1}^T \varepsilon_t \{\widehat{\Gamma}_t(\pi) - \widehat{\Gamma}'_t(\pi)\}\right|\right.\right. \cdot \left[\sum_{t=1}^T \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e(X_t, \pi(X_t))^2}\right]^{-1/2} \right\} \\
&\quad \left.\geq \frac{\xi}{8} \left|\{X_t, A_t, A'_t, Y_t, Y'_t\}_{t=1}^T\right| \left|\{X_t\}_{t=1}^T\right|\right],
\end{aligned}$$

where the inner conditional probability is over the randomness of the Rademacher random variables while conditioning on $\{X_t, A_t, A'_t, Y_t, Y'_t\}_{t=1}^T$. Taking a supremum over all possible realizations of $\{X_t, A_t, Y_t, A'_t, Y'_t\}_{t=1}^T$, we have

$$\begin{aligned}
(16) &\leq \sup_{x, a, a', y, y'} \mathbb{P}\left(\left|\sum_{t=1}^T \varepsilon_t \{\widehat{\gamma}_t(\pi) - \widehat{\gamma}'_t(\pi)\}\right|\right. \\
&\quad \left.\geq \frac{\xi}{8} \cdot \left\{\sum_{t=1}^T \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2}\right\}^{1/2} \text{ for all } \pi \in \Pi\right),
\end{aligned}$$

where for any x, a, a', y, y' , the probability is only over the Rademacher variables $\varepsilon_1, \dots, \varepsilon_T$ while holding other quantities at their realized values. This concludes the proof of Lemma B.1. $\square$

C.2 Proof of Lemma B.2

Proof of Lemma B.2. Fix any realizations x, a, a', y, y' and any $\pi \in \Pi$,

$$\begin{aligned}
& (\hat{\gamma}_t(\pi) - \hat{\gamma}'_t(\pi))^2 \\
&= \left(\frac{\mathbb{1}\{a_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \cdot \{y_t - \hat{\mu}(x_t, \pi(x_t))\} - \frac{\mathbb{1}\{a'_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \cdot \{y'_t - \hat{\mu}(x_t, \pi(x_t))\} \right)^2 \\
&\leq 2 \cdot \hat{\mu}(x_t, \pi(x_t))^2 \cdot \left(\frac{\mathbb{1}\{a_t = \pi(x_t)\} - \mathbb{1}\{a'_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \right)^2 \\
&\quad + 2 \cdot \left(y_t \cdot \frac{\mathbb{1}\{a_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} - y'_t \cdot \frac{\mathbb{1}\{a'_t = \pi(x_t)\}}{e(x_t, \pi(x_t))} \right)^2 \\
&\leq 4 \cdot \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2},
\end{aligned}$$

where the second line uses the triangle inequality, and the third line uses the fact that $\hat{\mu} \in [0, 1]$ and $y_t, y'_t \in [0, 1]$. Then by Hoeffding's inequality (c.f. Lemma F.3), over the randomness in ε one has

$$\begin{aligned}
& \mathbb{P}_\varepsilon \left(\left| \sum_{t=1}^T \varepsilon_t \cdot (\hat{\gamma}_t(\pi) - \hat{\gamma}'_t(\pi)) \right| \geq \frac{\xi}{8} \cdot \left\{ \sum_{t=1}^T \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2} \right\}^{1/2} \right) \\
&\leq 2 \cdot \exp \left(-\frac{\xi^2}{512} \right).
\end{aligned}$$

We note that fixing x, a, a', y, y' , the only difference the choice of $\pi \in \Pi$ impacts the above event is through $\pi(x_1), \dots, \pi(x_T)$. Letting $\Pi(x) = \{(\pi(x_1), \dots, \pi(x_T)) : \pi \in \Pi\}$, from Lemma F.1 we know $|\Pi(x)| \leq T^{\text{N-dim}(\Pi)} K^{\text{N-dim}(\Pi)}$, where $K = |\mathcal{A}|$ is the number of actions. Then, taking a union bound over $\Pi(x)$, we have

$$\begin{aligned}
& \mathbb{P}_\varepsilon \left(\left| \sum_{t=1}^T \varepsilon_t \cdot (\hat{\gamma}_t(\pi) - \hat{\gamma}'_t(\pi)) \right| \geq \frac{\xi}{8} \cdot \left\{ \sum_{t=1}^T \frac{(\mathbb{1}\{a_t = \pi(x_t)\} + \mathbb{1}\{a'_t = \pi(x_t)\})^2}{e(x_t, \pi(x_t))^2} \right\}^{1/2} \text{ for all } \pi \in \Pi \right) \\
&\leq 2|\Pi(x)| \cdot \exp \left(-\frac{\xi^2}{512} \right) \leq 2T^{\text{N-dim}(\Pi)} K^{2\text{N-dim}(\Pi)} \cdot \exp \left(-\frac{\xi^2}{512} \right).
\end{aligned}$$

Finally, taking $\xi = 16\sqrt{2\text{N-dim}(\Pi) \log(TK^2) + 2\log(2/\delta)}$ completes the proof. $\square$

C.3 Proof of Lemma B.3

Proof of Lemma B.3. For each $t \in [T]$, we let X'_t be an i.i.d. copy of X_t . Recall that $Q(\pi) = \mathbb{E}[Q(X_t, \pi)] = \mathbb{E}[Y(\pi(X))]$ is the policy value. For any fixed $\eta > 1/\sqrt{2}$, by Markov's inequality we have

$$\begin{aligned}
& \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X'_t, \pi(X'_t)) - Q(\pi) \right| \geq \eta\sqrt{T} \right) \\
&\leq \frac{1}{\eta^2 T} \cdot \mathbb{E} \left[\left(\sum_{t=1}^T \mu(X'_t, \pi(X'_t)) - Q(\pi) \right)^2 \right] = \frac{1}{\eta^2 T} \cdot \sum_{t=1}^T \mathbb{E} \left[(\mu(X'_t, \pi(X'_t)) - Q(\pi))^2 \right] \leq \frac{1}{\eta^2} \leq \frac{1}{2}, \quad (17)
\end{aligned}$$

where the second line uses the fact that $Q(X'_t, \pi)$ are i.i.d., and that $\mu(X'_t, \pi(X'_t)), Q(\pi) \in [0, 1]$.

Let $\pi^* = \operatorname{argmax}_{\pi \in \Pi} |\sum_{t=1}^T Q(X_t, \pi) - Q(\pi)|$, which is measurable with respect to $\{X_t\}_{t=1}^T$. Since $\{X_t\}_{t=1}^T$ are independent of $\{X'_t\}_{t=1}^T$, by (17) we have

$$\mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T \mu(X'_t, \pi^*(X'_t)) - Q(\pi^*) \right| < \eta\sqrt{T} \mid \{X_t\}_{t=1}^T \right) \geq \frac{1}{2}.$$

Then for any $\eta \geq 1/\sqrt{2}$, we have

$$\begin{aligned}
& \frac{1}{2} \cdot \mathbb{P} \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \mu(X_t, \pi(X_t)) - Q(\pi) \right| \geq 2\eta\sqrt{T} \right) \\
&= \frac{1}{2} \cdot \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X_t, \pi^*(X_t)) - Q(\pi^*) \right| \geq 2\eta\sqrt{T} \right) \\
&\leq \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X'_t, \pi^*(X'_t)) - Q(\pi^*) \right| \leq \eta\sqrt{T} \mid \{X_t\}_{t=1}^T \right) \\
&\quad \times \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X_t, \pi^*(X_t)) - Q(\pi^*) \right| \geq 2\eta\sqrt{T} \right) \\
&= \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X'_t, \pi^*(X'_t)) - Q(\pi^*) \right| \leq \eta\sqrt{T}, \left| \sum_{t=1}^T \mu(X_t, \pi^*(X_t)) - Q(\pi^*) \right| \geq 2\eta\sqrt{T} \right) \\
&\leq \mathbb{P} \left(\left| \sum_{t=1}^T \mu(X'_t, \pi^*(X'_t)) - \mu(X_t, \pi^*(X_t)) \right| \geq \eta\sqrt{T} \right) \\
&\leq \mathbb{P} \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \mu(X'_t, \pi(X'_t)) - \mu(X_t, \pi(X_t)) \right| \geq \eta\sqrt{T} \right), \tag{18}
\end{aligned}$$

where the next-to-last step is due to the triangle inequality. Let $\varepsilon_1, \dots, \varepsilon_T \sim \text{Unif}\{\pm 1\}$ be i.i.d. Rademacher random variables. By the exchangeability of X_t and X'_t ,

$$\begin{aligned}
(18) &= \mathbb{P} \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \varepsilon_t \cdot (\mu(X'_t, \pi(X'_t)) - \mu(X_t, \pi(X_t))) \right| \geq \eta\sqrt{T} \right) \\
&\leq 2\mathbb{P} \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \varepsilon_t \cdot \mu(X_t, \pi(X_t)) \right| \geq \eta\sqrt{T}/2 \right) \\
&= 2\mathbb{E} \left[\mathbb{P}_\varepsilon \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \varepsilon_t \cdot \mu(X_t, \pi(X_t)) \right| \geq \eta\sqrt{T}/2 \mid \{X_t\}_{t=1}^T \right) \right] \\
&\leq 2 \sup_x \mathbb{P}_\varepsilon \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \varepsilon_t \cdot \mu(x_t, \pi(x_t)) \right| \geq \eta\sqrt{T}/2 \right),
\end{aligned}$$

where $x = (x_1, \dots, x_T)$ denotes any realization of $\{X_t\}_{t=1}^T$. Above, $\mathbb{P}_\varepsilon$ denotes the probability only over the Rademacher random variables while holding the data as fixed. For any x , similar to our proof of Lemma B.2,

$$\begin{aligned}
& \mathbb{P}_\varepsilon \left(\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \varepsilon_t \cdot \mu(x_t, \pi(x_t)) \right| \geq \eta\sqrt{T}/2 \right) \\
&\leq |\Pi(x)| \cdot \mathbb{P}_\varepsilon \left(\left| \sum_{t=1}^T \varepsilon_t \cdot \mu(x_t, \pi(x_t)) \right| \geq \eta\sqrt{T}/2 \right) \\
&\leq 2T^{\text{N-dim}(\Pi)} K^{2\text{N-dim}(\Pi)} \cdot \mathbb{P}_\varepsilon \left(\left| \sum_{t=1}^T \varepsilon_t \cdot \mu(x_t, \pi(x_t)) \right| \geq \eta\sqrt{T}/2 \right) \\
&\leq 2T^{\text{N-dim}(\Pi)} K^{2\text{N-dim}(\Pi)} \cdot \exp(-\eta^2/8).
\end{aligned}$$

Above, the third line applies Natarajan's lemma (c.f. Lemma F.1) and the fourth line applies Hoeffding's inequality. Finally, taking $\eta = \sqrt{8(\text{N-dim}(\Pi) \log(TK^2) + \log(8/\delta))}$ and noting that $V(\pi) \geq \sqrt{T}$ concludes the proof of Lemma B.3. $\square$

C.4 Proof of Corollary 4.3

Proof of Corollary 4.3. When $C_* \leq 1/T$, the first term is dominated by the constant term, and the bound follows from the boundedness of $Q(\pi)$. We now focused on the case where $C_* > 1/T$. From Theorem 4.1, for any $\beta \geq 10\sqrt{2(\text{N-dim}(\Pi) \log(TK^2) + \log(16/\delta))}$, we know that $\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq 2\beta \cdot V(\pi^*)$ with probability at least $1 - \delta$. In the following, we are to bound $V(\pi^*)$ by controlling the three terms in Equation (7) separately.

Bound $V_s(\pi^)$.* Note that $D_t := \frac{\mathbb{1}\{A_t = \pi^*(X_t)\}}{e(X_t, \pi^*(X_t))^2}$, $t = 1, \dots, T$, are i.i.d. random variables with $\mathbb{E}[D_t] = \mathbb{E}[e(X, \pi^*(X))^{-1}]$. Recall that $C_* \leq \inf_x \mu(x, \pi^*(x))$ is the overlap for the optimal policy, so $|D_t| \leq 1/C_*^2$. We also know that

$$\text{Var}(D_t) \leq \mathbb{E}[D_t^2] = \mathbb{E}[e(X_t, \pi^*(X_t))^{-3}] \leq C_*^{-3},$$

Next, invoking Bernstein's inequality (c.f. Lemma F.4), we have for any $x > 0$ that

$$\mathbb{P}\left(\sum_{t=1}^T \frac{\mathbb{1}\{A_t = \pi^*(X_t)\}}{e(X_t, \pi^*(X_t))^2} - T \cdot \mathbb{E}[e(X, \pi^*(X))^{-1}] \geq x\right) \leq \exp\left(\frac{-x^2/2}{T/C_*^3 + x/(3C_*^2)}\right)$$

Take $x = 2\log(1/\delta)\sqrt{TC_*^{-3}}$. Since $C_*T > 1$, we know

$$\frac{x^2/2}{T/C_*^3 + x/(3C_*^2)} = \frac{2\log^2(1/\delta)}{1 + \frac{2}{3}\log(1/\delta)/\sqrt{TC_*}} \geq \log(1/\delta)$$

as long as $\log(1/\delta) \geq 1$. Therefore, with probability at least $1 - \delta$, we have

$$\begin{aligned} V_s(\pi^*) &\leq \frac{1}{T} \sqrt{T \cdot \mathbb{E}[e(X, \pi^*(X))^{-1}] + 2\log(1/\delta)\sqrt{T/C_*^3}} \\ &\leq \frac{1}{\sqrt{T}} \cdot \sqrt{C_*^{-1} + 2\log(1/\delta) \cdot C_*^{-3/2} \cdot T^{-1/2}} \leq \frac{\log(1/\delta)}{\sqrt{C_*T}}. \end{aligned}$$

where the last inequality uses the fact that $\sqrt{a+b} = \sqrt{a} \cdot \sqrt{1+b/a} \leq \sqrt{a} \cdot b/(2a)$ for any $a, b > 0$.

Bounding $V_p(\pi^)$.* Since $e(X_t, \pi^*(X_t)) \geq C_*$ almost surely, we directly have the upper bound

$$V_p(\pi^*) \leq \frac{\sqrt{T/C_*}}{T} = \frac{1}{\sqrt{C_*T}}.$$

Bounding $V_h(\pi^)$.* Finally, the boundedness of $e(X_t, \pi^*(X_t))$ gives

$$V_h(\pi^*) \leq \frac{1}{T} (T/C_*^3)^{1/4} \leq \frac{1}{(C_*T)^{3/4}}.$$

Combining the three bounds, when $C_* \geq 1/T$, we have that with probability at least $1 - \delta$ that

$$V(\pi^*) \leq \max\left\{\frac{\log(1/\delta)}{\sqrt{C_*T}}, \frac{1}{\sqrt{C_*T}}, \frac{1}{(C_*T)^{3/4}}\right\} = \frac{2\log(1/\delta)}{\sqrt{C_*T}}.$$

Further taking a union bound with the event $\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq 2\beta \cdot V(\pi^*)$ from Theorem 4.1, we know that with probability at least $1 - \delta$,

$$\begin{aligned} \mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) &\leq 2c \cdot \sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(2/\delta)} \cdot \frac{\log(1/\delta)}{\sqrt{C_*T}} \\ &\leq 2c \cdot \sqrt{\text{N-dim}(\Pi) \log(TK^2)} \cdot \frac{\log(2/\delta)^{3/2}}{\sqrt{C_*T}} \end{aligned}$$

$$\leq 2c \cdot \sqrt{\frac{\text{N-dim}(\Pi) \log(TK^2)}{C_* T}} \cdot \log(2/\delta)^{3/2},$$

which concludes the proof of Corollary 4.3. $\square$

C.5 Proof of Corollary 4.4

Proof of Corollary 4.4. Under the setup of Theorem 4.1, we know that with probability at least $1 - \delta$,

$$|\hat{Q}(\pi) - Q(\pi)| \leq \beta \cdot V(\pi), \quad \text{for all } \pi \in \Pi.$$

Since $\hat{\pi} = \operatorname{argmax}_{\pi \in \Pi} \{\hat{Q}(\pi) - \beta \cdot V(\pi)\}$, we see that

$$\begin{aligned} \hat{Q}(\hat{\pi}) &\geq \sup_{\pi \in \Pi} \{\hat{Q}(\pi) - \beta \cdot V(\pi) + \beta \cdot V(\hat{\pi})\} \\ &\geq \sup_{\pi \in \Pi} \{Q(\pi) - \beta \cdot V(\pi) - \beta \cdot V(\pi)\}, \end{aligned}$$

where the second inequality uses the uniform bound above and the fact that $V(\hat{\pi}) \geq 0$. Thus, we conclude the proof of Corollary 4.4. $\square$

C.6 Proof of Theorem 4.6

Proof of Theorem 4.6. Let $S = \{x_1, \dots, x_d\} \subseteq \mathcal{X}$ with $d := \text{N-dim}(\Pi)$ be the set that is N-shattered by Π (c.f. Definition 2.2). That is, there exists two functions $f_1, f_2: \mathcal{X} \rightarrow \mathcal{A}$ such that $f_1(x_i) \neq f_2(x_i)$ for all $i = 1, \dots, d$, and for every subset $S_0 \subseteq S$, there exists a policy $\pi \in \Pi$ such that $\pi(x) = f_1(x)$ for all $x \in S_0$ and $\pi(x) = f_2(x)$ for all $x \in S \setminus S_0$. We let $\mathcal{V} = \{\pm 1\}^d$ be an index set of size 2^d .

Let $\delta \in (0, 1/4)$ be a constant to be decided later. We consider a subset of instances $\mathcal{R}_0(C_*, T, \mathcal{A}, \Pi) = \{R_v\}_{v \in \mathcal{V}} \subseteq \mathcal{R}(C_*, T, \mathcal{A}, \Pi)$ indexed by all $v \in \mathcal{V}$. Each element $R_v = (C_v, e) \in \mathcal{R}_0(C_*, T, \mathcal{A}, \Pi)$ corresponds to $C_v = (\rho, \mu_v, \mathcal{A})$, where ρ is the uniform distribution over S , and the reward distribution μ_v is given by

$$\mathbb{P}(Y(a) | X = x_i) \sim \begin{cases} \text{Bern}(1/2), & a = f_1(x_i) \\ \text{Bern}(1/2 + v_i \cdot \delta), & a = f_2(x_i), \quad \forall x_i \in S. \\ \text{Bern}(0), & \text{otherwise.} \end{cases}$$

Since S is N-shattered by Π , for every $v \in \mathcal{V}$, there exists some $\pi_v^* \in \Pi$ such that $\pi_v^*(x_i) = f_1(x_i)$ for all $v_i < 0$ and $\pi_v^*(x_i) = f_2(x_i)$ for all $v_i > 0$, which is the optimal policy for C_v by definition. Since $\delta \in (0, 1/4)$, we know that for any $\pi \in \Pi$,

$$\mathcal{L}(\pi; C_v, \Pi) \geq \frac{1}{d} \sum_{i=1}^d \delta \cdot \mathbb{1}\{\pi(x_i) \neq \pi_v^*(x_i)\}. \quad (19)$$

Also, we define the fixed behavior policy by

$$e(x_i, a) = \begin{cases} C_*, & a = f_1(x_i) \\ C_*, & a = f_2(x_i), \\ (1 - 2C_*)/(K - 2), & \text{otherwise.} \end{cases} \quad \forall x_i \in S.$$

For any $v \in \mathcal{V}$, we let $\mathbb{P}_v(\cdot)$ and $\mathbb{E}_v[\cdot]$ denote the joint distribution of data under (C_v, e) . Then for any (data-dependent) policy $\hat{\pi}$, we have

$$\sup_{R_v \in \mathcal{R}_0(C_*, T)} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; C_v, \Pi)] \geq \frac{1}{2^d} \sum_{v \in \mathcal{V}} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; C_v, \Pi)] \geq \frac{1}{2^d} \sum_{v \in \mathcal{V}} \frac{1}{d} \sum_{i=1}^d \delta \cdot \mathbb{E}_v[\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_v^*(x_i)\}], \quad (20)$$

where the last equality uses (19). Now fix any $v \in \mathcal{V}$, and for any $i \in [d]$, we denote $M_i[v] \in \{\pm 1\}^d$ as the vector that only differs from v in entry i , i.e., $(M_i[v])_i = -v_i$ and $(M_i[v])_j = v_j$ for all $j \neq i$. Thus

$$\begin{aligned}
(20) &\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}} \mathbb{E}_v [\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_v^*(x_i)\}] \\
&= \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \{ \mathbb{E}_v [\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_v^*(x_i)\}] + \mathbb{E}_{M_i[v]} [\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_{M_i[v]}^*(x_i)\}] \} \\
&= \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \{ \mathbb{E}_v [\mathbb{1}\{\hat{\pi}(x_i) \neq f_2(x_i)\}] + \mathbb{E}_{M_i[v]} [\mathbb{1}\{\hat{\pi}(x_i) \neq f_1(x_i)\}] \} \\
&\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \{ \mathbb{E}_v [\mathbb{1}\{\hat{\pi}(x_i) \neq f_2(x_i)\}] + 1 - \mathbb{E}_{M_i[v]} [\mathbb{1}\{\hat{\pi}(x_i) \neq f_2(x_i)\}] \} \\
&\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} (1 - \text{TV}(\mathbb{P}_v, \mathbb{P}_{M_i[v]})), \tag{21}
\end{aligned}$$

where the second line is an equivalent decomposition of $v \in \mathcal{V}$, the third line uses the fact that $\pi_v^*(x_i) = f_2(x_i)$ for $v_i > 0$ and $\pi_v^*(x_i) = f_1(x_i)$ for $v_i < 0$, and $\text{TV}(P, Q)$ is the Total-Variation distance between distributions P, Q . Using the relationship between TV distance and KL divergence (c.f. (Tsybakov, 2008, Lemma 2.6)),

$$(1 - \text{TV}(\mathbb{P}_v, \mathbb{P}_{M_i[v]})) \geq \frac{1}{2} \exp(-\text{KL}(\mathbb{P}_v \| \mathbb{P}_{M_i[v]})), \tag{22}$$

where $\text{KL}(\mathbb{P}_v \| \mathbb{P}_{M_i[v]})$ denotes the KL divergence of $\{X_t, Y_t, A_t\}$ under R_v versus under $R_{M_i[v]}$. By construction, for any j , the conditional log-likelihood of reward $y \in \{0, 1\}$ is

$$\begin{aligned}
&\log \mathbb{P}_v(y | X_t = x_j, A_t = a) \\
&= \log(1/2) \mathbb{1}\{a = f_1(x_j)\} + (y \log(1/2 + \delta) + (1 - y)(1 - \log(1/2 + \delta))) \cdot \mathbb{1}\{v_j > 0, a = f_2(x_j)\} \\
&\quad + (y \log(1/2 - \delta) + (1 - y)(1 - \log(1/2 - \delta))) \cdot \mathbb{1}\{v_j < 0, a = f_2(x_j)\} \\
&= \log(1/2) \mathbb{1}\{a = f_1(x_j)\} + (1 - y) \mathbb{1}\{a = f_2(x_j)\} \\
&\quad + (2y - 1) \log(1/2 + \delta) \cdot \mathbb{1}\{v_j > 0, a = f_2(x_j)\} + (2y - 1) \log(1/2 - \delta) \cdot \mathbb{1}\{v_j < 0, a = f_2(x_j)\}.
\end{aligned}$$

Note that v and $M_i[v]$ only differ in entry i . We then have

$$\log \left(\frac{\mathbb{P}_v}{\mathbb{P}_{M_i[v]}}(x_j, y, a) \right) = (2y - 1) \log \left(\frac{1/2 + \delta}{1/2 - \delta} \right) \mathbb{1}\{a = f_2(x_j), j = i\}.$$

Therefore, the KL divergence can be bounded as

$$\text{KL}(\mathbb{P}_v \| \mathbb{P}_{v'}) \leq \sum_{t=1}^T \mathbb{E} [\mathbb{P}_v(A_t = f_2(x_i), X_t = x_i) \mathbb{E}_v[2Y - 1 | A_t, X_t]] \log \left(\frac{1/2 + \delta}{1/2 - \delta} \right) \leq \frac{T}{d} \cdot 12\delta^2 C_*, \tag{23}$$

where the last inequality uses the fact that $\log(1 + 2\delta) \leq 2\delta$ and $\log(1 - 2\delta) \geq -4\delta$ for $\delta \in (0, 1/4)$. Combining (21), (22), and (23), we have

$$(21) \geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \frac{1}{2} \exp(-12\delta^2 C_* T/d) = \delta \cdot \exp(-12\delta^2 C_* T/d).$$

Finally, taking $\delta = \sqrt{d/(24C_*T)}$, we arrive at

$$\begin{aligned} & \sup_{(\mathcal{C}, e) \in \mathcal{R}(C_*, T, \mathcal{A}, \Pi)} \mathbb{E}_v [\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi)] \\ & \geq \sup_{R_v \in \mathcal{R}_0(C_*, T, \mathcal{A}, \Pi)} \mathbb{E}_v [\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] \geq \frac{\exp(-1/2)}{\sqrt{24}} \sqrt{\frac{d}{C_*T}} \geq 0.12 \sqrt{\frac{d}{C_*T}}, \end{aligned}$$

which concludes the proof of Theorem 4.6. $\square$

D Proof of adaptive results

D.1 Proof of Lemma B.4

Proof of Lemma B.4. First, note that $\log(896T^2 \cdot V(\pi)^2) \leq 2\log(30T \cdot V(\pi))$. We define the event of interest as

$$\begin{aligned} \mathcal{E} &:= \left\{ \sup_{\pi \in \Pi} \left| \frac{1}{T} \sum_{t=1}^T \frac{\hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi) \sqrt{\log(896T^2 \cdot V(\pi)^2)}} \right| \geq \xi \right\} \\ &= \left\{ \left| \frac{1}{T} \sum_{t=1}^T \frac{\hat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t))}{V(\pi^\dagger) \sqrt{\log(896T^2 \cdot V(\pi^\dagger)^2)}} \right| \geq \xi \right\}, \end{aligned}$$

where we let

$$\pi^\dagger = \operatorname{argmax}_{\pi \in \Pi} \left\{ \left| \frac{1}{T} \sum_{t=1}^T \frac{\hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t))}{V(\pi) \sqrt{\log(896T^2 \cdot V(\pi)^2)}} \right| \right\}.$$

Note that $\pi^\dagger$ is measurable with respect to $\{X_t, A_t, Y_t\}_{t=1}^T$. Based on the symmetric copies, we also define

$$V'_s(\pi) = \frac{1}{T} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2} \right)^{1/2}.$$

Next, we define two additional events

$$\begin{aligned} \mathcal{E}_1 &= \left\{ \frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}'_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| \geq 2V_p(\pi^\dagger) \sqrt{\log(896T^2 \cdot V_p(\pi^\dagger)^2)} \right\}, \\ \mathcal{E}_2 &= \left\{ V'_s(\pi^\dagger) \geq 2 \cdot \max \{V_h(\pi^\dagger), V_p(\pi^\dagger)\} \right\}. \end{aligned}$$

The following lemma controls the conditional probability of $\mathcal{E}_1$ and $\mathcal{E}_2$, whose proof is in Appendix E.2.

Lemma D.1. *Under the above notations, we have*

$$\mathbb{P}(\mathcal{E}_1 | \{X_t, A_t, Y_t\}_{t=1}^T) \leq 1/4, \quad \text{and} \quad \mathbb{P}(\mathcal{E}_2 | \{X_t, A_t, Y_t\}_{t=1}^T) \leq 1/4.$$

Since $\mathcal{E}$ is measurable with respect to $\{X_t, A_t, Y_t\}_{t=1}^T$, by the tower property we have

$$\begin{aligned} \mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c \cap \mathcal{E} | \{X_t\}_{t=1}^T) &= \mathbb{E} \left[\mathbb{E}[\mathbb{1}_{\{\mathcal{E}_1^c \cap \mathcal{E}_2^c\}} \mathbb{1}_{\{\mathcal{E}\}} | \{X_t, A_t, Y_t\}_{t=1}^T] | \{X_t\}_{t=1}^T \right] \\ &= \mathbb{E} \left[\mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c | \{X_t, A_t, Y_t\}_{t=1}^T) \cdot \mathbb{1}_{\{\mathcal{E}\}} | \{X_t\}_{t=1}^T \right] \\ &\geq \frac{1}{2} \cdot \mathbb{P}(\mathcal{E} | \{X_t\}_{t=1}^T), \end{aligned}$$

where the last line uses Lemma D.1 with a union bound. We now proceed to bound the left-handed side. For any $\xi \geq 4$, on the event $\mathcal{E} \cap \mathcal{E}_1^c$, the triangle inequality implies that

$$\begin{aligned} \frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}'_t(\pi^\dagger) - \hat{\Gamma}_t(\pi^\dagger) \right| &\geq \frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| - \frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}'_t(\pi^\dagger) - \mu(X_t, \pi^\dagger(X_t)) \right| \\ &\geq \xi \cdot V(\pi^\dagger) \sqrt{\log(896T^2 \cdot V(\pi^\dagger)^2)} - 2V_p(\pi^\dagger) \geq \frac{\xi}{2} \cdot V(\pi^\dagger) \sqrt{\log(896T^2 \cdot V(\pi^\dagger)^2)}, \end{aligned} \quad (24)$$

where we used the fact that $\xi \geq 4$ and $V(\pi^\dagger) \geq V_p(\pi^\dagger) \geq 1$. Since $V(\pi^\dagger) \geq V'_s(\pi^\dagger)/2$ on the event $\mathcal{E}_2^c$, we know that on the event $\mathcal{E} \cap \mathcal{E}_1^c \cap \mathcal{E}_2^c$, we additionally have

$$\frac{\xi}{2} \cdot V(\pi^\dagger) \geq \frac{\xi}{4} \cdot (V(\pi^\dagger) + V'_s(\pi^\dagger)/2) \geq \frac{\xi}{8} \cdot (V_s(\pi^\dagger) + V'_s(\pi^\dagger)). \quad (25)$$

Taking $a, b \in \mathbb{R}^T$ whose t -th entries are given by $a_t = \mathbb{1}\{A_t = \pi^\dagger(X_t)\}/e_t(X_t, \pi^\dagger(X_t) | \mathcal{H}_t)$ and $b_t = \mathbb{1}\{A'_t = \pi^\dagger(X_t)\}/e_t(X_t, \pi^\dagger(X_t) | \mathcal{H}_t)$, the triangle inequality implies

$$V_s(\pi^\dagger) + V'_s(\pi^\dagger) = \frac{\|a\|_2 + \|b\|_2}{T} \geq \frac{\|a+b\|_2}{T} = \frac{1}{T} \left(\sum_{t=1}^T \hat{U}_t(\pi) \right)^{1/2}, \quad (26)$$

where recall that

$$\hat{U}_t(\pi) = \frac{(\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2}.$$

Combining (24), (25) and (26), we know that

$$\begin{aligned} &\mathbb{P}(\mathcal{E}_1^c \cap \mathcal{E}_2^c \cap \mathcal{E} \mid \{X_t\}_{t=1}^T) \\ &\leq \mathbb{P} \left(\frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}'_t(\pi^\dagger) - \hat{\Gamma}_t(\pi^\dagger) \right| \geq \frac{\xi}{8T} \cdot \sqrt{\sum_{t=1}^T \hat{U}_t(\pi^\dagger)} \sqrt{\log \left(56 \sum_{t=1}^T \hat{U}_t(\pi^\dagger) \mid \{X_t\}_{t=1}^T \right)} \right) \\ &\leq \mathbb{P} \left(\sup_{\pi \in \Pi} \frac{|\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \hat{\Gamma}_t(\pi)|}{\sqrt{\sum_{t=1}^T \hat{U}_t(\pi)} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(\pi))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T \right). \end{aligned} \quad (27)$$

Next, note that the quantities $\hat{\Gamma}_t(\pi)$, $\hat{\Gamma}'_t(\pi)$, $V_p(\pi)$, and $\hat{U}_t(\pi)$ only depend on $\pi \in \Pi$ through the vector $\pi(X_{1:T}) := (\pi(X_1), \dots, \pi(X_T))$. For clarity we use the notation $\hat{\Gamma}_t(\pi(X_{1:T}))$, $\hat{\Gamma}'_t(\pi(X_{1:T}))$, $V_p(\pi(X_{1:T}))$, and $\hat{U}_t(\pi(X_{1:T}))$ interchangeably to emphasize such dependence. Recall that

$$\Pi(X) = \{(\pi(X_1), \dots, \pi(X_T)) \in \mathcal{A}^T : \pi \in \Pi\}.$$

We then have

$$\begin{aligned} (27) &= \mathbb{P} \left(\sup_{a_{1:T} \in \Pi(X)} \frac{|\sum_{t=1}^T \hat{\Gamma}'_t(a_{1:T}) - \hat{\Gamma}_t(a_{1:T})|}{\sqrt{\sum_{t=1}^T \hat{U}_t(a_{1:T})} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(a_{1:T}))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T \right) \\ &\leq \sum_{a_{1:T} \in \Pi(X)} \mathbb{P} \left(\frac{|\sum_{t=1}^T \hat{\Gamma}'_t(a_{1:T}) - \hat{\Gamma}_t(a_{1:T})|}{\sqrt{\sum_{t=1}^T \hat{U}_t(a_{1:T})} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(a_{1:T}))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T \right) \\ &\leq |\Pi(X)| \cdot \sup_{a_{1:T} \in \Pi(X)} \mathbb{P} \left(\frac{|\sum_{t=1}^T \hat{\Gamma}'_t(a_{1:T}) - \hat{\Gamma}_t(a_{1:T})|}{\sqrt{\sum_{t=1}^T \hat{U}_t(a_{1:T})} \cdot \sqrt{\log(56 \sum_{t=1}^T \hat{U}_t(a_{1:T}))}} \geq \frac{\xi}{8} \mid \{X_t\}_{t=1}^T \right) \end{aligned}$$

$$\leq |\Pi(X)| \cdot \sup_{\pi \in \Pi} \mathbb{P} \left(\frac{|\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \hat{\Gamma}_t(\pi)|}{\sqrt{\sum_{t=1}^T \hat{U}_t(\pi) \cdot \log(56 \sum_{t=1}^T \hat{U}_t(\pi))}} \geq \frac{\xi}{8} \middle| \{X_t\}_{t=1}^T \right),$$

where the second line uses a union bound, and the last line is because every $a_{1:T} \in \Pi(X)$ corresponds to a fixed choice of $\pi \in \Pi$. This completes the proof of Lemma B.4. $\square$

D.2 Proof of Lemma B.6

Proof of Lemma B.6. Fix any $\pi \in \Pi$. Since we always condition on $\{X_t\}_{t=1}^T$ and have fixed π , only for this proof, we suppress the dependence on π and X_t , and write $\hat{\Gamma}_t$, $\hat{\Gamma}'_t$, $\hat{U}_t$, $\gamma_t(z)$, $\gamma'_t(z)$, and $\hat{u}_t(z)$ instead of the notations used in Section B.2.

First, recall that $\mathcal{H}_T = \{X_t, A_t, Y_t\}_{t=1}^{T-1}$, so $\mathcal{H}_T \cup \{X_T\} \supseteq \{X_t\}_{t=1}^T$. By the tower property, we have

$$\begin{aligned} & \mathbb{P} \left(\frac{|\sum_{t=1}^T \{\hat{\Gamma}'_t - \hat{\Gamma}_t\}|}{\sqrt{\sum_{t=1}^T \hat{U}_t \cdot \log(56 \sum_{t=1}^T \hat{U}_t)}} \geq \frac{\xi}{8} \middle| \{X_t\}_{t=1}^T \right) \\ &= \mathbb{E} \left[\mathbb{P} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{\sqrt{\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T \cdot \log(56[\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T])}} \geq \frac{\xi}{8} \middle| \mathcal{H}_T \cup \{X_T\} \right) \middle| \{X_t\}_{t=1}^T \right] \\ &= \mathbb{E} \left[\mathbb{P} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{\sqrt{\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T \cdot \log(56[\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T])}} \geq \frac{\xi}{8} \middle| \mathcal{H}_T \cup \{X_T\} \right) \middle| \{X_t\}_{t=1}^T \right]. \end{aligned} \quad (28)$$

To see why the last line is true, simply note that conditional on $\{X_t, A_t, Y_t\}_{t=1}^{T-1} \cup \{X_T\}$, the two random vectors have the same distribution:

$$(\varepsilon_T \cdot \{\hat{\Gamma}'_T(\pi) - \hat{\Gamma}_T(\pi)\}, \hat{U}_T(\pi)) \stackrel{d}{=} (\{\hat{\Gamma}'_T(\pi) - \hat{\Gamma}_T(\pi)\}, \hat{U}_T(\pi)) \mid \{X_t, A_t, Y_t\}_{t=1}^{T-1} \cup \{X_T\}.$$

The above is because conditional on $\{X_t, A_t, Y_t\}_{t=1}^{T-1} \cup \{X_T\}$, (A_T, Y_T) and (A'_T, Y'_T) are independent and have the same distribution by construction, and that $\hat{U}_T = (\mathbb{1}\{A_t = \pi(X_t)\} + \mathbb{1}\{A'_t = \pi(X_t)\})^2 / e_t(X_t, \pi(X_t) \mid \mathcal{H}_t)^2$ is invariant to the permutation of A_t and A'_t . Hereafter for notational simplicity, we define

$$g(z) = \sqrt{z \cdot \log(56z)}.$$

Symmetrization for the $t = T$. In the following, we are to apply the symmetrization trick to turn (28) into a tail probability over ε_T , where the symmetrization argument is more delicate than in the i.i.d. case. Applying the tower property again to the inner probability in (28), we have

$$\begin{aligned} & \mathbb{P} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{\sqrt{\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T \cdot \log(56[\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T])}} \geq \frac{\xi}{8} \middle| \mathcal{H}_T \cup \{X_T\} \right) \\ &= \mathbb{E} [P_T(\mathcal{H}_T, X_T, A_T, Y_T, A'_T, Y'_T, e_T, \hat{\mu}_T) \mid \mathcal{H}_T \cup \{X_T\}] \end{aligned} \quad (29)$$

Here, we define

$$\begin{aligned} & P_T(\mathcal{H}_T, X_T, A_T, Y_T, A'_T, Y'_T, e_T, \hat{\mu}_T) \\ &:= \mathbb{P}_\varepsilon \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T)} \geq \frac{\xi}{8} \right), \end{aligned}$$

where $\mathbb{P}_\varepsilon$ denotes the probability over $\varepsilon_T \sim \text{Unif}\{\pm 1\}$ while treating everything else as fixed values; $e_T = e_T(\cdot, \cdot \mid \mathcal{H}_T)$ is the sampling rule at time T and $\hat{\mu}_T$ is the estimator for μ that both depend on $\mathcal{H}_T$. More

specifically, by the distribution of ε_T ,

$$\begin{aligned} & P_T(\mathcal{H}_T, X_T, A_T, Y_T, A'_T, Y'_T, e_T, \hat{\mu}_T) \\ &= \frac{1}{2} \cdot \mathbb{1} \left\{ \frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T)} \geq \frac{\xi}{8} \right\} + \frac{1}{2} \cdot \mathbb{1} \left\{ \frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} - \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T)} \geq \frac{\xi}{8} \right\}, \end{aligned}$$

which is a random variable with respect to $\mathcal{H}_T$ and $X_T, A_T, Y_T, A'_T, Y'_T, e_T, \hat{\mu}_T$.

Taking supremum over all possible realization of A_T, A'_T, Y_T, Y'_T and $e_T(\cdot, \cdot), \hat{\mu}_T(\cdot, \cdot)$ in the potentially larger set $\mathcal{Z}_T := \mathcal{Z}$, we have

$$\begin{aligned} & \mathbb{P} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\Gamma}'_T - \hat{\Gamma}_T\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{U}_T)} \geq \frac{\xi}{8} \middle| \mathcal{H}_T \cup \{X_T\} \right) \\ & \leq \sup_{z_T \in \mathcal{Z}_T} P_T(\mathcal{H}_T, X_T, a_T, y_T, a'_T, y'_T, f_T, h_T) \middle| \mathcal{H}_T \cup \{X_T\}, \end{aligned}$$

where $z_T = (a_T, y_T, a'_T, y'_T, f_T, h_T) \in \mathcal{Z}_T = \mathcal{Z}$ is any realization of A_T, A'_T, Y_T, Y'_T and $e_T(\cdot, \cdot), \hat{\mu}_T(\cdot, \cdot)$. Now we recall the definitions (suppressing the dependence on π and $\{X_t\}_{t=1}^T$)

$$\begin{aligned} \hat{\gamma}_T(z_T) &= h_T(X_T, \pi(X_T)) + \frac{\mathbb{1}\{a_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))} \cdot (y_T - h_T(X_T, \pi(X_T))), \\ \hat{\gamma}'_T(z_T) &= h_T(X_T, \pi(X_T)) + \frac{\mathbb{1}\{a'_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))} \cdot (y'_T - h_T(X_T, \pi(X_T))), \\ \hat{u}_T(z_T) &= \frac{(\mathbb{1}\{a_T = \pi(X_T)\} + \mathbb{1}\{a'_T = \pi(X_T)\})^2}{f_T(X_T, \pi(X_T))^2}. \end{aligned}$$

Returning to (28) and (29) and recalling the definition of P_T , we know that

$$(28) \leq \mathbb{E} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\gamma}'_T(z_T) - \hat{\gamma}_T(z_T)\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \middle| \{X_t\}_{t=1}^T \right]. \quad (30)$$

where $\mathbb{P}_{\varepsilon}$ denotes the probability over ε_T while conditioning on everything else. Hence, for any $z_T \in \mathcal{Z}_T$, the inner probability in (30) depends on $\{X_t, A_t, Y_t, A'_t, Y'_t\}_{t=1}^{T-1} \cup \{X_T\}$. So far, we have turned (28) into an expectation of the tail probability for ε_T . We will continue to do this for $t = T - 1$.

Recursive symmetrization for $t = T - 1$. We now continue the recursive symmetrization trick for $t = T - 1$, and obtain an upper bound for

$$\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_T \cdot \{\hat{\gamma}'_T(z_T) - \hat{\gamma}_T(z_T)\}|}{g(\sum_{t=1}^{T-1} \hat{U}_t + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right), \quad (31)$$

which is a random variable determined by $\{\hat{\Gamma}'_t - \hat{\Gamma}_t\}_{t=1}^{T-1} \cup \{\hat{U}_t\}_{t=1}^{T-1} \cup \{X_T\}$. For convenience we will write

$$D_T(\varepsilon_T, z_T) := \varepsilon_T \cdot \{\hat{\gamma}'_T(z_T) - \hat{\gamma}_T(z_T)\}.$$

Similar to the previous step, we know by the construction of (A'_{T-1}, Y'_{T-1}) that

$$(\varepsilon_{T-1} \cdot \{\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}\}, \hat{U}_{T-1}) \stackrel{d}{=} (\{\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}\}, \hat{U}_{T-1}) \middle| \mathcal{H}_{T-1} \cup \{X_{T-1}, X_T\}.$$

Since (31) is a random variable that can be viewed as a function of $\{\hat{\Gamma}'_t - \hat{\Gamma}_t\}_{t=1}^{T-1} \cup \{\hat{U}_t\}_{t=1}^{T-1} \cup \{X_T\}$, the above equivalence in distribution leads to

$$(31) \stackrel{d}{=} \sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_{T-1} \cdot (\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right),$$

where the above two distributions are both conditional on $\{X_t, A_t, Y_t\}_{t=1}^{T-2} \cup \{X_{T-1}, X_T\}$. Define the σ -field $\mathcal{F}_{T-1} := \sigma(\{X_t, A_t, Y_t\}_{t=1}^{T-2} \cup \{X_{T-1}, X_T\})$. We have

$$\begin{aligned} & \mathbb{E} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-1} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \middle| \mathcal{F}_{T-1} \right] \\ &= \mathbb{E} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_{T-1} \cdot (\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \middle| \mathcal{F}_{T-1} \right] \\ &= \mathbb{E} \left[\mathbb{E}_{\varepsilon_{T-1}} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_{T-1} \cdot (\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \middle| \mathcal{F}_{T-1} \right] \right], \end{aligned}$$

where the second line uses the equality in conditional distribution, and the third line uses the tower property with $\mathbb{E}_{\varepsilon_{T-1}}$ denoting the expectation over ε_{T-1} while conditioning on everything else. In particular, we can rewrite the inner expectation as

$$\begin{aligned} & \mathbb{E}_{\varepsilon_{T-1}} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + \varepsilon_{T-1} \cdot (\hat{\Gamma}'_{T-1} - \hat{\Gamma}_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \right] \\ &= P_{T-1}(\mathcal{H}_{T-2}, X_{(T-1):T}, A_{T-1}, Y_{T-1}, A'_{T-1}, Y'_{T-1}, e_{T-1}, \hat{\mu}_{T-1}), \end{aligned} \quad (32)$$

where $P_{T-1}(\dots)$ is a fixed function, and e_{T-1} and $\hat{\mu}_{T-1}$ are the sampling propensity and regression estimator at time $T-1$. Therefore, we can write

$$\begin{aligned} (32) &= \mathbb{E} \left[P_{T-1}(\mathcal{H}_{T-2}, X_{(T-1):T}, A_{T-1}, Y_{T-1}, A'_{T-1}, Y'_{T-1}, e_{T-1}, \hat{\mu}_{T-1}) \middle| \mathcal{F}_{T-1} \right] \\ &\leq \mathbb{E} \left[\sup_{z_{T-1} \in \mathcal{Z}_{T-1}} P_{T-1}(\mathcal{H}_{T-2}, X_{(T-1):T}, a_{T-1}, y_{T-1}, a'_{T-1}, y'_{T-1}, f_{T-1}, h_{T-1}) \middle| \mathcal{F}_{T-1} \right] \end{aligned} \quad (33)$$

where $z_{T-1} = (a_{T-1}, y_{T-1}, a'_{T-1}, y'_{T-1}, f_{T-1}, h_{T-1}) \in \mathcal{Z}_{T-1} := \mathcal{Z}$ denotes any possible realization of $(A_{T-1}, Y_{T-1}, A'_{T-1}, Y'_{T-1}, e_{T-1}, \hat{\mu}_{T-1})$. By the definition of P_{T-1} , (33) is upper bounded by

$$\sup_{z_{T-1} \in \mathcal{Z}_{T-1}} \mathbb{E}_{\varepsilon_{T-1}} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + D_{T-1}(\varepsilon_{T-1}, z_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{u}_{T-1}(z_{T-1}) + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \right],$$

where for any $z_{T-1} = (a_{T-1}, y_{T-1}, a'_{T-1}, y'_{T-1}, f_{T-1}, h_{T-1}) \in \mathcal{Z}_{T-1} := \mathcal{Z}$, we write

$$\begin{aligned} D_{T-1}(\varepsilon_{T-1}, z_{T-1}) &= \varepsilon_{T-1} \cdot \{\hat{\gamma}'_{T-1}(z_{T-1}) - \hat{\gamma}_{T-1}(z_{T-1})\}, \\ \text{and } \hat{u}_{T-1}(z_{T-1}) &= \frac{(\mathbf{1}\{a_{T-1} = \pi(X_{T-1})\} + \mathbf{1}\{a'_{T-1} = \pi(X_{T-1})\})^2}{f_{T-1}(X_{T-1}, \pi(X_{T-1}))^2}. \end{aligned}$$

Returning to (30), we have (30) upper bounded by

$$\mathbb{E} \left[\sup_{z_{T-1} \in \mathcal{Z}_{T-1}} \mathbb{E}_{\varepsilon_{T-1}} \left[\sup_{z_T \in \mathcal{Z}_T} \mathbb{P}_{\varepsilon_T} \left(\frac{|\sum_{t=1}^{T-2} \{\hat{\Gamma}'_t - \hat{\Gamma}_t\} + D_{T-1}(\varepsilon_{T-1}, z_{T-1}) + D_T(\varepsilon_T, z_T)|}{g(\sum_{t=1}^{T-2} \hat{U}_t + \hat{U}_{T-1} + \hat{u}_T(z_T))} \geq \frac{\xi}{8} \right) \right] \middle| \{X_t\}_{t=1}^T \right],$$

which completes the recursive symmetrization for $t = T-1$.

Continuing the recursive symmetrization. Using similar notations $D_t(\varepsilon_t, z_t)$, $\hat{u}_t(z_t)$ for $z_t \in \mathcal{Z}_t := \mathcal{Z}$, we can continue the above recursive symmetrization for $t = T-2, T-3, \dots, 1$. This gives

$$(30) \leq \sup_{z_1 \in \mathcal{Z}_1} \mathbb{E}_{\varepsilon_1} \sup_{z_2 \in \mathcal{Z}_2} \mathbb{E}_{\varepsilon_2} \cdots \sup_{z_T \in \mathcal{Z}_T} \mathbb{E}_{\varepsilon_T} \mathbf{1} \left\{ \sup_{\pi \in \Pi} \left| \frac{\sum_{t=1}^T D_t(\varepsilon_t, z_t)}{g(\sum_{t=1}^T \hat{u}_t(z_t))} \right| \geq \frac{\xi}{8} \right\}, \quad (34)$$

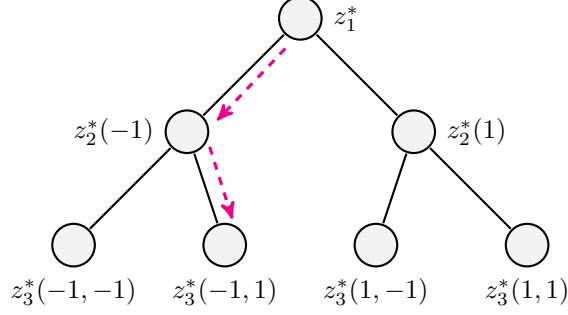


Figure 14: The optimal $\mathcal{Z}$ -valued tree of depth $T = 3$. The pink path shows a realization $\varepsilon_1 = -1$ and $\varepsilon_2 = 1$.

where $\mathbb{E}_{\varepsilon_t}$ denotes the expectation over the randomness in ε_t while conditioning on everything else.

We note that in (34), for every $t > 1$, the choice of $z_t \in \mathcal{Z}_t$ that attains the supremum depends on the realization of $\varepsilon_1, \dots, \varepsilon_{t-1}$. Now let $z_1^* \in \mathcal{Z}_1$ denote the (fixed) element in $\mathcal{Z}_1$ that attains the supremum in (34). It is fixed since we have conditioned on $\{X_t\}_{t=1}^T$ and fixed some π . Then the element that attains the second supremum $z_2 \in \mathcal{Z}_2$ depends on ε_1 ; we define it as $z_2^*(\varepsilon_1) \in \mathcal{Z}_2$ to emphasize such dependence. Further, the element $z_3 \in \mathcal{Z}_3$ that achieves the third supremum depends on $\varepsilon_1, \varepsilon_2$, and we denote it by $z_3^*(\varepsilon_{1:2})$, etc. So on and so forth, for every $t \in [T]$, we can write $z_t^*(\varepsilon_{1:(t-1)})$ that attain the supremum $z_t \in \mathcal{Z}_t$ which depends on $\varepsilon_1, \dots, \varepsilon_{t-1}$. That is, we can use a $\mathcal{Z}$ -valued tree (c.f. Definition B.5) to represent the $2^T - 1$ elements $z_1^*, z_2^*(\pm 1), z_3^*(\{\pm 1\}^2), \dots, z_T^*(\{\pm 1\}^{T-1})$. Figure 14 is a visualization.

Recall that $\mathbb{T}_T(\mathcal{Z})$ is the set of all $\mathcal{Z}$ -valued trees of depth T , and the optimal tree

$$\mathbf{z}^* = (z_1^*, z_2^*(\pm 1), z_3^*(\{\pm 1\}^2), \dots, z_T^*(\{\pm 1\}^{T-1})) \in \mathbb{T}_T(\mathcal{Z}),$$

as illustrated in Figure 14. With this notation, we can rewrite (34) as

$$(34) = \mathbb{P}_\varepsilon \left(\left| \frac{\sum_{t=1}^T \varepsilon_t \cdot \{\hat{\gamma}_t(\pi; X_t, z_t^*(\varepsilon_{1:(t-1)})) - \hat{\gamma}'_t(\pi; X_t, z_t^*(\varepsilon_{1:(t-1)}))\}}{g(\sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t^*(\varepsilon_{1:(t-1)})))} \right| \geq \frac{\xi}{8} \cdot (\log T)^{\alpha/2} \right),$$

where $\mathbb{P}_\varepsilon$ is over the randomness of $\varepsilon_{1:T}$ treating everything else as fixed, and we have returned to the original notations for $\hat{\gamma}_t, \hat{\gamma}'_t$ and $\hat{u}_t$. We emphasize that, above, the only randomness in $z_t^*(\varepsilon_{1:(t-1)})$ is in the realization of $\varepsilon_{1:(t-1)}$, while the image set $z_t^*(\{\pm 1\}^{t-1})$ is fixed for all $t \in [T]$. Finally, taking the supremum over all $\mathbf{z} \in \mathbb{T}_T(\mathcal{Z})$ concludes our proof. $\square$

D.3 Proof of Lemma B.7

Proof of Lemma B.7. As before, we condition on $\{X_t\}_{t=1}^T$ throughout the proof. For any $\mathbf{z} = \{z_t(\{\pm 1\}^{t-1})\}_{t=1}^T \in \mathbb{T}_T(\mathcal{Z})$, we define two processes

$$S_t := \sum_{s=1}^t \varepsilon_s \cdot \{\hat{\gamma}_s(\pi; X_s, z_s(\varepsilon_{1:(s-1)})) - \hat{\gamma}'_s(\pi; X_s, z_s(\varepsilon_{1:(s-1)}))\},$$

$$M_t := \sum_{s=1}^t 4 \cdot \hat{u}_s(\pi; X_s, z_s(\varepsilon_{1:(s-1)})), \quad t \in [T],$$

and $S_0 = M_0 = 0$. Next, for any $\lambda > 0$, we define

$$L_t(\lambda) := \exp \left(\lambda S_t - \frac{\lambda^2}{2} M_t \right), \quad t \in [T].$$

It is clear that $\{S_t\}$ and $\{M_t\}$ are adapted to the filtration $\{\mathcal{G}_t\}$ where we define $\mathcal{G}_t := \sigma(\varepsilon_{1:t})$. Also, we can check that for any $\lambda > 0$, $L_T(\lambda)$ is a super-martingale with respect to $\{\mathcal{G}_t\}$ with $L_0(\lambda) = 1$. To see this, we note that

$$\begin{aligned}
& \mathbb{E}[L_T(\lambda) \mid \mathcal{G}_{T-1}] \\
&= L_{T-1}(\lambda) \cdot \exp\left(-\frac{\lambda^2}{2} \cdot 4\widehat{u}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)}))\right) \\
&\quad \times \mathbb{E}\left[\exp\left(\lambda \varepsilon_T \cdot \{\widehat{\gamma}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)})) - \widehat{\gamma}'_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)}))\}\right) \mid \mathcal{G}_{T-1}\right] \\
&\leq L_{T-1}(\lambda) \cdot \exp\left(-\frac{\lambda^2}{2} \cdot 4\widehat{u}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)}))\right. \\
&\quad \left.+ \frac{\lambda^2}{2} \cdot \left\{\widehat{\gamma}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)})) - \widehat{\gamma}'_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)}))\right\}^2\right),
\end{aligned}$$

where the last step is because $(e^t + e^{-t})/2 \leq e^{t^2/2}$ for any $t \in \mathbb{R}$. Meanwhile, writing $z_T(\varepsilon_{1:(T-1)}) = (a_T, a'_T, y_T, y'_T, f_T, h_T)$, it holds deterministically that

$$\begin{aligned}
& \left\{\widehat{\gamma}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)})) - \widehat{\gamma}'_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)}))\right\}^2 \\
&= \left\{\frac{y_T \mathbf{1}\{a_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))} - \frac{y'_T \mathbf{1}\{a'_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))}\right. \\
&\quad \left.+ h_T(X_T, \pi(X_T)) \cdot \left(\frac{\mathbf{1}\{a_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))} - \frac{\mathbf{1}\{a'_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))}\right)\right\}^2 \\
&\leq 2 \cdot \left\{y_T \cdot \frac{\mathbf{1}\{a_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))} - y'_T \cdot \frac{\mathbf{1}\{a'_T = \pi(X_T)\}}{f_T(X_T, \pi(X_T))}\right\}^2 + 2\widehat{u}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)})) \\
&\leq 4\widehat{u}_T(\pi; X_T, z_T(\varepsilon_{1:(T-1)})),
\end{aligned}$$

where the first inequality uses the fact that $(a+b)^2 \leq 2a^2 + 2b^2$ for any $a, b \in \mathbb{R}$. Collectively, we have

$$\mathbb{E}[L_T(\lambda) \mid \mathcal{G}_{T-1}] \leq L_{T-1}(\lambda),$$

and hence $L_T(\lambda)$ is a super-martingale with

$$\mathbb{E}[L_T(\lambda)] = \mathbb{E}[\exp(\lambda A - \lambda^2 B^2/2)] \leq 1, \quad \text{where } A = S_T, \ B = M_T^{1/2}.$$

Now invoking Lemma F.2 for A, B and taking $y = 1$, we know that for any $x \geq \sqrt{2}$,

$$\begin{aligned}
\exp(-x^2/2) &\geq \mathbb{P}\left(\frac{|A|}{\sqrt{(B^2 + 1)(1 + 0.5 \log(B^2 + 1))}} \geq x\right) \\
&\geq \mathbb{P}\left(\frac{|A|}{\sqrt{2B^2 \cdot (1 + 0.5 \log(2B^2))}} \geq x\right), \\
&\geq \mathbb{P}\left(\frac{|A|}{\sqrt{B^2 \cdot \log(14B^2)}} \geq x\right),
\end{aligned} \tag{35}$$

where the second line uses the fact that $B^2 + 1 \leq 2B^2$ since $B \geq 1$, and the third line uses the fact that $1 + \log(2B^2)/2 = \log(2e^2 B^2)/2 \leq \log(14B^2)/2$. In (35), plugging in the expression of A and B we obtain

$$\mathbb{P}_\varepsilon \left(\frac{|\sum_{t=1}^T \varepsilon_t \cdot \{\hat{\gamma}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1})) - \hat{\gamma}'_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))\}|}{2\sqrt{\sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))} \sqrt{\log[56 \sum_{t=1}^T \hat{u}_t(\pi; X_t, z_t^*(\varepsilon_{1:t-1}))]}} \geq x \right) \leq e^{-x^2/2}$$

for any $x \geq \sqrt{2}$, which concludes the proof of Lemma B.7. $\square$

D.4 Proof of Lemma B.8

Proof of Lemma B.8. Following exactly the same arguments as in the proof of Lemma B.3, we have that

$$\sup_{\pi \in \Pi} \left| \sum_{t=1}^T \mu(X_t, \pi(X_t)) - Q(\pi) \right| \leq \sqrt{T} \cdot \sqrt{8(\text{N-dim}(\Pi) \log(TK^2) + \log(16/\delta))}$$

with probability at least $1 - \delta/2$. Noting that $V(\pi) \geq \sqrt{T}$ deterministically completes the proof of Lemma B.8. $\square$

D.5 Proof of Corollary 5.5

Proof of Corollary 5.5. From Theorem 5.1, it holds with probability at least $1 - \delta$ that $\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) \leq 2\beta \cdot V(\pi^*)$ if we choose any $\beta \geq 67 \cdot (\log T)^{\alpha/2} \cdot \sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(16/\delta)}$. We now proceed to bound $V(\pi^*)$ by the three terms in Equation (7) separately.

Bounding $V_s(\pi^)$.* We consider martingale differences $D_t = \frac{\mathbb{1}\{A_t = \pi^*(X_t)\}}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)^2} - \frac{1}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)}$ and the filtration $\mathcal{F}_{t-1} = \sigma(\mathcal{H}_t, X_t)$. Then $\{D_t\}$ is adapted to $\{\mathcal{F}_t\}$ and $\mathbb{E}[D_t | \mathcal{F}_{t-1}] = 0$. Moreover, Assumption 5.4 implies

$$M_t := \sum_{j=1}^t \mathbb{E}[D_j^2 | \mathcal{F}_{j-1}] \leq \sum_{j=1}^t \frac{1}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)^3} \leq \sum_{j=1}^t g_t^{-3} \leq \frac{\bar{c}^{-3}}{1 + 3\gamma} (T + 1)^{1+3\gamma}$$

for all $t \in [T]$. Also, $D_t \leq g_t^{-2} \leq R := T^{2\gamma}$. Then invoking Freedman's inequality (c.f. Lemma F.4) with $\sigma^2 = \frac{\bar{c}^{-3}}{1+3\gamma} (T + 1)^{1+3\gamma}$ and $R = T^{2\gamma}$, for any $x \geq 0$, we have

$$\mathbb{P} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A_t = \pi^*(X_t)\}}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)^2} - \sum_{t=1}^T \frac{1}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)} \geq x \right) \leq \exp \left(\frac{-x^2/2}{\sigma^2 + Rx/3} \right).$$

Now we take $x = 2 \max\{1, \sqrt{\bar{c}^{-3}/(1 + 3\gamma)}\} \cdot (T + 1)^{(1+3\gamma)/2} \cdot \log(1/\delta)$. Then since $(1 + 3\gamma)/2 \geq 2\gamma$, we know $x \geq 2\sigma \log(1/\delta)$ and $x \geq 2R \log(1/\delta)$. Hence

$$\frac{-x^2/2}{\sigma^2 + Rx/3} \leq \frac{-2 \log(1/\delta)^2}{1 + 2/3 \cdot \log(1/\delta)} \leq -\log(1/\delta).$$

That is, with probability at least $1 - \delta$, we have

$$\begin{aligned} V_s(\pi^*) &\leq \frac{1}{T} \left\{ \sum_{t=1}^T \frac{1}{e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)} + 2 \max \left\{ 1, \frac{\bar{c}^{-3/2}}{\sqrt{1 + 3\gamma}} \right\} \cdot (T + 1)^{(1+3\gamma)/2} \cdot \log(1/\delta) \right\}^{1/2} \\ &\leq \frac{1}{T} \left\{ \sum_{t=1}^T \bar{c}^{-1} \cdot t^\gamma + 2 \max \left\{ 1, \frac{\bar{c}^{3/2}}{\sqrt{1 + 3\gamma}} \right\} \cdot (T + 1)^{(1+3\gamma)/2} \cdot \log(1/\delta) \right\}^{1/2} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{1}{T} \left\{ \frac{\bar{c}^{-1} \cdot (T+1)^{1+\gamma}}{1+\gamma} + 2 \max \left\{ 1, \frac{\bar{c}^{-3/2}}{\sqrt{1+3\gamma}} \right\} \cdot (T+1)^{(1+3\gamma)/2} \cdot \log(1/\delta) \right\}^{1/2} \\
&\leq \sqrt{\frac{2\bar{c}^{-1} + 4 \max\{1, \bar{c}^{-3/2}\} \cdot \log(1/\delta) \cdot T^{(\gamma-1)/2}}{T^{1-\gamma}}}.
\end{aligned}$$

Here the fourth line uses the fact that $(T+1)^{1+\gamma}/(1+\gamma) \leq T^{1+\gamma} \cdot \frac{2^{1+\gamma}}{1+\gamma} \leq 2T^{1+\gamma}$ and

$$(T+1)^{(1+3\gamma)/2} \leq T^{(1+3\gamma)/2} \cdot \frac{2^{(1+3\gamma)/2}}{\sqrt{1+3\gamma}} \leq 2T^{(1+3\gamma)/2}$$

for $\gamma \in (0, 1)$.

Bounding $V_p(\pi^)$.* By the lower bound for $e_t(X_t, \pi^*(X_t) | \mathcal{H}_t)$, we have the deterministic upper bound

$$V_p(\pi^*) \leq \frac{1}{T} \sqrt{\sum_{t=1}^T \bar{c}^{-1} \cdot t^\gamma} \leq \sqrt{\frac{\bar{c}^{-1}}{T^2} \cdot \frac{(T+1)^{\gamma+1}}{\gamma+1}} \leq \sqrt{\frac{2}{\bar{c} \cdot T^{1-\gamma}}}.$$

Bounding $V_h(\pi^)$.* Finally, we have the deterministic upper bound

$$V_h(\pi^*) \leq \frac{1}{T} \left(\sum_{t=1}^T \bar{c}^{-3} \cdot t^{3\gamma} \right)^{1/4} = \left(\frac{1}{\bar{c}^3 \cdot T^4} \cdot \frac{(T+1)^{1+3\gamma}}{1+3\gamma} \right)^{1/4} \leq \left(\frac{2 \cdot T^{1+3\gamma}}{\bar{c}^3 \cdot T^4} \right)^{1/4} \leq \frac{2}{\bar{c}^{3/4} \cdot T^{3(1-\gamma)/4}}.$$

Putting the three terms together, we know that with probability at least $1 - \delta$,

$$\begin{aligned}
V(\pi^*) &\leq \max \left\{ \sqrt{\frac{2/\bar{c} + 4 \max\{1, \bar{c}^{-3/2}\} \cdot \log(1/\delta) \cdot T^{(\gamma-1)/2}}{T^{1-\gamma}}}, \sqrt{\frac{2\bar{c}^{-1}}{T^{1-\gamma}}}, \frac{2\bar{c}^{-3/4}}{T^{3(1-\gamma)/4}} \right\} \\
&\leq T^{-(1-\gamma)/2} \cdot \max \left\{ \sqrt{2/\bar{c} + 4 \max\{1, \bar{c}^{-3/2}\} \cdot \log(1/\delta) \cdot T^{(\gamma-1)/2}}, 2\bar{c}^{-3/4} T^{-(1-\gamma)/4} \right\} \\
&\leq T^{-(1-\gamma)/2} \cdot \max \left\{ 2/\sqrt{\bar{c}}, 4 \max\{1, \bar{c}^{-3/4}\} \cdot \log(1/\delta) \cdot T^{-\frac{1-\gamma}{4}} \right\} \\
&\leq 4T^{-(1-\gamma)/2} \cdot \max \left\{ 1/\sqrt{\bar{c}}, \max\{1, \bar{c}^{-3/4}\} \cdot \log(1/\delta) \cdot T^{-\frac{1-\gamma}{4}} \right\} \\
&\leq \frac{4 \log(1/\delta)}{T^{(1-\gamma)/2} \cdot \max\{1, \bar{c}^{3/4}\}}.
\end{aligned}$$

Here we used the fact that $\sqrt{x} \leq \max\{1, x^{3/4}\}$ for $x > 0$ and $T^{-(1-\gamma)/4} \leq 1$. Further applying a union bound, we know that with probability at least $1 - 2\delta$, since $TK^2 \geq 8/\delta$,

$$\begin{aligned}
\mathcal{L}(\hat{\pi}; \mathcal{C}, \Pi) &\leq 2c \cdot (\log T)^{\alpha/2} \cdot \sqrt{\text{N-dim}(\Pi) \log(TK^2) + \log(8/\delta)} \cdot \frac{4 \log(1/\delta)}{T^{(1-\gamma)/2} \cdot \max\{1, \bar{c}^{3/4}\}} \\
&\leq 2c \cdot (\log T)^{\alpha/2} \cdot \sqrt{2\text{N-dim}(\Pi) \log(TK^2)} \cdot \frac{4 \log(1/\delta)}{T^{(1-\gamma)/2} \cdot \max\{1, \bar{c}^{3/4}\}} \\
&\leq 12c \cdot \sqrt{\frac{\text{N-dim}(\Pi)}{T^{1-\gamma}}} \cdot \frac{(\log(TK^2))^{(1+\alpha)/2} \log(1/\delta)}{\max\{1, \bar{c}^{3/4}\}},
\end{aligned}$$

which concludes the proof of Corollary 5.5. $\square$

D.6 Proof of Theorem 5.7

Proof of Theorem 5.7. This proof is similar to that of Theorem 4.6, and also similar to that in Zhan et al. (2021).

Let $S = \{x_1, \dots, x_d\} \subseteq \mathcal{X}$ with $d := \text{N-dim}(\Pi)$ be the set that is N-shattered by Π . Then we know that there exists two functions $f_1, f_2: \mathcal{X} \rightarrow \mathcal{A}$ such that $f_1(x_i) \neq f_2(x_i)$ for all $i = 1, \dots, d$, and for every subset $S_0 \subseteq S$, there exists a policy $\pi \in \Pi$ such that $\pi(x) = f_1(x)$ for all $x \in S_0$ and $\pi(x) = f_2(x)$ for all $x \in S \setminus S_0$. We let $\mathcal{V} = \{\pm 1\}^d$ be an index set of size 2^d .

Let $\delta \in (0, 1/4)$ be a constant to be decided later. We consider a subset of instances $\mathcal{R}_0(\bar{c}, \gamma, T) = \{R_v\}_{v \in \mathcal{V}} \subseteq \mathcal{R}(\bar{c}, \gamma, T)$ indexed by all $v \in \mathcal{V}$. Each element $R_v = (\mathcal{C}_v, e) \in \mathcal{R}_0(\bar{c}, \gamma, T)$ corresponds to $\mathcal{C}_v = (\rho, \mu_v, \mathcal{A})$, where ρ is the uniform distribution over S , and the reward distribution μ_v is given by

$$\mathbb{P}(Y(a) | X = x_i) \sim \begin{cases} \text{Bern}(1/2), & a = f_1(x_i) \\ \text{Bern}(1/2 + v_i \cdot \delta), & a = f_2(x_i), \quad \forall x_i \in S. \\ \text{Bern}(0), & \text{otherwise.} \end{cases}$$

Since S is N-shattered by Π , for every $v \in \mathcal{V}$, there exists some $\pi_v^* \in \Pi$ such that $\pi_v^*(x_i) = f_1(x_i)$ for all $v_i < 0$ and $\pi_v^*(x_i) = f_2(x_i)$ for all $v_i > 0$, which is the optimal policy for $\mathcal{C}_v$ by definition. Since $\delta \in (0, 1/4)$, similar to the proof of Theorem 4.1, we know

$$\mathcal{L}(\pi; \mathcal{C}_v, \Pi) \geq \frac{1}{d} \sum_{i=1}^d \delta \cdot \mathbb{1}\{\pi(x_i) \neq \pi_v^*(x_i)\}.$$

Also, we define the fixed behavior policy by

$$e_t(x_i, a) = \begin{cases} \bar{c} \cdot t^{-\gamma}, & a = f_1(x_i) \\ \bar{c} \cdot t^{-\gamma}, & a = f_2(x_i), \\ (1 - 2\bar{c} \cdot t^{-\gamma})/(K - 2), & \text{otherwise.} \end{cases} \quad \forall x_i \in S.$$

For any $v \in \mathcal{V}$, we let $\mathbb{P}_v(\cdot)$ and $\mathbb{E}_v[\cdot]$ denote the joint distribution of data under $(\mathcal{C}_v, e)$. Now fix any $v \in \mathcal{V}$, and for any $i \in [d]$, we denote $M_i[v] \in \{\pm 1\}^d$ as the vector that only differs from v in entry i , i.e., $(M_i[v])_i = -v_i$ and $(M_i[v])_j = v_j$ for all $j \neq i$. Then, following exactly the same arguments as in (20) and (21) in the proof of Theorem 4.1, we know that for any (data-dependent) policy $\hat{\pi}$, we have

$$\begin{aligned} \sup_{R_v \in \mathcal{R}_0(\bar{c}, \gamma, T)} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] &\geq \frac{1}{2^d} \sum_{v \in \mathcal{V}} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] \geq \frac{1}{2^d} \sum_{v \in \mathcal{V}} \frac{1}{d} \sum_{i=1}^d \delta \cdot \mathbb{E}_v[\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_v^*(x_i)\}] \\ &\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}} \mathbb{E}_v[\mathbb{1}\{\hat{\pi}(x_i) \neq \pi_v^*(x_i)\}] \\ &\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} (1 - \text{TV}(\mathbb{P}_v, \mathbb{P}_{M_i[v]})) \\ &\geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \frac{1}{2} \exp(-\text{KL}(\mathbb{P}_v \| \mathbb{P}_{M_i[v]})), \end{aligned}$$

where $\text{KL}(\mathbb{P}_v \| \mathbb{P}_{M_i[v]})$ denotes the KL divergence of $\{X_t, Y_t, A_t\}$ under R_v versus under $R_{M_i[v]}$. Again, following the arguments in the proof of Theorem 4.6, we know that

$$\log \left(\frac{\mathbb{P}_v}{\mathbb{P}_{M_i[v]}}(x_j, y, a) \right) = (2y - 1) \log \left(\frac{1/2 + \delta}{1/2 - \delta} \right) \mathbb{1}\{a = f_2(x_j), j = i\}.$$

Therefore, by our construction, the KL divergence can be bounded as

$$\begin{aligned} \text{KL}(\mathbb{P}_v \parallel \mathbb{P}_{v'}) &\leq \sum_{t=1}^T \mathbb{E}[\mathbb{P}_v(A_t = f_2(x_i), X_t = x_i) \mathbb{E}_v[2Y - 1 \mid A_t, X_t]] \log \left(\frac{1/2 + \delta}{1/2 - \delta} \right) \\ &\leq \frac{12\delta^2 \bar{c}}{d} \cdot \sum_{t=1}^T t^{-\gamma} \leq \frac{12\delta^2 \bar{c}}{d} \cdot \frac{T^{1-\gamma}}{1-\gamma}, \end{aligned}$$

where we use the fact that $\log(1 + 2\delta) \leq 2\delta$ and $\log(1 - 2\delta) \geq -4\delta$ for $\delta \in (0, 1/4)$. Combining the above results, we have

$$\sup_{R_v \in \mathcal{R}_0(\bar{c}, \gamma, T)} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] \geq \frac{\delta}{d \cdot 2^d} \sum_{i=1}^d \sum_{v \in \mathcal{V}: v_i=1} \frac{1}{2} \exp \left(- \frac{12\delta^2 \bar{c}}{d} \cdot \frac{T^{1-\gamma}}{1-\gamma} \right) = \delta \cdot \exp \left(- \frac{12\delta^2 \bar{c}}{d} \cdot \frac{T^{1-\gamma}}{1-\gamma} \right).$$

Finally, taking $\delta = \sqrt{\frac{(1-\gamma)d}{24\bar{c} \cdot T^{1-\gamma}}} \leq 1/4$, we arrive at

$$\begin{aligned} \sup_{R_v \in \mathcal{R}_0(\bar{c}, \gamma, T)} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] &\geq \sup_{R_v \in \mathcal{R}_0(\bar{c}, \gamma, T)} \mathbb{E}_v[\mathcal{L}(\hat{\pi}; \mathcal{C}_v, \Pi)] \\ &\geq \frac{\exp(-1/2)}{\sqrt{24\bar{c}/(1-\gamma)}} \sqrt{\frac{d}{T^{1-\gamma}}} \geq 0.12 \sqrt{\frac{(1-\gamma)d}{\bar{c}T^{1-\gamma}}}, \end{aligned}$$

which concludes the proof of Theorem 5.7. $\square$

E Proof of supporting lemmas

E.1 Proof of Lemma C.1

Proof of Lemma C.1. Fix any $\pi \in \Pi$ and any $\eta > 0$. Since (Y'_t, A'_t) are independent copies of (Y_t, A_t) conditional on X_t , it is easy to check that

$$\mathbb{E}[\hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t)) \mid \{X_t, A_t, Y_t\}_{t=1}^T] = 0, \quad (36)$$

and $\{\hat{\Gamma}'_t(\pi)\}_{t=1}^T$ are mutually independent conditional on $\{X_t, A_t, Y_t\}_{t=1}^T$. Then by Chebyshev's inequality,

$$\begin{aligned} &\mathbb{P} \left(\left| \sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t)) \right| \geq \eta T \mid \{X_t, A_t, Y_t\}_{t=1}^T \right) \\ &\leq \frac{1}{\eta^2 T^2} \mathbb{E} \left[\left(\sum_{t=1}^T \{\hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\} \right)^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T \right] \\ &= \frac{1}{\eta^2 T^2} \sum_{t=1}^T \mathbb{E} \left[\{\hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\}^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T \right]. \end{aligned}$$

In view of (36), each term in the last summation can be bounded as

$$\begin{aligned} &\mathbb{E} \left[\{\hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\}^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T \right] = \text{Var}(\hat{\Gamma}'_t(\pi) \mid X_t) \\ &= \text{Var} \left(\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))} (Y'_t - \hat{\mu}(X_t, \pi(X_t))) \mid X_t \right) \\ &\leq \mathbb{E} \left[\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} (Y'_t - \hat{\mu}(X_t, \pi(X_t)))^2 \mid X_t \right] \leq \frac{1}{e(X_t, \pi(X_t))}, \end{aligned}$$

where the last inequality uses that fact that $Y'_t \in [0, 1]$ and $\hat{\mu}(X_t, \pi(X_t)) \in [0, 1]$. We then have

$$\mathbb{P}\left(\left|\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right| \geq \eta T \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \leq \frac{1}{\eta^2 T^2} \sum_{t=1}^T \frac{1}{e(X_t, \pi(X_t))} = \frac{V_p(\pi)^2}{\eta^2}.$$

Taking $\eta = 2V_p(\pi)$ (this is measurable with respect to $\{X_t, A_t, Y_t\}_{t=1}^T$) in the above inequality leads to

$$\mathbb{P}\left(\frac{1}{T} \left|\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right| \geq 2V_p(\pi) \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \leq \frac{1}{4}.$$

We then proceed to prove the second inequality. For any $\pi \in \Pi$ and any $\zeta > 0$,

$$\begin{aligned} & \mathbb{P}\left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} \geq \zeta + V_p(\pi)^2 T^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \\ & \leq \mathbb{P}\left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} - \mathbb{E}\left[\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} \mid \{X_t, A_t, Y_t\}_{t=1}^T\right] \geq \zeta \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \\ & \leq \frac{1}{\zeta^2} \sum_{t=1}^T \text{Var}\left(\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2} \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \\ & \leq \frac{1}{\zeta^2} \sum_{t=1}^T \mathbb{E}\left[\left(\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e(X_t, \pi(X_t))^2}\right)^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T\right] \\ & = \frac{1}{\zeta^2} \sum_{t=1}^T \frac{1}{e(X_t, \pi(X_t))^3} = \frac{V_h(\pi)^4 T^4}{\zeta^2}, \end{aligned}$$

where the second inequality is because $\{A'_t\}_{t=1}^T$ are mutually independent conditional on $\{X_t, A_t, Y_t\}_{t=1}^T$. Taking $\zeta = 2V_h(\pi)^2 T^2$, we have

$$\mathbb{P}\left(V'_s(\pi)^2 \geq 4 \cdot \max(V_p(\pi)^2, V_h(\pi)^2) \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \leq \frac{1}{4},$$

which concludes the proof of Lemma C.1. $\square$

E.2 Proof of Lemma D.1

Proof of Lemma D.1. Fix $\pi \in \Pi$ and $\eta > 0$. By Chebyshev's inequality, we have

$$\begin{aligned} & \mathbb{P}\left(\frac{1}{T} \left|\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right| \geq \eta \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \\ & \leq \frac{1}{\eta^2 T^2} \mathbb{E}\left[\left(\sum_{t=1}^T \hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right)^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T\right]. \end{aligned}$$

Since $\{A'_t, Y'_t\}_{t=1}^T$ are mutually independent conditional on $\{X_t, A_t, Y_t\}_{t=1}^T$, the right-hand side of the above can be written as

$$\begin{aligned} & \frac{1}{\eta^2 T^2} \sum_{t=1}^T \mathbb{E}\left[\left(\hat{\Gamma}'_t(\pi) - \mu(X_t, \pi(X_t))\right)^2 \mid \{X_t, A_t, Y_t\}_{t=1}^T\right] \\ & = \frac{1}{\eta^2 T^2} \sum_{t=1}^T \text{Var}\left(\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) \mid \mathcal{H}_t)} \mid \{X_t, A_t, Y_t\}_{t=1}^T\right) \end{aligned}$$

$$\begin{aligned}
&\leq \frac{1}{\eta^2 T^2} \sum_{t=1}^T \mathbb{E} \left[\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2} \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right] \\
&\leq \frac{1}{\eta^2 T^2} \sum_{t=1}^T \frac{1}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} \leq \frac{V_p(\pi)^2}{\eta^2},
\end{aligned}$$

where the step is because $\mathbb{E}[\hat{\Gamma}_t(\pi) | \{X_t, A_t, Y_t\}_{t=1}^T] = \mu(X_t, \pi(X_t))$ and the last step uses the fact that $g_t(x, \pi(x)) \leq e_t(x, \pi(x) | \mathcal{H}_t)$. Taking $\eta = 2V_p(\pi)$, we have

$$\mathbb{P} \left(\frac{1}{T} \left| \sum_{t=1}^T \hat{\Gamma}_t(\pi) - \mu(X_t, \pi(X_t)) \right| \geq 2V_p(\pi) \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right) \leq \frac{1}{4}.$$

We proceed to bound the conditional probability of $\mathcal{E}_2$. For any $\pi \in \Pi$ and $\zeta > 0$ that may depend on $\{X_t, A_t, Y_t\}_{t=1}^T$, since $(\zeta + V_p(\pi))^2 \geq \zeta^2 + V_p(\pi)^2$, we have

$$\begin{aligned}
&\mathbb{P}(V'_s(\pi) \geq \zeta + V_p(\pi) | \{X_t, A_t, Y_t\}_{t=1}^T) \\
&\leq \mathbb{P} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2} \geq \zeta^2 T^2 + T^2 V_p(\pi)^2 \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right)
\end{aligned} \tag{37}$$

Note that by construction

$$\mathbb{E} \left[\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)^2} \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right] = \sum_{t=1}^T \frac{1}{e_t(X_t, \pi(X_t) | \mathcal{H}_t)} = T^2 V_p(\pi)^2.$$

Then by Chebyshev's inequality, we have

$$\begin{aligned}
(37) &\leq \mathbb{P} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t))^2} - \mathbb{E} \left[\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t))^2} \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right] \geq \zeta^2 T^2 \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right) \\
&\leq \frac{1}{\zeta^4 T^4} \cdot \text{Var} \left(\sum_{t=1}^T \frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t))^2} \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right) \\
&= \frac{1}{\zeta^4 T^4} \cdot \sum_{t=1}^T \text{Var} \left(\frac{\mathbb{1}\{A'_t = \pi(X_t)\}}{e_t(X_t, \pi(X_t))^2} \middle| \{X_t, A_t, Y_t\}_{t=1}^T \right) \\
&\leq \frac{1}{\zeta^4 T^4} \cdot \sum_{t=1}^T \frac{1}{e_t(X_t, \pi(X_t))^3} \leq \frac{V_h(\pi)^4}{\zeta^4},
\end{aligned}$$

where in the next-to-last step we use the independence between $\{A'_t\}_{t=1}^T$ conditional on $\{X_t, A_t, Y_t\}_{t=1}^T$. Taking $\zeta = 2V_h(\pi)$, we have

$$\mathbb{P}(V'_s(\pi) \geq 2 \cdot \max\{V_h(\pi), V_p(\pi)\} | \{X_t, A_t, Y_t\}_{t=1}^T) \leq \frac{1}{4},$$

for any $\pi \in \Pi$. Now that $\pi^\dagger$ only depends on $\{X_t, A_t, Y_t\}_{t=1}^T$, hence the two inequalities also hold for $\pi^\dagger$. Therefore, we conclude the proof of Lemma D.1. $\square$

F Auxiliary lemmas

The following Natarajan's lemma characterizes the number of distinct realizations of $\pi(x)$ for $\pi \in \Pi$, which allows us to turn the uniform bound over Π to that over a finite set with polynomially growing size.

Lemma F.1 (Natarajan (1989)). Let $\Pi := \{\pi : S \mapsto \mathcal{A}\}$ for a finite set $S \in \mathcal{X}^{|S|}$. Then

$$|\Pi| \leq |\mathcal{X}|^{\text{N-dim}(\Pi)} |\mathcal{A}|^{2\text{N-dim}(\Pi)}.$$

The following self-normalized concentration inequality is from Corollary 2.2 of De la Pena et al. (2004).

Lemma F.2 (Self-normalized concentration). Suppose two random variables A and $B > 0$ satisfies that $\mathbb{E}[\exp(\lambda A - \lambda^2 B^2/2)] \leq 1$ for all $\lambda \in \mathbb{R}$. Then for all $x \geq \sqrt{2}$ and all $y > 0$, it holds that

$$\mathbb{P}\left(\frac{|A|}{\sqrt{(B^2 + y)(1 + 0.5 \log(1 + B^2/y))}} \geq x\right) \leq \exp(-x^2/2).$$

The lemma below describes Hoeffding's inequality (Hoeffding, 1994).

Lemma F.3 (Hoeffding's inequality). Let $X_1, \dots, X_n$ be independent random variables on $\mathbb{R}$ such that $a_i \leq X_i \leq b_i$ almost surely. If $S_n = \sum_{i=1}^n X_i$, then for all $x > 0$,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq x) \leq \exp\left(-\frac{2x^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

The next lemma is adapted from Freedman's inequality (Freedman, 1975), with Bernstein's inequality as a special case.

Lemma F.4 (Freedman's inequality). Consider a real-valued martingale $\{Y_k\}_{k \geq 0}$ adapted to a filtration $\{\mathcal{F}_k\}$, and its difference sequence $\{X_k = Y_k - Y_{k-1}\}_{k \geq 1}$. Assume the difference sequence is uniformly bounded: $X_k \leq R$ almost surely for $k \geq 1$. Define the predictable quadratic variation process of the martingale: $M_k := \sum_{j=1}^k \mathbb{E}[X_j^2 | \mathcal{F}_{j-1}]$. Then for all $x \geq 0$ and $\sigma^2 > 0$, we have

$$\mathbb{P}(\exists k \geq 0: Y_k \geq x \text{ and } M_k \leq \sigma^2) \leq \exp\left(\frac{-x^2/2}{\sigma^2 + Rx/3}\right).$$

As a special case, if $Y_0 = 0$ and $Y_k = \sum_{j=1}^k X_j$ for $k = 1, \dots, n$, where $\{X_k\}_{k=1}^n$ are independent mean-zero random variables, it becomes Bernstein's inequality with $\sigma^2 = \sum_{k=1}^n \text{Var}(X_k)$.